

INTRODUCTION TO ORDINARY DIFFERENTIAL EQUATIONS

with Scientific Applications



First Edition

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Introduction to Ordinary Differential Equations with Scientific Applications

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Preface

This book started off as an online tutorial series about ordinary differential equations (ODEs) posted on my personal webpage, <https://benjamath.com/>. Eventually, as the content got richer and more refined, I decided to organize it into a well-designed book to present the theories of ODEs in a more formal and coherent fashion. This book discusses introductory ODEs comprehensively, from first-order/second-order equations, up to series solutions and special functions, with the hope to offer a full learning experience for keen students. It also comes with many scientific applications, particularly in Physics and Earth Sciences, which further help students to make connections between abstract theories and real-world problems. With carefully chosen examples, figures, and exercises, students can understand the material and concepts in this book effectively. This book can also serve as a quick reference and question bank for instructors. Overall, this book is suitable for undergraduate students majoring in Mathematics, Engineering, Physics, and Earth Sciences who already have basic familiarity with Calculus and are taking a first course in ordinary differential equations. Specifically, it is expected that the readers are comfortable with prerequisite knowledge up to Multivariable Calculus (including, but not limited to, integrating by parts and partial derivatives). For US students, this will be equivalent to Calc III.

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This book heavily references "Differential Equations with Applications and Historical Notes" by Simmons [9] and "Mathematical Methods for Physics and Engineering: A Comprehensive Guide" by Riley, Hobson, and Bence [7]. Readers are encouraged to read them as well.

The full solution manual and errata are available at https://github.com/BenjaminGor/Intro_to_ODE_sol_err.

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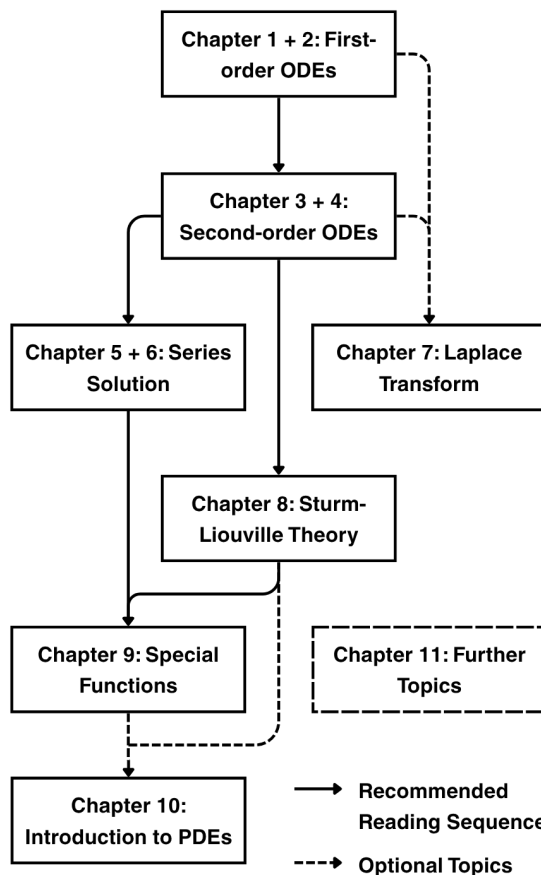
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How to Read this Book

Although each section may be taken out as a standalone and used for quick reference, this book is designed to be a complete course because the later chapters are strongly based on the earlier ones. The recommended reading flow is illustrated below:



Starred (sub)sections are optional and may be skipped without heavily impacting later content.

Table of Symbols



y	Dependent variable, Independent variable (Chapter 10)
x, t	Independent variable (space, time)
u, v	Transformed dependent variable
w, z	Transformed independent variable
i	The imaginary number, Index
j	Index
k	Index, Constant
l, m, n	Index, Parameter
a, b, c	Constant (a and b are usually the lower/upper limit of an interval)
a_n, b_n, c_n	The n -th coefficient (in a series)
A, B, C	Integration constant
C_m^n	Combination (n choose m)
f, g, h	Function
F, G, M, N, P, Q, R	Function
p, q, r	Index, Function
s	Index, Parameter
I	Integral
$\alpha, \beta, \gamma, \dots$	Constant
δ	Dirac Delta
ε	Infinitesimally small quantity
λ	Eigenvalue
μ	Integrating factor, Parameter
ν	Parameter
φ	Eigenfunction
ρ	Weighting
τ, ξ	Dummy integration variable
Ω	Domain

y', y''	The first/second-order derivative of y (with respect to some independent variable implicitly)
$y^{(n)}$	The n -th derivative of y (with respect to some independent variable implicitly)
$\frac{dy}{dx}, \frac{d^2y}{dx^2}, \frac{d^3y}{dx^3}, \dots$	The first/second/third-order derivative of y with respect to x
$\frac{d^n y}{dx^n}$	The n th derivative of y with respect to x
$\int y dx$	The indefinite integral of y with respect to x
$\int_a^b y dx$	The definite integral of y from $x = a$ to $x = b$
$y _{x=a}$	y evaluated at $x = a$
y, f , etc.	Vector (Boldface)
$\frac{\partial u}{\partial x}$ or u_x	u partially differentiated with respect to x
$\frac{\partial^2 u}{\partial x^2}$ or u_{xx}	u partially differentiated with respect to x twice
$\frac{\partial^2 u}{\partial x \partial t}$ or u_{xt}	u partially differentiated with respect to x then t
$\mathcal{L}$	Linear operator
$\mathcal{L}$	Laplace Transform
x', z', τ' , etc.	Dummy integration variables (to be distinguished from y' or similar derivatives)
$ $	Absolute value
$\ \ $	Norm
$\langle \cdot, \cdot \rangle$	Inner product
$\lfloor \rfloor$	Floor function
Σ, Π	Summation and Product
(a, b)	Segment (excluding the end-points a, b)
$[a, b]$	Interval (including the end-points a, b)
$\{ \}$	Set
$\rightarrow$	Change of variable, Replaced by; or Taking a limit to
$\Rightarrow$	Implying that
$\equiv$	Identically equal to
$:=$	Defined as
$\therefore$	Therefore

Basic First-order Ordinary Differential Equations

Introduction Ordinary Differential Equations (ODEs) are equations in which the *unknown, dependent* variable $y = y(x)$ depends on only one *independent* variable x , and consist of terms that involve functions of x , y , and **ordinary derivatives** solely. In an ordinary derivative, y is differentiated any number of times, but only with respect to x . These include $\frac{dy}{dx}$, $\frac{d^2y}{dx^2}$, $\frac{d^3y}{dx^3}$, (or denoted by y' , y'' , $y^{(3)}$, ...) and so on. Some examples of ODEs are

$$\frac{dy}{dx} + (\sin x)y = e^{-x} \quad (1.1a)$$

$$\frac{d^2y}{dx^2} + 4y = 0 \quad (1.1b)$$

$$\frac{dy^3}{dx^3} + x\left(\frac{dy}{dx}\right)^2 + 2 \ln y = 1 \quad (1.1c)$$

ODEs appear very frequently in both daily life and science applications. For daily life, a possible scenario may be modeling the investments and revenues/expenses of a small company, where the evolution of its finances can be symbolically written as

$$\frac{dC}{dt} = G(C, t) - L(C, t) \quad (1.2)$$

Here, the time t takes the place of the independent variable; $C(t)$ denotes the capital possessed by the company at some time t , and is the dependent variable. Then, $\frac{dC}{dt}$ represents the rate of change in the capital of the company. $G(C, t)$ and $L(C, t)$ contain various sources of capital gain/loss and are functions of both the capital C and time t . For example, one of the terms in $L(C, t)$ may be about the purchase of raw materials, which can be modulated by the financial status (C) and current season (t).

■ Chapter 1 – Basic First-order Ordinary Differential Equations

In Atmospheric Sciences, we also have the energy budget E on the Earth's surface, which can be expressed as

$$\frac{dE}{dt} = \text{Shortwave Radiation} + \text{Longwave Radiation} + \text{Sensible Heat Flux} + \text{Latent Heat Flux} \quad (1.3)$$

Meanwhile, in Physics, a classic example will be the swinging motion of a pendulum, which can be cast into the following ODE:

$$\frac{d^2\theta}{dt^2} + \frac{g}{L} \sin \theta = 0 \quad (1.4)$$

where θ is the angle of the pendulum to the vertical, g is the gravitational acceleration, and L is the length of the string.

In this chapter, we are going to begin with the simplest class of ODEs: **First-order ODEs**. As their name suggests, there will be at most first-order derivatives (differentiated only once). We will first tackle the most common types of first-order ODEs, such as separable equations, linear equations, and exact equations.

1.1. Three Basic Classifications of ODEs

Before anything, we need to know that there are three basic classifications of ODEs: *Order*, *Linearity*, and *Homogeneity*. The **order** of an ODE refers to the highest order of ordinary derivative(s) appearing in the equation.

Definition 1.1.

(Order of an ODE) The order of an ODE (in the unknown y) is equal to the unique positive integer $n \geq 1$, where

$$n = \max(m) \text{ such that } y^{(m)} \text{ appears in the ODE} \quad (1.5)$$

where $y^{(m)}$ denotes the m -th derivative of y .

For example, (below, the highest-order derivative is highlighted in red)

1. $\frac{dy}{dx} + 2y = 3 \sin^3 x$ is first-order;
2. $2 \frac{d^2y}{dx^2} + x^5 \frac{dy}{dx} - y^3 = 1$ is second-order;
3. $\left(\frac{d^2y}{dx^2}\right)^5 + x^4 \frac{d^3y}{dx^3} + 7y^5 = e^{-x}$ is third-order.

We usually arrange the terms of an ODE in descending order of derivatives.

An ODE is called **linear** if, within each additive term, only at most one of the dependent variable y or its ordinary derivatives $\frac{dy}{dx}, \frac{d^2y}{dx^2}, \frac{d^3y}{dx^3}, \dots$ is present, with a power of 1. Otherwise, when the dependent variable and derivatives appear as a product together, have some power other than 1, or are embedded in other non-linear functions, it is called **non-linear**. A more formal definition is given by

Definition 1.2.

(Linearity of an ODE) An ODE is known as linear if it can be written in the form of

$$a_0(x)y + a_1(x)y' + a_2(x)y'' + \cdots + a_n(x)y^{(n)} = f(x) \quad (1.6)$$

where n is a positive integer, and $a_m(x)$, for $0 \leq m \leq n$, as well as $f(x)$, are functions of the independent variable x only. $a_m(x)$ is sometimes referred to as the *linear coefficient* of $y^{(m)}$.

Note that the linear coefficients are named relative to y and need not be linear in x . We require $a_n(x) \neq 0$ for an n -th order linear ODE.

For instance,

1. $x^2 \frac{d^3y}{dx^3} + 5y = -2x$, and
2. $3 \frac{dy}{dx} + (\cos^2 x)y = e^{-x} \tan x$

are both linear, while

1. $y \frac{dy}{dx} = 3x$,
2. $\frac{dy}{dx} - \cos y = 2$, and
3. $\left(\frac{dy}{dx}\right)^2 + 4y^5 = 0$

are all non-linear (the non-linear terms are highlighted in red).

Finally, a **homogeneous** ODE is a linear ODE that has no term involving the independent variable x only (including any non-zero constant as well). Otherwise, it is known as **non-homogeneous/inhomogeneous**. Bear in mind that the notion of homogeneity only applies to linear ODEs in particular, and is not so meaningful for non-linear ODEs.

Definition 1.3.

(Homogeneity of an ODE) A linear ODE as given by Definition 1.2 is called homogeneous if $f(x) \equiv 0$ in (1.6).

■ Chapter 1 – Basic First-order Ordinary Differential Equations

For example, $e^x \frac{dy}{dx} - x^2 y = 0$ is homogeneous, but $\frac{dy}{dx} = \alpha x + 1$, with α being a constant, is always non-homogeneous even when $\alpha = 0$.

The $f(x)$ term on the R.H.S. in (1.6) is often referred to as the *source term/forcing term/inhomogeneous part*, while the L.H.S. of (1.6) alone is called the *homogeneous part*, often implying an associated ODE with the same homogeneous part but no inhomogeneous part. An alternative definition of linearity then can be made with reference to the homogeneous part of an ODE.

Definition 1.4. (**Linearity of an ODE (Alternative)**) An ODE is linear if, for all pairs of numbers α, β and two distinct¹ solutions y_1, y_2 for its homogeneous part, $\alpha y_1 + \beta y_2$ is also a solution for the homogeneous part.

This is not very hard to see, using the linearity of derivatives themselves:

$$\begin{aligned} & a_0(x)(\alpha y_1 + \beta y_2) + a_1(x)(\alpha y_1 + \beta y_2)' + \cdots + a_n(x)(\alpha y_1 + \beta y_2)^{(n)} \\ &= \alpha \left(a_0(x)y_1 + a_1(x)y_1' + \cdots + a_n(x)y_1^{(n)} \right) + \beta \left(a_0(x)y_2 + a_1(x)y_2' + \cdots + a_n(x)y_2^{(n)} \right) \\ &= \alpha(0) + \beta(0) = 0 \end{aligned} \tag{1.7}$$

It may be checked that the three ODEs (1.1) shown in the beginning are

1. First-order, linear, non-homogeneous;
2. Second-order, linear, homogeneous; and
3. Third-order, non-linear.

In Table 1.1 on the next page, we classify six more ODEs for demonstration.

¹How we define "distinct" will be made clear in Chapter 3.

	Order	Linearity	Homogeneity
$\frac{d^2y}{dx^2} - 2\frac{dy}{dx} - 3y = 0$	Second-order	Linear	Homogeneous
$\frac{dy}{dx} + x^4y = 2$	First-order	Linear	Non-homogeneous
$y\frac{d^3y}{dx^3} + 4x^2y = e^{-x}$	Third-order	Non-linear	
$\frac{dy}{dx} = x(1 - xe^{-y})$	First-order	Non-linear	
$\frac{dy}{dx} + (\cos x)y + e^{-x^2} = 0$	First-order	Linear	Non-homogeneous
$\cot\left(\frac{d^2y}{dx^2}\right) = e^{-x}$	Second-order	Linear ²	Non-homogeneous

Table 1.1.: Three basic classifications: order, linearity, and homogeneity, for some selected ODEs.

1.2. Separable First-order ODEs

Now we are going to study the first major kind of first-order ODEs.

Definition 1.5.

(Separable First-order ODEs) Separable ODEs are first-order ODEs that can be written in the form of

$$\frac{dy}{dx} = f(x)g(y) \quad (1.8)$$

where $f(x)$ is some function of x only, while $g(y)$ is some function of y only.

1.2.1. Separation of Variables

The reason why this kind of ODEs is called separable is pertinent to their solution method, **Separation of Variables**. Heuristically, we move $g(y)$ and group it with dy and do the same for dx and $f(x)$, in order to separate the parts containing x and y respectively. This involves treating the derivative $\frac{dy}{dx}$ like a fraction informally, and the rigorous justification is out of our scope. Subsequently, we can integrate both sides together:

$$\frac{dy}{dx} = f(x)g(y)$$

²Just take $\cot^{-1}$ on both sides.

$$\begin{aligned}\frac{dy}{g(y)} &= f(x)dx \\ \int \frac{dy}{g(y)} &= \int f(x)dx\end{aligned}\tag{1.9}$$

1.2.2. Initial Condition

The full solution for (1.8) also requires an **initial condition (I.C.)** to determine the value of the integration constant that emerges at the end of (1.9). In general, every first-order ODE requires exactly one initial condition.

Definition 1.6. (**Initial Condition for First-order ODEs**) An initial condition for a first-order ODE consists of an imposed value of y at some point $x = a$, denoted by $y(a)$.

An ODE that comes with some initial condition(s) is together known as an **initial value problem**. Now let's illustrate this with a small example.

Example 1.1. Solve the following separable first-order ODE

$$\frac{dy}{dx} = \frac{x^2}{y}\tag{1.10}$$

with the initial condition $y = 1$ when $x = 0$, succinctly written as $y(0) = 1$.

Solution. Doing Separation of Variables as in (1.9) gives

$$\begin{aligned}\frac{dy}{dx} &= \frac{x^2}{y} \\ \int y dy &= \int x^2 dx \\ \frac{1}{2}y^2 &= \frac{1}{3}x^3 + C \\ y &= \pm\sqrt{2}\sqrt{\frac{1}{3}x^3 + C}\end{aligned}\tag{1.11}$$

where a constant $+C$ appears after the indefinite integration. Given the I.C. $y(0) = 1$, we will have to take the positive root in the solution (1.11), and substitution then yields

$$(1) = \sqrt{2}\sqrt{\frac{1}{3}(0)^3 + C}$$

$$\begin{aligned}
 1 &= \sqrt{2C} \\
 C &= \frac{1}{2}
 \end{aligned} \tag{1.12}$$

$$\therefore y = \sqrt{2} \sqrt{\frac{1}{3}x^3 + \frac{1}{2}} = \sqrt{\frac{2}{3}x^3 + 1}, \quad x > \left(-\frac{3}{2}\right)^{\frac{1}{3}} \tag{1.13}$$

□

1.2.3. Example: Radioactive Decay

A recurring physical example of separable ODEs is *radioactive decay*, which we now discuss.

Example 1.2.

(Radioactive Decay) The radioactive decay process can be described by the following separable ODE:

$$\frac{dN}{dt} = -kN \tag{1.14}$$

where $N(t)$ is the number of radioactive atoms at time t , and $k > 0$ is the decay constant, which indicates the probability of an individual isotope atom to decay per unit time and has the unit of s^{-1} . So, on average, there will be kN radioactive atoms that decay over a unit time interval. This implies the form of the ODE, where the L.H.S., $\frac{dN}{dt}$, symbolizes the corresponding rate of change in the number of radioactive atoms. Derive its solution.

Solution. Applying Separation of Variables to (1.14) leads to

$$\begin{aligned}
 \frac{dN}{dt} &= -kN \\
 \int \frac{dN}{N} &= - \int k dt \\
 \ln |N| &= -kt + C \\
 N &= e^C e^{-kt} \quad (N > 0)
 \end{aligned} \tag{1.15}$$

If the initial isotope concentration is N_0 when $t = 0$ (I.C.), then

$$\begin{aligned}
 (N_0) &= e^C e^{-k(0)} \\
 e^C &= N_0 \\
 \therefore N &= N_0 e^{-kt}
 \end{aligned} \tag{1.16}$$

The full solution (1.16) for the radioactive process is thus an *exponential decay*. From this, we can infer the time required for the radioactive isotope to halve in amount, called the *half-life*, $\tau_{1/2}$:

$$\frac{\frac{1}{2}N}{N} = \frac{N_0 e^{-k(t+\tau_{1/2})}}{N_0 e^{-kt}}$$

$$\frac{1}{2} = e^{-k\tau_{1/2}}$$

$$\tau_{1/2} = \frac{\ln 2}{k} \quad (1.17)$$

Hence, there exists an inverse relationship between the half-life and decay constant. For example, the carbon-14 isotope is known to have a half-life of about 5730 years, and the corresponding decay constant is therefore

$$k = \frac{\ln 2}{5730 \text{ yrs}} = \frac{0.693147}{1.81 \times 10^{11} \text{ s}} \approx 3.83 \times 10^{-12} \text{ s}^{-1} \quad (1.18)$$

A schematic plot for the exponential decay is given as Figure 1.1 below. □

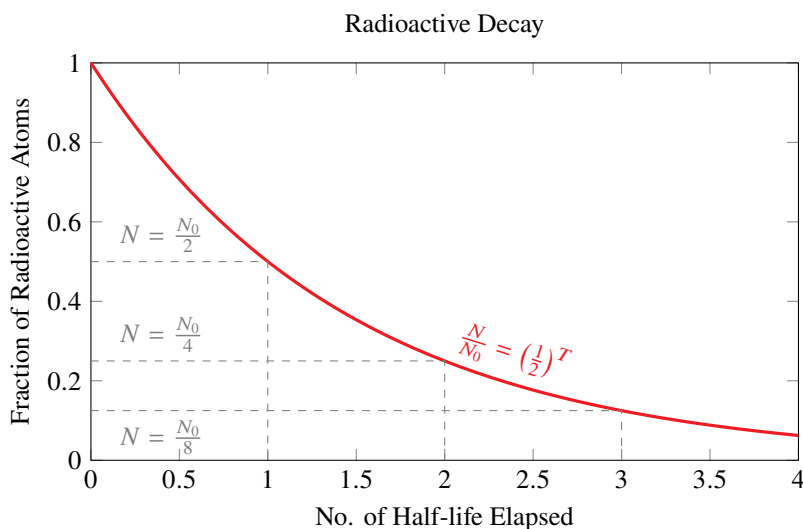


Figure 1.1.: Exponential radioactive decay: the x-axis indicates how many half-life, T , have passed, and the y-axis is the fraction of the remaining radioactive atoms.

1.2.4. Example: Logistic Population Model

Another example commonly seen in scientific literature is the *logistic population model*.

Example 1.3.

(Logistic Population Model) The rate of change in a population may be modelled by the separable ODE:

$$\frac{dN}{dt} = KN(M - N) \quad (1.19)$$

where $N = N(t)$ represents the population at time t , K is the natural growth factor, and M is the maximum sustainable population. Then, KN represents the birth capacity, and $M - N$ acts like a soft barrier to limit the population increase, which corresponds to real-life restrictions like food consumption. Given the I.C. at $t = 0$ is $N(0) = \frac{M}{4}$: the initial population is a quarter of the maximum population, find the corresponding solution. How about if $N(0) = \frac{M}{3}$ or $\frac{M}{6}$?

Solution. Again, by Separation of Variables, we have

$$\begin{aligned} \frac{dN}{dt} &= KN(M - N) \\ \int \frac{dN}{N(M - N)} &= \int K dt \\ \int \frac{1}{M} \left(\frac{1}{N} + \frac{1}{M - N} \right) dN &= \int K dt \end{aligned} \quad (1.20)$$

where we have utilized the *partial fractions* $\frac{1}{N(M-N)} = \frac{1}{M} \left(\frac{1}{N} + \frac{1}{M-N} \right)$. Now integrate both sides to get

$$\begin{aligned} \frac{1}{M} (\ln |N| - \ln |M - N|) &= Kt + C \\ \ln \left(\frac{N}{M - N} \right) &= MKt + MC \quad (M, N > 0) \\ \frac{N}{M - N} &= e^{MKt + MC} = Ae^{MKt} \end{aligned} \quad (1.21)$$

where we let $A = e^{MC}$. Then

$$\begin{aligned} N &= (M - N)Ae^{MKt} \\ N(1 + Ae^{MKt}) &= MAe^{MKt} \\ N &= \frac{MAe^{MKt}}{1 + Ae^{MKt}} \end{aligned} \quad (1.22)$$

Plugging in the I.C. $N(0) = \frac{M}{4}$ now gives

$$\left(\frac{M}{4} \right) = \frac{MA(1)}{1 + A(1)} = \frac{MA}{1 + A}$$

$$\begin{aligned}\frac{1}{4} &= \frac{A}{1+A} \\ A &= \frac{1}{3}\end{aligned}\tag{1.23}$$

Therefore, the full solution is

$$N = \frac{M\left(\frac{1}{3}\right)e^{MKt}}{1 + \left(\frac{1}{3}\right)e^{MKt}} = \frac{Me^{MKt}}{3 + e^{MKt}}\tag{1.24}$$

In a similar vein, if $N(0) = \frac{M}{3}$, we will obtain $A = \frac{1}{2}$ and $N = \frac{Me^{MKt}}{2+e^{MKt}}$. Else if $N(0) = \frac{M}{6}$, then $A = \frac{1}{5}$ and $N = \frac{Me^{MKt}}{5+e^{MKt}}$. The solution curves, which have the shape of *sigmoids*, are plotted in Figure 1.2 for the case $M = 100, K = 0.001$ as follows. It can be readily seen that the population always approaches the saturation capacity M . $\square$

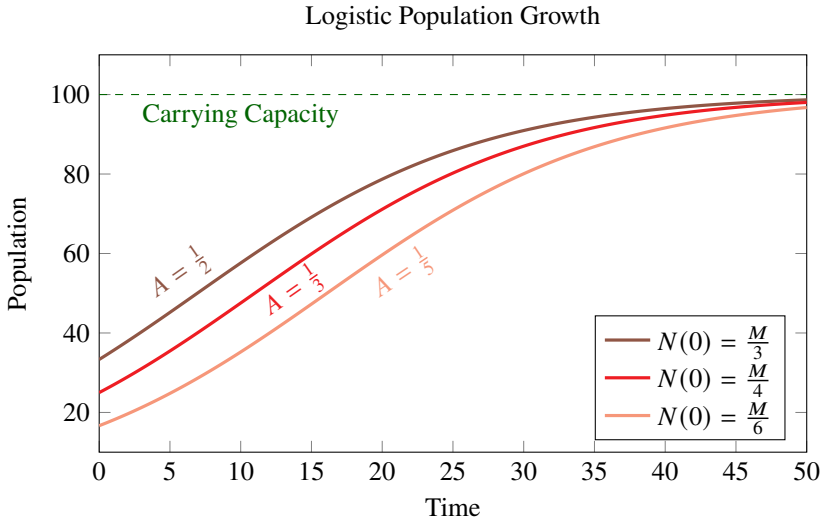


Figure 1.2.: *Logistic Population Model: the x-axis is the time t and the y-axis is the current population N . Here $M = 100, K = 0.001$.*

1.3. First-order Linear ODEs

The next major kind of first-order ODEs to be introduced is

Definition 1.7.

(First-order Linear ODEs) First-order linear ODEs are first-order ODEs that can be written in the form of

$$\frac{dy}{dx} + P(x)y = Q(x) \quad (1.25)$$

where $P(x), Q(x)$ are some functions of the independent variable x . In particular, $Q(x)$ here is the source/inhomogeneous term.

The linear part refers to the $P(x)y$ term. Examples of first-order linear ODEs are:

1. $\frac{dy}{dx} + (\cos x)y = e^{-x^2}$;
2. $\frac{dy}{dx} - 3y = 0$; and
3. $\frac{dy}{dx} - x^2 e^x y = 1 - \sqrt{x}$.

1.3.1. Integrating Factor

The solution method is via an **integrating factor**. We set out to find this integrating factor $\mu(x)$ such that multiplying it on both sides of (1.25) gives

$$\mu(x) \frac{dy}{dx} + \mu(x)P(x)y = \mu(x)Q(x) \quad (1.26)$$

and we require that the two terms on the L.H.S. can be grouped into a single derivative:

$$\mu(x) \frac{dy}{dx} + \mu(x)P(x)y = \frac{d}{dx}(\mu(x)y) \quad (1.27)$$

On the other hand, by *Product Rule*, we have

$$\frac{d}{dx}(\mu(x)y) = \mu(x) \frac{dy}{dx} + \frac{d\mu(x)}{dx}y \quad (1.28)$$

Comparing (1.27) and (1.28), for it to work, we need

$$\frac{d\mu(x)}{dx} = \mu(x)P(x) \quad (1.29)$$

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This is a separable ODE. Subsequently, we can use Separation of Variables, described in the last section, to arrive at

$$\begin{aligned}\int \frac{d\mu(x)}{\mu(x)} &= \int P(x)dx \\ \ln |\mu(x)| &= \int P(x)dx \\ \mu(x) &= \exp\left(\int P(x)dx\right)\end{aligned}\tag{1.30}$$

This will be the form of integrating factor we need (here, the integration constant and absolute sign do not matter), and combining (1.26) and (1.27) will lead to the desired solution:

$$\begin{aligned}\frac{d}{dx}(\mu(x)y) &= \mu(x)Q(x) \\ \mu(x)y &= \int \mu(x)Q(x)dx \\ y &= \frac{1}{\mu(x)} \int \mu(x)Q(x)dx + C\end{aligned}\tag{1.31}$$

Bear in mind that the integration constant is now required at the very end.

Example 1.4.

Solve the first-order linear ODE

$$\frac{dy}{dx} + 4xy = e^{-2x^2}\tag{1.32}$$

with the initial condition $y(0) = -1$.

Solution. Here $P(x) = 4x$, and by (1.30), the integrating factor is hence

$$\mu(x) = \exp\left(\int P(x)dx\right) = \exp\left(\int 4x dx\right) = e^{2x^2}\tag{1.33}$$

By multiplying this $\mu(x)$ on both sides of (1.32), combining the L.H.S. into a single derivative, and then integrating both sides, we obtain

$$\begin{aligned}e^{2x^2} \frac{dy}{dx} + 4xe^{2x^2}y &= 1 \\ \frac{d}{dx}(e^{2x^2}y) &= 1 \\ e^{2x^2}y &= \int dx = x + C \\ y &= xe^{-2x^2} + Ce^{-2x^2}\end{aligned}\tag{1.34}$$

The first (red) term is called the *particular solution*, produced by the non-homogeneous source term $Q(x)$, and no integration constant is involved. Meanwhile, the second (green) term is known as the *complementary solution*, induced by the homogeneous part, and accompanied by an integration constant.

Plugging in the provided I.C., we have

$$\begin{aligned}(-1) &= 0(1) + C(1) \\ C &= -1\end{aligned}\tag{1.35}$$

So the full solution will be

$$y = xe^{-2x^2} - e^{-2x^2} = (x - 1)e^{-2x^2}\tag{1.36}$$

We can also opt to incorporate the initial condition at the outset by utilizing a definite integral instead. Then, we will write something along the lines of

$$\begin{aligned}\frac{d}{dx}(e^{2x^2}y) &= 1 \\ [e^{2x^2}y]_0^x &= \int_0^x dx = [x]_0^x \\ e^{2x^2}y(x) - e^{2(0)^2}y(0) &= e^{2x^2}y - (1)(-1) = (x) - (0) \\ e^{2x^2}y + 1 &= x \\ y &= xe^{-2x^2} - e^{-2x^2}\end{aligned}\tag{1.37}$$

□

1.3.2. Example: Falling Object

Another example is the simple physics of a downward-falling object subjected to gravity as well as linear air resistance.

Example 1.5.

(Falling Object and Air Resistance) Denote the downward velocity of a falling object at time t as $v(t)$, its acceleration can be modelled to follow

$$\frac{dv}{dt} = g - \alpha v\tag{1.38}$$

where α is a constant representing the magnitude of air resistance and has a unit of s^{-1} . Express the velocity $v(t)$ as well as the downward displacement $s(t) = \int v(t)dt$ in terms of the time elapsed t since it is released from rest. Also, if the object reaches 90% of the terminal speed at $t = 10$ s, find the value of α .

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Solution. To tackle the problem, we move the αv term from R.H.S. to L.H.S. to transform the ODE into the linear form:

$$\frac{dv}{dt} + \alpha v = g \quad (1.39)$$

From (1.30), the integrating factor is

$$\mu(t) = \exp\left(\int \alpha dt\right) = e^{\alpha t} \quad (1.40)$$

Multiplying this to (1.39), we have

$$\begin{aligned} e^{\alpha t} \frac{dv}{dt} + \alpha e^{\alpha t} v &= g e^{\alpha t} \\ \frac{d}{dt}(e^{\alpha t} v) &= g e^{\alpha t} \\ e^{\alpha t} v &= g \int e^{\alpha t} dt \\ e^{\alpha t} v &= g \left(\frac{1}{\alpha} e^{\alpha t} + C \right) = \frac{g}{\alpha} e^{\alpha t} + gC \\ v &= \frac{g}{\alpha} + gC e^{-\alpha t} \end{aligned} \quad (1.41)$$

The initial condition is naturally $v(0) = 0$ as the object is released from rest, so

$$\begin{aligned} (0) &= \frac{g}{\alpha} + gC(1) \\ C &= -\frac{1}{\alpha} \end{aligned} \quad (1.42)$$

and the full solution is thus

$$v = \frac{g}{\alpha} - \frac{g}{\alpha} e^{-\alpha t} = \frac{g}{\alpha} (1 - e^{-\alpha t}) \quad (1.43)$$

and

$$s = \int v dt = \int \left(\frac{g}{\alpha} - \frac{g}{\alpha} e^{-\alpha t} \right) dt = \frac{g}{\alpha} t + \frac{g}{\alpha^2} (e^{-\alpha t} - 1) \quad (1.44)$$

(The integration constant for the expression of the displacement s is determined to match $s(0) = 0$.) It is not hard to see that the terminal speed is $\frac{g}{\alpha}$ as $t \rightarrow \infty$. To reach 90% of the terminal speed at $t = 10$ s, we require

$$\begin{aligned} 1 - e^{-\alpha(10)} &= 0.9 \\ e^{-10\alpha} &= 0.1 \\ -10\alpha &= \ln 0.1 \\ \alpha &= \frac{\ln 10}{10} \end{aligned} \quad (1.45)$$

□

1.3.3. Linear ODEs and Integration by Parts

In this part, we will briefly touch on how *integration by parts (I.B.P.)* in elementary Calculus may be needed in solving linear ODEs.

Example 1.6.

Solve the following first-order linear ODE

$$\frac{dy}{dx} + 2y = \sin x \quad (1.46)$$

with the I.C. of $y(0) = \frac{1}{2}$.

Solution. Here $P(x) = 2$ and hence, by (1.30), the integrating factor is $\exp(\int P(x)dx) = \exp(\int 2dx) = e^{2x}$. Multiplying it on both sides leads to

$$\begin{aligned} e^{2x} \frac{dy}{dx} + 2e^{2x}y &= e^{2x} \sin x \\ \frac{d}{dx}(e^{2x}y) &= e^{2x} \sin x \\ e^{2x}y &= \int e^{2x} \sin x dx \end{aligned} \quad (1.47)$$

Set $I = \int e^{2x} \sin x dx$, and this integral on the R.H.S. can be dealt with using integration by parts. By applying I.B.P. twice, we can obtain

$$\begin{aligned} I &= \int e^{2x} \sin x dx = \frac{1}{2} \int \sin x d(e^{2x}) \\ &= \frac{1}{2} [e^{2x} \sin x] - \frac{1}{2} \int e^{2x} d(\sin x) && \text{(first I.B.P.)} \\ &= \frac{1}{2} [e^{2x} \sin x] - \frac{1}{2} \int e^{2x} \cos x dx \\ &= \frac{1}{2} [e^{2x} \sin x] - \frac{1}{4} \int \cos x d(e^{2x}) \\ &= \frac{1}{2} [e^{2x} \sin x] - \frac{1}{4} [e^{2x} \cos x] + \frac{1}{4} \int e^{2x} d(\cos x) && \text{(second I.B.P.)} \\ &= \frac{1}{2} [e^{2x} \sin x] - \frac{1}{4} [e^{2x} \cos x] - \frac{1}{4} \int e^{2x} \sin x dx \\ &= \frac{1}{2} [e^{2x} \sin x] - \frac{1}{4} [e^{2x} \cos x] - \frac{1}{4} I \end{aligned} \quad (1.48)$$

Therefore

$$I = \frac{1}{2} [e^{2x} \sin x] - \frac{1}{4} [e^{2x} \cos x] - \frac{1}{4} I$$

$$\begin{aligned}\frac{5}{4}I &= \frac{1}{2}[e^{2x} \sin x] - \frac{1}{4}[e^{2x} \cos x] \\ I &= \frac{2}{5}e^{2x} \sin x - \frac{1}{5}e^{2x} \cos x + C\end{aligned}\quad (1.49)$$

(Don't forget the integration constant here!) Substituting (1.49) back into (1.47) gives

$$\begin{aligned}e^{2x}y &= \frac{2}{5}e^{2x} \sin x - \frac{1}{5}e^{2x} \cos x + C \\ y &= \frac{2}{5} \sin x - \frac{1}{5} \cos x + Ce^{-2x}\end{aligned}\quad (1.50)$$

Using the I.C. then yields the answer:

$$\begin{aligned}\left(\frac{1}{2}\right) &= \frac{2}{5}(0) - \frac{1}{5}(1) + C(1) \\ C &= \frac{7}{10}\end{aligned}\quad (1.51)$$

$$\therefore y = \frac{2}{5} \sin x - \frac{1}{5} \cos x + \frac{7}{10}e^{-2x}\quad (1.52)$$

□

1.4. Exact First-order ODEs

Sometimes, a first-order ODE may be in a specifically convenient form that allows solving for it by a straightforward integration.

Definition 1.8.

(Exact first-order ODEs) Exact ODEs are first-order ODEs that can be written in the form of

$$M(x, y)dx + N(x, y)dy = 0\quad (1.53)$$

where $M(x, y), N(x, y)$ are some functions of x and y , such that

$$M = \frac{\partial f}{\partial x}\quad (1.54a)$$

$$N = \frac{\partial f}{\partial y}\quad (1.54b)$$

are the first-order partial derivatives for another function $f(x, y)$.

A simple example is

$$y \cos x \, dx + \sin x \, dy = 0\quad (1.55)$$

with $f(x, y) = y \sin x$. By *Chain Rule* for multiple variables, the ODE (1.53) can be rewritten into

$$\frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy = df = 0 \quad (1.56)$$

where df is called the *exact/total differential* and f is known as the *potential function*. Integration then yields a family of *solution/level curves*:

$$\begin{aligned} \int df &= 0 \\ f(x, y) &= C \end{aligned} \quad (1.57)$$

generated by changing the values of the integration constant C . Figure 1.3 below is a schematic illustrating how the level curves, in addition to the potential, behave. The overall change in potential is

$$\Delta f = \frac{\partial f}{\partial x} \Delta x + \frac{\partial f}{\partial y} \Delta y = 0 \quad (1.58)$$

Along a level curve, as the displacement becomes infinitesimally small, it becomes $df = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy$ which still equals zero as required by (1.56). This means that the potential conserves along any solution curve of an exact ODE.

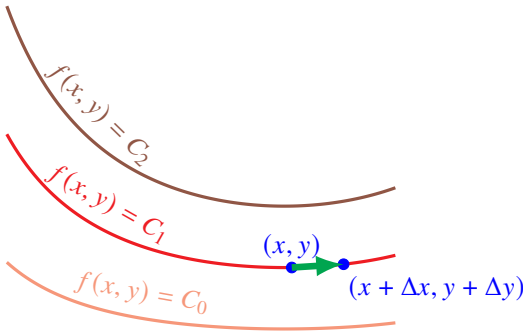


Figure 1.3.: The family of level curves for the case of (1.55) where $f(x, y) = y \sin x$.

1.4.1. Definition and Requirement for Exactness

We emphasize that for an ODE like (1.53) to be exact, we need (1.54), i.e. $M = \frac{\partial f}{\partial x}$, $N = \frac{\partial f}{\partial y}$, to hold. Due to *Clairaut's Theorem* in elementary Calculus, the order of partial derivatives can be interchanged under mild conditions, such that

Properties 1.9.

(Exactness Condition) The requirement for exactness of a first-order ODE in the form of (1.53) is that

$$\begin{aligned}\frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right) &= \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right) \\ \frac{\partial M}{\partial y} &= \frac{\partial N}{\partial x}\end{aligned}\tag{1.59}$$

Example 1.7.

Show that

$$3y \, dx + (3x + 2y) \, dy = 0\tag{1.60}$$

is exact and hence solve for the general solution.

Solution. We first check exactness following Properties 1.9 above:

$$\begin{aligned}\frac{\partial M}{\partial y} &= \frac{\partial(3y)}{\partial y} \\ &= 3 = \frac{\partial(3x + 2y)}{\partial x} = \frac{\partial N}{\partial x}\end{aligned}\tag{1.61}$$

Then we can integrate $M = 3y$ with respect to x and $N = 3x + 2y$ with respect to y to retrieve f . Note that by (1.54), we have, again using elementary Calculus

$$\begin{aligned}M &= \frac{\partial f}{\partial x} \\ \int M(x, y) \, dx &= \int \frac{\partial f}{\partial x} \, dx = f(x, y) + g(y)\end{aligned}\tag{1.62a}$$

$$\begin{aligned}N &= \frac{\partial f}{\partial y} \\ \int N(x, y) \, dy &= \int \frac{\partial f}{\partial y} \, dy = f(x, y) + h(x)\end{aligned}\tag{1.62b}$$

where integrating with respect to x will produce an integration "constant" $g(y)$ that depends on y , and similarly for the case of y and $h(x)$. Subsequently,

$$\int M \, dx = \int 3y \, dx = 3xy + g(y)\tag{1.63a}$$

$$\int N \, dy = \int (3x + 2y) \, dy = 3xy + y^2 + h(x)\tag{1.63b}$$

By comparing these two expressions, we conclude that a possible, consistent choice for the potential function is

$$f(x, y) = 3xy + y^2\tag{1.64}$$

and the general solution will be

$$3xy + y^2 = C \quad (1.65)$$

For example, if the I.C. is $y(1) = 1$, then we have $C = 4$. $\square$

1.4.2. Possible Integrating Factors to Retrieve Exact ODEs

However, sometimes we will also encounter first-order ODEs that resemble (1.53), but fail to satisfy (1.54): hence they are *non-exact*, and it is not possible to directly apply the procedure above.

One of the remedies is using a suitable integrating factor μ , and the problem becomes how to decide its form. Assuming that upon multiplying μ to (1.53), the equation is transformed into an exact one, such that we have

$$\mu M(x, y)dx + \mu N(x, y)dy = 0 \quad (1.66)$$

now with the requirement in Properties 1.9 fulfilled:

$$\frac{\partial}{\partial y}(\mu M(x, y)) = \frac{\partial}{\partial x}(\mu N(x, y)) \quad (1.67)$$

By the Product Rule, we have

$$\begin{aligned} M \frac{\partial \mu}{\partial y} + \mu \frac{\partial M}{\partial y} &= N \frac{\partial \mu}{\partial x} + \mu \frac{\partial N}{\partial x} \\ M \frac{\partial \mu}{\partial y} - N \frac{\partial \mu}{\partial x} &= \mu \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) \\ \frac{1}{\mu} \left(M \frac{\partial \mu}{\partial y} - N \frac{\partial \mu}{\partial x} \right) &= \frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \end{aligned} \quad (1.68)$$

Now further suppose that μ depends on x only, $\mu = \mu(x)$, then (1.68) becomes

$$\begin{aligned} -\frac{1}{\mu} \left(N \frac{d\mu}{dx} \right) &= \frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \\ \frac{1}{\mu} \frac{d\mu}{dx} &= \frac{\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}}{-N} \end{aligned} \quad (1.69)$$

where, in particular, the partial derivative $\frac{\partial \mu}{\partial x}$ becomes an ordinary one $\frac{d\mu}{dx}$. Since the L.H.S. is assumed to depend on x only, the R.H.S. also has to be in x only (we will have to check that for consistency in the end), so we can let

$$g(x) = \frac{\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}}{-N} \quad (1.70)$$

and (1.69) becomes a separable ODE in the form of

$$\begin{aligned}\int \frac{d\mu}{\mu} &= \int \frac{\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}}{-N} dx = \int g(x) dx \\ \ln |\mu| &= \int g(x) dx \\ \mu &= \exp\left(\int g(x) dx\right)\end{aligned}\tag{1.71}$$

(the absolute sign does not matter here) (1.71) is then a possible integrating factor that can be multiplied to the non-exact ODE to make it exact, if our premise that (1.70) and hence μ are a function of x only, is indeed true. By a similar argument, if

$$h(y) = \frac{\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}}{M}\tag{1.72}$$

is a function of y only, then $\exp(\int h(y) dy)$ can be an integrating factor. The quantity $\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}$ is sometimes known as the *curl*.

Example 1.8.

Transform

$$y \ln y \, dx + (x + y) dy = 0\tag{1.73}$$

into an exact ODE and solve it.

Solution. This ODE is non-exact as the required partial derivatives are not equal:

$$\frac{\partial M}{\partial y} = \frac{\partial (y \ln y)}{\partial y} = \ln y + 1 \neq 1 = \frac{\partial (x + y)}{\partial x} = \frac{\partial N}{\partial x}\tag{1.74}$$

Since $\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} = -\ln y$, we are tempted to try (1.72):

$$\begin{aligned}h &= \frac{\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}}{M} \\ &= \frac{-\ln y}{y \ln y} = -\frac{1}{y}\end{aligned}\tag{1.75}$$

which indeed depends on y only. So the integrating factor is

$$\mu = \exp\left(-\int \frac{1}{y} dy\right) = \exp(-\ln |y|) = \exp(\ln(y^{-1})) = \frac{1}{y}\tag{1.76}$$

(again, the absolute value does not matter here) Multiplying by this form, the equation then becomes exact:

$$\ln y \, dx + \left(\frac{x}{y} + 1\right) dy = 0\tag{1.77}$$

with

$$\frac{\partial M}{\partial y} = \frac{1}{y} = \frac{\partial N}{\partial x} \quad (1.78)$$

Now we integrate the new M and N with respect to x and y , yielding

$$\int M dx = \int \ln y dx = x \ln y + p(y) \quad (1.79a)$$

$$\int N dy = \int \left(\frac{x}{y} + 1 \right) dy = x \ln y + y + q(x) \quad (1.79b)$$

where we now use p, q to denote the integration constants. The potential function, as well as the general solution, is subsequently seen to be

$$f(x, y) = x \ln y + y = C \quad (1.80)$$

□

1.4.3. Identification of Special Differentials

Notice that we have simple formulas for some recurring differentials:

$$d\left(\frac{x}{y}\right) = \frac{ydx - xdy}{y^2} \quad (1.81a)$$

$$d(xy) = xdy + ydx \quad (1.81b)$$

$$d(x^2 + y^2) = 2(xdx + ydy) \quad (1.81c)$$

$$d\left(\tan^{-1}\left(\frac{x}{y}\right)\right) = \frac{ydx - xdy}{x^2 + y^2} \quad (1.81d)$$

$$d\left(\ln\left(\frac{x}{y}\right)\right) = \frac{ydx - xdy}{xy} \quad (1.81e)$$

where the roles of x and y may be interchanged. So, we can rewrite and solve some ODEs by grouping terms into these forms. Particularly, if the expression $ydx - xdy$ appears, it can be divided by x^2 , xy , y^2 , or $x^2 + y^2$, respectively, to produce other new differentials ((a), (d), (e) above).

Example 1.9.

Solve the ODE

$$x^2 y dy - (x dy - y dx) = 0 \quad (1.82)$$

Solution. We can utilize (1.81a) with x, y swapped:

$$d\left(\frac{y}{x}\right) = \frac{x dy - y dx}{x^2} \quad (1.83)$$

Dividing both sides of (1.82) by x^2 and using (1.83), we arrive at the solution:

$$\begin{aligned} ydy - \frac{xdy - ydx}{x^2} &= 0 \\ d\left(\frac{1}{2}y^2\right) - d\left(\frac{y}{x}\right) &= 0 \\ \frac{1}{2}y^2 - \frac{y}{x} &= C \end{aligned} \quad (1.84)$$

where we have integrated the two differentials. $\square$

1.5. Solution Curves Plotting

In this section, we will talk about various aspects related to plotting the solution curves.

1.5.1. Slope Fields and Solution Curves

Apart from the family of solution curves themselves, we may further visualize the *slope fields*: the local slopes $\frac{dy}{dx}$ at equally spaced grid points. We will use the *Lotka-Volterra predator-prey model* as an illustration.

Example 1.10.

(Lotka-Volterra Predator-prey Model) Denote the populations of the prey and predator in a simple biological system by x and y , respectively. The rate of change in these two populations can be described by the set of ODEs:

$$\begin{cases} \frac{dx}{dt} = \alpha x - \beta xy \\ \frac{dy}{dt} = -\gamma y + \delta xy \end{cases} \quad (1.85a)$$

$$(1.85b)$$

where $\alpha, \beta, \gamma, \delta > 0$ are fixed parameters. The α term represents the birth rate of the prey, the β term represents the death of the prey due to the predator, while the γ term indicates the natural death of the predator, and the δ term indicates the growth of the predator due to the consumption of the prey. Solve for the relationship between x and y under the regime $\alpha = 1.0, \beta = 0.5, \gamma = 0.8, \delta = 0.3$, and plot the corresponding solution curves, as well as the slope fields.

Solution. Dividing (1.85b) by (1.85a), we have

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{-\gamma y + \delta xy}{\alpha x - \beta xy} = -\frac{y}{x} \left(\frac{\gamma - \delta x}{\alpha - \beta y} \right) \quad (1.86)$$

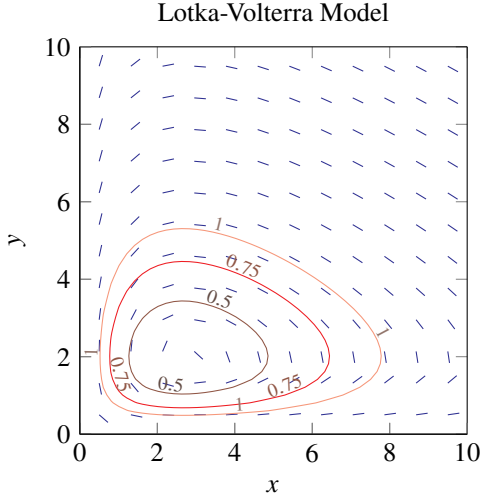


Figure 1.4.: The solution curves (red) and slope fields (blue) for the Lotka-Volterra predator-prey model in Example 1.10 with the setting $\alpha = 1.0, \beta = 0.5, \gamma = 0.8, \delta = 0.3$, and $C = 0.5, 0.75, 1$.

By doing so, we effectively eliminate the original independent variable t , and x now inherits the role of independent variable for y .³ The corresponding slope fields are plotted in blue in Figure 1.4. This is a separable ODE and may be solved as follows.

$$\begin{aligned} \int \frac{\alpha - \beta y}{y} dy &= - \int \frac{\gamma - \delta x}{x} dx \\ \alpha \ln |y| - \beta y &= -\gamma \ln |x| + \delta x - C \\ \beta y - \alpha \ln y + \delta x - \gamma \ln x &= C \end{aligned} \quad (x, y > 0) \quad (1.87)$$

The solution curves when $C = 0.5, 0.75, 1$ are plotted in red in Figure 1.4. Once an initial condition is specified, only one corresponding solution curve will remain. $\square$

³Under the hood, the time t is the independent variable while both x and y are dependent variables. In such a scenario, x and y form the *phase space*, and Figure 1.4 actually shows the *phase plane*, which is different from the solution curves of x or y itself in terms of t .

1.5.2. Autonomous ODE Systems

A special case is when the ODE does not explicitly depend on the independent variable x , or in other words, only involves the dependent variable y and its ordinary derivatives.

Definition 1.10.

(Autonomous ODEs) An ODE is called **autonomous** if it can be written as

$$f(y, y', y'', \dots) = 0 \quad (1.88)$$

We may also express (1.88) as

$$f(y(x), y'(x), y''(x), \dots) = 0 \quad (1.89)$$

to emphasize that $y, y', y'', \dots$ are still functions of x , even though x does not explicitly appear in (1.88). Now, if we replace x by $x - C$, then (1.88) retains the same form since there is no explicit x dependence. Suppose

$$y = g(x) \quad (1.90)$$

is a solution to (1.88), then after the same change of variable

$$y = g(x - C) \quad (1.91)$$

must also be a solution to (1.88). This means that the family of solution curves to the problem (1.88) is simply generated by a horizontal translation (by any value of C). We then say that such an autonomous system is *invariant under space/time translation*. Example 1.3 and Figure 1.2 about the logistic population model perfectly demonstrate this property, where the solution curves are just horizontally translated sigmoids.

In an autonomous system, the values of the dependent variable y such that all the involved derivatives vanish, $y' = y'' = \dots = 0$ (determined by (1.88)), are known as **equilibrium points**. y will retain the same value at an equilibrium point indefinitely. When y is close enough to a given equilibrium point, and if y tends (attracted) to this equilibrium point, then it will be called a *stable* equilibrium point. Otherwise, if nearby y moves away (repelled) from the equilibrium point, then it is an *unstable* equilibrium point. In the logistic population model example, $N = M$ is a stable equilibrium while $N = 0$ is an unstable equilibrium.

1.5.3. Orthogonal Trajectories

Returning to first-order ODEs that have the general form of

$$f(x, y, \frac{dy}{dx}) = 0 \quad (1.92)$$

and suppose the set of solutions is

$$g(x, y; C) = 0 \quad (1.93)$$

where C is the integration constant, acting as a *parameter* to control the appearance of the solution curves as explained before. A related problem is to find the **orthogonal trajectories** with respect to (1.93), which are another family of curves always *orthogonal* (perpendicular) to those indicated by (1.93). An illustration is given as Figure 1.5 below.

Remember that in high-school geometry, for a curve $y(x)$ which has a slope of $\frac{dy}{dx}$ locally, another curve to cross it perpendicularly needs to have a negative reciprocal slope of $\frac{-1}{\frac{dy}{dx}} = -\frac{dx}{dy}$. For any orthogonal trajectory, this needs to hold everywhere. Therefore, to derive orthogonal trajectories, we simply replace $\frac{dy}{dx}$ by $-\frac{dx}{dy}$ in (1.92) and solve this newly modified ODE:

$$f(x, y, -\frac{dx}{dy}) = 0 \quad (1.94)$$

Sometimes we are given (1.93) only, and we can obtain (1.92) by an implicit differentiation of (1.93) against x and eliminate C by using (1.93) again.

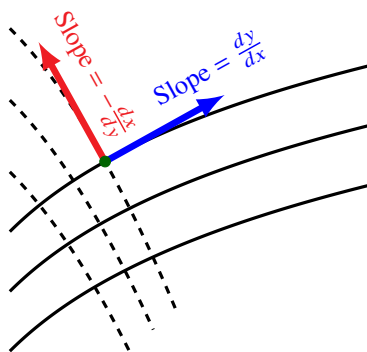


Figure 1.5.: A family of solution curves (solid) and its orthogonal trajectories (dashed).

Example 1.11.

The equation for the family of circles centered at the origin is

$$x^2 + y^2 = r^2 \quad (1.95)$$

where r is the radius. Find their orthogonal trajectories.

Solution. The corresponding ODE is, by differentiating (1.95) with respect to x :

$$\begin{aligned} 2x + 2y \frac{dy}{dx} &= 0 \\ \frac{dy}{dx} &= -\frac{x}{y} \end{aligned} \tag{1.96}$$

To obtain the orthogonal trajectories, replace $\frac{dy}{dx}$ by $-\frac{dx}{dy}$ as suggested above and solve the new equation:

$$\begin{aligned} -\frac{dx}{dy} &= -\frac{x}{y} \\ \frac{dx}{x} &= \frac{dy}{y} \\ \int \frac{dx}{x} &= \int \frac{dy}{y} \\ \ln |x| &= \ln |y| + C \\ x &= Ay \end{aligned} \tag{1.97}$$

where $A = \pm e^C$. So the orthogonal trajectories of circles are the family of straight lines through the origin, each with a slope of A . (see Figure 1.6 below) $\square$

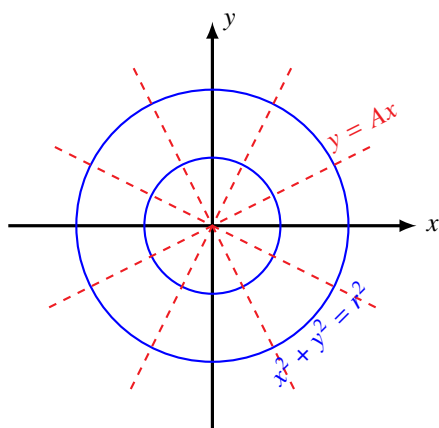


Figure 1.6.: The circles (blue) and their orthogonal trajectories (red) in Example 1.11.

1.6. Exercises

1.1) Decide whether the following ODEs are first-order. If so, determine whether they are written in the form of separable or linear form (or neither/both).

(a) $\frac{dy}{dx} + 3xy = \sin x$;

(b) $\frac{dy}{dx} = e^{-x} \cos y$;

(c) $\frac{d^2y}{dx^2} + 4y = 0$;

(d) $\frac{dy}{dx} = x(1 - xe^{-y})$;

(e) $\frac{dy}{dx} + (\cos x)y = e^{-x^2}$;

(f) $\frac{dy}{dx} + xy^2 = \frac{1}{x^2}$;

(g) $\left(\frac{dy}{dx}\right)^2 + y = 3$;

(h) $\frac{dy}{dx} = \sqrt{\pi}$.

1.2) Solve

$$\frac{dy}{dx} = e^{-x}y^3 \quad (1.98)$$

with the I.C. $y(0) = 1$.

1.3) Solve

$$(\csc x) \frac{dy}{dx} = y + 1 \quad (1.99)$$

with the I.C. $y(0) = 2$.

1.4) Solve

$$\frac{dy}{dx} + 2xy = e^{-x^2} \quad (1.100)$$

with the I.C. $y(0) = 1$.

1.5) Solve

$$\frac{dy}{dx} - \frac{y}{2x} = 1 \quad (1.101)$$

with the I.C. $y(1) = 3$.

1.6) Show that

$$(2x \tan y + 1)dx + (x^2 \sec^2 y)dy = 0 \quad (1.102)$$

is exact and solve it under the I.C. $y(1) = 1$.

1.7) Derive the solution to

$$y(e^{xy} + y)dx + x(e^{xy} + 2y)dy = 0 \quad (1.103)$$

with the I.C. $y(0) = 1$.

1.8) Convert

$$(1 - y^2 \tan x)dx + 2ydy = 0 \quad (1.104)$$

to the exact form and solve for its general solution.

1.9) Solve

$$ydx - xdy = x^3ydx \quad (1.105)$$

with the I.C. $y(1) = 2$.

1.10) Solve

$$\frac{dy}{dx} + \frac{2y}{x} = e^x \quad (1.106)$$

where $y(1) = 1$.

1.11) Find the full solution to

$$\frac{dy}{dx} + \frac{2x}{x^2 + 1}y = \cos(2x) \quad (1.107)$$

where $y(0) = 1$.

Basic Second-order Ordinary Differential Equations

Introduction We now move one order higher and begin our study of **second-order ODEs**, in which obviously the highest order of ordinary derivatives appearing is of second-order, $\frac{d^2y}{dx^2}$ or $\frac{d^2y}{dt^2}$ (or y''). They are important because many physical laws involve acceleration (second-order) rather than just the velocity (first-order), such as Newton's Second Law. For arbitrary second-order ODEs, the solutions are usually hard to find analytically. However, if we focus on **second-order linear ODEs**, which take the general form of

$$\frac{d^2y}{dx^2} + P(x)\frac{dy}{dx} + Q(x)y = f(x) \quad (3.1)$$

where $P(x)$, $Q(x)$, and $f(x)$ are some functions of the independent variable x only, then they often possess some convenient properties, and arriving at a satisfactory solution becomes possible. The most outstanding example of second-order ODEs is perhaps the class of harmonic oscillators, including pendulum motion, spring-mass system, the Schrödinger equation in quantum mechanics, and atmospheric gravity waves:

$$\frac{d^2y}{dx^2} + \omega^2 y = f(x) \quad (3.2)$$

Here, ω is a constant that represents the frequency. When $f(x) = 0$, the equation (3.1) is known as homogeneous, just like the first-order counterpart. This source/forcing term $f(x)$ on the R.H.S. can be viewed to induce a response in $y(x)$ via the structure of that differential equation on the L.H.S. For example, in Atmospheric Sciences, the external forcing by sea surface temperature (SST) can produce a response in the atmospheric circulation lying above the ocean.

3.1. Some Basics about Second-order ODEs

In this section, we are going to discuss some prerequisite concepts about second-order ODEs.

3.1.1. Existence and Uniqueness of Solutions (Second-order)

An essential thing to do before we get our hands on solving second-order ODEs is to establish the existence and uniqueness of the solutions. For brevity, we assume that the second-order ODE in question is linear as in (3.1), then we can easily rewrite it into a system of two first-order ODEs by letting $z = \frac{dy}{dx}$:

$$\begin{cases} \frac{dy}{dx} = z \\ \frac{dz}{dx} = \frac{d^2y}{dx^2} = -P(x)z - Q(x)y + f(x) \end{cases} \quad (3.3a)$$

$$\frac{dz}{dx} = \frac{d^2y}{dx^2} = -P(x)z - Q(x)y + f(x) \quad (3.3b)$$

The Picard-Lindelöf Theorem 2.8 for a system of ODEs given at the end of the last chapter, where $\mathbf{y} = (y, z) = (y, \frac{dy}{dx})$, then immediately leads to the existence and uniqueness of solutions for (3.3) and hence equivalently (3.1).

As the order of equations increases to 2, there will now be two possible, "different" solutions, accompanied by respective integration constants to complete. How we define "different" in this setting will be explained very soon. In view of (3.3) and the Picard-Lindelöf Theorem, this also means that we need two initial conditions, one for y and one for $z = \frac{dy}{dx}$ to specify the full solution.

Definition 3.1.

(Initial Conditions for Second-order ODEs) A complete set of initial conditions for a second-order ODE consists of imposed values of both y and y' at some point $x = a$, denoted by $y(a)$ and $y'(a)$.

This may be compared to Definition 1.6. Moreover, in the case of second-order linear ODEs, Definition 1.4 still applies. So, if $y_1(x)$ and $y_2(x)$ are two solutions to the homogeneous part of (3.1), then linearity holds, i.e. any linear combination in the form of $\alpha y_1 + \beta y_2$ is still a solution. Particularly, the constant multiple of a solution is also a solution, but should be regarded to be the same one, as there is no new information provided.

3.1.2. Wronskian for Testing Linear Independence

To make the notion of "different" solutions clear, we now introduce the concept of **linear independence**.

Definition 3.2.

(Linear Independence) A set of functions $y_1, y_2, \dots, y_n$ are **linearly independent** if the equation

$$c_1 y_1 + c_2 y_2 + \dots + c_n y_n = 0 \quad (3.4)$$

has only the trivial solution $c_1 = c_2 = \dots = c_n = 0$. Otherwise, they are known as **linearly dependent**.

In the case of two functions y_1, y_2 , linear independence reduces to the usual statement that each of them cannot be expressed as a constant multiple of the other.¹ For example, $\sin x$ and $\cos x$ are linearly independent as it is not possible to write $\sin x$ as $k \cos x$ with a constant k and vice versa; but $2e^x$ and $-3e^x$ will be linearly dependent because $-\frac{3}{2}(2e^x) = -3e^x$. To test the linear independence between two functions $y_1(x)$ and $y_2(x)$, we can calculate the **Wronskian determinant**.

Definition 3.3.

(Wronskian) The Wronskian of two functions $y_1(x)$ and $y_2(x)$ is defined as the determinant

$$W[y_1, y_2](x) = \begin{vmatrix} y_1(x) & y_2(x) \\ y_1'(x) & y_2'(x) \end{vmatrix} = y_1(x)y_2'(x) - y_1'(x)y_2(x) \quad (3.5)$$

Properties 3.4.

If the Wronskian (3.5) of two functions is not identically equal to 0 for all values of x , then they must be linearly independent.

To show that, we can prove the contrapositive, i.e. Linearly dependent functions $y_1(x)$ and $y_2(x) = ky_1(x)$ always have a Wronskian of 0:

$$\begin{aligned} W[y_1, ky_1](x) &= \begin{vmatrix} y_1(x) & ky_1(x) \\ y_1'(x) & [ky_1]'(x) \end{vmatrix} = \begin{vmatrix} y_1(x) & ky_1(x) \\ y_1'(x) & ky_1'(x) \end{vmatrix} \\ &= k \begin{vmatrix} y_1(x) & y_1(x) \\ y_1'(x) & y_1'(x) \end{vmatrix} = k(0) \equiv 0 \end{aligned} \quad (3.6)$$

because a determinant that contains two identical columns will always yield zero. Using the sine

¹If $y_2 = ky_1$ for some k , then we have $-ky_1 + y_2 = 0$, and by Definition 3.2, (3.4) has the non-trivial solution $c_1 = -k, c_2 = 1$, hence they are linearly dependent.

and cosine example above, their Wronskian is

$$\begin{aligned} W[\sin, \cos](x) &= \begin{vmatrix} \sin x & \cos x \\ (\sin x)' & (\cos x)' \end{vmatrix} \\ &= \begin{vmatrix} \sin x & \cos x \\ \cos x & -\sin x \end{vmatrix} = -\sin^2 x - \cos^2 x = -1 \neq 0 \end{aligned} \quad (3.7)$$

so they are indeed linearly independent by Properties 3.4.

We can further strengthen this result about the Wronskian if we are working with second-order linear ODEs.

Theorem 3.5.

If $y_1(x)$ and $y_2(x)$ are two solutions to the homogeneous part of (3.1):

$$\frac{d^2 y}{dx^2} + P(x) \frac{dy}{dx} + Q(x)y = 0 \quad (3.8)$$

over an interval $[a, b]$, then their Wronskian $W[y_1, y_2](x)$ will either be identically zero or never vanish within the same interval.

To prove that, we first find the derivative of the Wronskian:

$$W = y_1 y_2' - y_1' y_2 \quad (3.9)$$

$$W' = y_1' y_2' + y_1 y_2'' - y_1'' y_2 - y_1' y_2' = y_1 y_2'' - y_1'' y_2 \quad (3.10)$$

As $y_1(x)$ and $y_2(x)$ are solutions to (3.8), we have

$$y_1'' + P y_1' + Q y_1 = 0 \quad (3.11a)$$

$$y_2'' + P y_2' + Q y_2 = 0 \quad (3.11b)$$

Multiplying the first equation by y_2 and the second equation by y_1 , then doing a subtraction between them, will give

$$\begin{aligned} (y_1 y_2'' + P y_1 y_2' + Q y_1 y_2) - (y_1'' y_2 + P y_1' y_2 + Q y_1 y_2) &= 0 \\ (y_1 y_2'' - y_1'' y_2) + P(y_1 y_2' - y_1' y_2) &= 0 \\ \frac{dW}{dx} + PW &= 0 \end{aligned} \quad (3.12)$$

where we have recalled (3.9) and (3.10). This relation is known as *Abel's Identity*. The general solution to (3.12), a first-order linear ODE, is just

$$W(x) = C \exp\left(-\int P(x) dx\right) \quad (3.13)$$

Since the exponential factor is always positive and never zero, W will either be identically zero (if $C = 0$) or never zero (if $C \neq 0$). The implication is that for a second-order linear ODE, the location where the value of W is taken does not matter: The Wronskian $W[y_1, y_2](x)$ being zero/non-zero at a point $x = x_0$ implies that it will always be zero/non-zero throughout all x .

3.1.3. Complementary and Particular Solutions

The idea of complementary and particular solutions, as pointed out in Example 1.4, transfers to second-order ODEs. Assume that we have found two linearly independent complementary solutions $y_1(x)$ and $y_2(x)$ that satisfy the homogeneous part of (3.1):

$$\frac{d^2y}{dx^2} + P(x)\frac{dy}{dx} + Q(x)y = 0 \quad (3.14)$$

Then by linearity,

$$y_c(x) = c_1y_1(x) + c_2y_2(x) \quad (3.15)$$

will be a valid general solution to 3.14, where c_1, c_2 are constants to be determined as follows. Given two initial conditions at some reference location $x = a$, we have

$$\begin{cases} y(a) = A = c_1y_1(a) + c_2y_2(a) \\ y'(a) = B = c_1y_1'(a) + c_2y_2'(a) \end{cases} \quad (3.16a) \quad (3.16b)$$

Then by *Cramer's Rule* from Linear Algebra, we have

$$c_1 = \frac{Ay_2'(a) - By_2(a)}{W[y_1, y_2](a)} \quad (3.17a)$$

$$c_2 = \frac{By_1(a) - Ay_1'(a)}{W[y_1, y_2](a)} \quad (3.17b)$$

As long as we possess a choice of linearly independent solutions y_1 and y_2 , we can determine the unique solution y_c that satisfies the initial conditions via (3.17), which is non-trivial as we have shown that the Wronskian in the denominator will be non-zero, $W \neq 0$.

Now let's consider the non-homogeneous case. Everything is still the same as before, except that we also have a particular solution $y_p(x)$ that satisfies (3.1) with the source term $f(x)$ present. Then, the general solution will be the sum of the complementary part $y_c(x)$ and particular part $y_p(x)$:

$$y = y_c(x) + y_p(x) = c_1y_1(x) + c_2y_2(x) + y_p(x) \quad (3.18)$$

Substituting this into the L.H.S. of (3.1) indeed gives

$$\frac{d^2}{dx^2}(y_c + y_p) + p(x)\frac{d}{dx}(y_c + y_p) + q(x)(y_c + y_p)$$

$$\begin{aligned}
 &= \left(\frac{d^2 y_c}{dx^2} + p(x) \frac{dy_c}{dx} + q(x) y_c \right) + \left(\frac{d^2 y_p}{dx^2} + p(x) \frac{dy_p}{dx} + q(x) y_p \right) \\
 &= (0) + f(x) = f(x)
 \end{aligned} \tag{3.19}$$

showing that (3.18) will be the required form of the full solution. Note that we have to take into account the particular part and determine a new appropriate set of integration constants for the given I.C.s (see Exercise 3.20).

3.2. Second-order Constant-coefficient Homogeneous ODEs

We now proceed to see how we actually solve second-order ODEs and will first address the most fundamental type of them.

Definition 3.6.

(Second-order Constant-coefficient Homogeneous ODEs) **Second-order constant-coefficient homogeneous ODEs** are second-order linear ODEs in the form of

$$\frac{d^2 y}{dx^2} + \alpha \frac{dy}{dx} + \beta y = 0 \tag{3.20}$$

where the coefficients α, β are some constants and the R.H.S. is zero.

3.2.1. Ansatz, Three Cases for Auxiliary Equation

Inspired by the first-order counterpart:

$$\frac{dy}{dx} + ky = 0 \tag{3.21}$$

which can be treated by either Separation of Variables or an integrating factor, and is easily seen to have the solution of

$$y = Ae^{-kx} \tag{3.22}$$

we are motivated to make a similar guess e^{rx} as the test solution (or called *ansatz*) to (3.20), where r is some constant to be determined as follows. Substituting this into (3.20) produces

$$\begin{aligned}
 \frac{d^2}{dx^2}(e^{rx}) + \alpha \frac{d}{dx}(e^{rx}) + \beta(e^{rx}) &= 0 \\
 r^2 e^{rx} + \alpha r e^{rx} + \beta e^{rx} &= 0 \\
 e^{rx}(r^2 + \alpha r + \beta) &= 0 \\
 r^2 + \alpha r + \beta &= 0
 \end{aligned} \tag{3.23}$$

3.2. Second-order Constant-coefficient Homogeneous ODEs ■

The last line follows because the exponential function is never zero. The relation (3.23) is known as the **auxiliary equation**. By the derivation above, provided that the auxiliary equation is satisfied, e^{rx} will be a valid solution to (3.20). The remaining problem is then to compute the roots r_1, r_2 of the auxiliary equation. It is essentially a quadratic equation, the behavior of which can be categorized into three cases, a well-known fact from high-school algebra. These three cases will then lead to correspondingly different forms of the solutions.

Two Real Roots

The first case is that the auxiliary equation (3.23) has two *distinct* real roots $r = r_1, r_2$, when the *discriminant* is greater than zero, $\alpha^2 - 4\beta > 0$. Then as aforementioned, $y_1 = e^{r_1x}$ and $y_2 = e^{r_2x}$ will be two possible solutions. They are linearly independent, seen by computing the Wronskian (3.5) and using Properties 3.4:

$$\begin{aligned} W[y_1, y_2](x) &= \begin{vmatrix} e^{r_1x} & e^{r_2x} \\ (e^{r_1x})' & (e^{r_2x})' \end{vmatrix} = \begin{vmatrix} e^{r_1x} & e^{r_2x} \\ r_1 e^{r_1x} & r_2 e^{r_2x} \end{vmatrix} \\ &= e^{r_1x}(r_2 e^{r_2x}) - e^{r_2x}(r_1 e^{r_1x}) \\ &= (r_2 - r_1)e^{(r_1+r_2)x} \neq 0 \end{aligned} \quad (r_1 \neq r_2) \quad (3.24)$$

So, by linearity, the general solution will be simply

$$y = c_1 e^{r_1x} + c_2 e^{r_2x} \quad (3.25)$$

Example 3.1.

Solve the second-order constant-coefficient ODE

$$\frac{d^2y}{dx^2} + 3\frac{dy}{dx} + 2y = 0 \quad (3.26)$$

where the initial conditions are $y(0) = 3, y'(0) = -5$.

Solution. Here, the auxiliary equation (3.23) is solved as

$$\begin{aligned} r^2 + 3r + 2 &= 0 \\ (r + 1)(r + 2) &= 0 \\ \therefore r &= -1 \text{ or } -2 \end{aligned} \quad (3.27)$$

Therefore, the general solution is

$$y = c_1 e^{-x} + c_2 e^{-2x} \quad (3.28)$$

where

$$y' = -c_1 e^{-x} - 2c_2 e^{-2x} \quad (3.29)$$

Substituting the I.C.s then gives the full answer:

$$\begin{cases} 3 = c_1 e^{-(0)} + c_2 e^{-2(0)} = c_1 + c_2 \\ -5 = -c_1 e^{-(0)} - 2c_2 e^{-2(0)} = -c_1 - 2c_2 \end{cases}$$

$$\therefore c_1 = 1, c_2 = 2 \quad (3.30)$$

$$\therefore y = e^{-x} + 2e^{-2x} \quad (3.31)$$

□

One Repeated Real Root

The next case for the auxiliary equation (3.23) is a repeated real root, let's say $r = r_1$, that solves

$$r^2 + \alpha r + \beta = (r - r_1)^2 = 0 \quad (3.32)$$

Expanding the square shows that $r^2 + \alpha r + \beta = r^2 - 2r_1 r + r_1^2$ and it must be $\alpha = -2r_1$ and $\beta = r_1^2$ (so $\alpha^2 = 4\beta$). By the same logic before, we know that $y_1 = e^{r_1 x}$ will be one possible solution. However, since the root is now a repeated one, we don't automatically get two linearly independent solutions as in the above case of two real roots.

To complete our solution, notice that with a constant k , $ky_1 = ke^{r_1 x}$ is also a solution, though it is clearly not linearly independent from y_1 . Our trick is to replace k by x , hoping that by increasing the degree by one, the new form

$$y_2 = xe^{r_1 x} \quad (3.33)$$

will remain a valid solution, and at the same time address the linear independence issue. This indeed works, because by (3.32), the ODE will read

$$\frac{d^2 y}{dx^2} - 2r_1 \frac{dy}{dx} + r_1^2 y = 0 \quad (3.34)$$

and substituting (3.33) shows that

$$\begin{aligned} & \frac{d^2 y_2}{dx^2} - 2r_1 \frac{dy_2}{dx} + r_1^2 y_2 \\ &= \frac{d^2}{dx^2} (xe^{r_1 x}) - 2r_1 \frac{d}{dx} (xe^{r_1 x}) + r_1^2 (xe^{r_1 x}) \\ &= \frac{d}{dx} (e^{r_1 x} + xr_1 e^{r_1 x}) - 2r_1 (e^{r_1 x} + xr_1 e^{r_1 x}) + xr_1^2 e^{r_1 x} \\ &= (2r_1 e^{r_1 x} + xr_1^2 e^{r_1 x}) - 2r_1 e^{r_1 x} - 2xr_1^2 e^{r_1 x} + xr_1^2 e^{r_1 x} \equiv 0 \end{aligned} \quad (3.35)$$

3.2. Second-order Constant-coefficient Homogeneous ODEs ■

Hence (3.34) is solved. And their linear independence can be easily confirmed by computing the Wronskian:

$$\begin{aligned} W[y_1, y_2](x) &= \begin{vmatrix} e^{r_1 x} & x e^{r_1 x} \\ (e^{r_1 x})' & (x e^{r_1 x})' \end{vmatrix} = \begin{vmatrix} e^{r_1 x} & x e^{r_1 x} \\ r_1 e^{r_1 x} & e^{r_1 x} + x r_1 e^{r_1 x} \end{vmatrix} \\ &= e^{2r_1 x} + x r_1 e^{2r_1 x} - x r_1 e^{2r_1 x} \\ &= e^{2r_1 x} \neq 0 \end{aligned} \quad (3.36)$$

and we are done by Properties 3.4 again. Other than guessing and direct verification, a more mathematical way leading to the ansatz (3.33) is to write the ODE (3.34) in the form of operators:

$$\left(\frac{d^2}{dx^2} - 2r_1 \frac{d}{dx} + r_1^2 \right) y = \left(\frac{d}{dx} - r_1 \right)^2 y = 0 \quad (3.37)$$

It is already known that $y_1 = e^{r_1 x}$ solves (3.37). Plugging this into the equation, and by further taking a partial derivative with respect to the parameter r_1 , we have

$$\begin{aligned} \frac{\partial}{\partial r_1} \left[\left(\frac{d}{dx} - r_1 \right)^2 (e^{r_1 x}) \right] &= 0 \\ \left(\frac{d}{dx} - r_1 \right)^2 (x e^{r_1 x}) - 2 \left(\frac{d}{dx} - r_1 \right) (e^{r_1 x}) &= 0 \\ \left(\frac{d}{dx} - r_1 \right)^2 (x e^{r_1 x}) - 2(r_1 e^{r_1 x} - r_1 e^{r_1 x}) &= 0 \\ \left(\frac{d}{dx} - r_1 \right)^2 (x e^{r_1 x}) - 2(0) &= \left(\frac{d}{dx} - r_1 \right)^2 (x e^{r_1 x}) = 0 \end{aligned} \quad (3.38)$$

So $y_2 = x e^{r_1 x}$ also solves (3.37). As an extra challenge, you may try to use the same method to argue that $x^2 e^{r_1 x}$ is *not* a valid solution in this case.

Example 3.2.

Solve the second-order constant-coefficient ODE

$$\frac{d^2 y}{dx^2} - 2 \frac{dy}{dx} + y = 0 \quad (3.39)$$

with the I.C.s of $y(0) = 1, y'(0) = 3$.

Solution. The auxiliary equation in this question is

$$\begin{aligned} r^2 - 2r + 1 &= 0 \\ (r - 1)^2 &= 0 \end{aligned} \quad (3.40)$$

so $r = 1$ is a repeated real root. According to the explanation above, two linearly independent solutions will be $y_1 = e^x$ and $y_2 = x e^x$, and so the general solution should take the form of

$$y = c_1 e^x + c_2 x e^x \quad (3.41)$$

and hence

$$y' = c_1 e^x + c_2 (x e^x + e^x) \quad (3.42)$$

Plugging in the I.C.s then leads to

$$\begin{cases} c_1(1) + c_2(0) = 1 \\ c_1(1) + c_2(0 + 1) = 3 \end{cases}$$

$$\therefore c_1 = 1, c_2 = 2 \quad (3.43)$$

$$\therefore y = e^x + 2x e^x \quad (3.44)$$

□

Two Complex Roots

The third, final case for the auxiliary equation is the presence of two complex roots. For this to happen, the quadratic auxiliary equation (3.23) will have a negative discriminant:

$$\Delta = \alpha^2 - 4\beta < 0 \quad (3.45)$$

and thus the complex conjugate roots are

$$\begin{aligned} r_{\pm} &= \frac{-\alpha \pm \sqrt{\alpha^2 - 4\beta}}{2} \\ &= \frac{-\alpha}{2} \pm \frac{\sqrt{4\beta - \alpha^2}}{2} i \end{aligned} \quad (3.46)$$

The above suggests that the two solutions to the second-order constant-coefficient homogeneous ODE will be

$$y_1, y_2 = \exp\left(\left(-\frac{\alpha}{2} \pm \frac{\sqrt{4\beta - \alpha^2}}{2} i\right)x\right) \quad (3.47)$$

To handle the complex exponential, we will make use of the famous *Euler's Formula*:

$$e^{i\theta} = \cos \theta + i \sin \theta \quad (3.48)$$

So that (3.47) becomes

$$\begin{aligned} y_1, y_2 &= \exp\left(-\frac{\alpha}{2}x\right) \left(\cos\left(\frac{\sqrt{4\beta - \alpha^2}}{2}x\right) \pm i \sin\left(\frac{\sqrt{4\beta - \alpha^2}}{2}x\right)\right) \\ &= e^{-\frac{\alpha}{2}x} (\cos(qx) \pm i \sin(qx)) \end{aligned} \quad (3.49)$$

where we have set $q = \frac{\sqrt{4\beta - \alpha^2}}{2}$.

3.2. Second-order Constant-coefficient Homogeneous ODEs ■

The problem with this formulation is that the two proposed solutions are complex-valued while the original ODE has real coefficients, and it would be more desirable to have real-valued solutions instead. Now, it is instructive to denote the ODE using an operator, i.e. $\mathcal{L}[y] = 0$, and notice that we can simply decompose it into real and complex parts after plugging in (3.49):

$$\begin{aligned}\mathcal{L}\left[e^{-\frac{q}{2}x}(\cos(qx) \pm i \sin(qx))\right] &= 0 \\ \mathcal{L}\left[e^{-\frac{q}{2}x} \cos(qx)\right] \pm i \mathcal{L}\left[e^{-\frac{q}{2}x} \sin(qx)\right] &= 0\end{aligned}\quad (3.50)$$

Since $\mathcal{L}$ is a real operator as the ODE has real coefficients, by equating the real and complex parts to zero, we must have $\mathcal{L}\left[e^{-\frac{q}{2}x} \cos(qx)\right] = \mathcal{L}\left[e^{-\frac{q}{2}x} \sin(qx)\right] = 0$, and hence we now have two real-valued solutions instead:

$$\begin{cases} y_1 = e^{-\frac{q}{2}x} \cos(qx) \\ y_2 = e^{-\frac{q}{2}x} \sin(qx) \end{cases}\quad (3.51a)$$

$$(3.51b)$$

If we denote the two complex roots for the auxiliary equation as $r_{\pm} = p \pm iq$, then by the same construction, the solutions will be $y_1, y_2 = e^{px} \cos(qx), e^{px} \sin(qx)$. We can subsequently check their linear independence by the Wronskian test as before:

$$\begin{aligned}W[y_1, y_2](x) &= \begin{vmatrix} e^{px} \cos(qx) & e^{px} \sin(qx) \\ (e^{px} \cos(qx))' & (e^{px} \sin(qx))' \end{vmatrix} \\ &= \begin{vmatrix} e^{px} \cos(qx) & e^{px} \sin(qx) \\ pe^{px} \cos(qx) - qe^{px} \sin(qx) & pe^{px} \sin(qx) + qe^{px} \cos(qx) \end{vmatrix} \\ &= (pe^{2px} \sin(qx) \cos(qx) + qe^{2px} \cos^2(qx)) \\ &\quad - (pe^{2px} \sin(qx) \cos(qx) - qe^{2px} \sin^2(qx)) \\ &= qe^{2px} \neq 0 \quad (q \neq 0) \end{aligned}\quad (3.52)$$

Example 3.3.

Solve the ODE

$$\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + 5y = 0 \quad (3.53)$$

with the I.C.s $y(0) = -1, y'(0) = 3$.

Solution. The auxiliary equation of this second-order constant-coefficient ODE yields

$$\begin{aligned}r^2 + 2r + 5 &= 0 \\ r &= \frac{-2 \pm \sqrt{(2)^2 - 4(1)(5)}}{2}\end{aligned}$$

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$$= -1 \pm \frac{\sqrt{-16}}{2} = -1 \pm \frac{4i}{2} = -1 \pm 2i \quad (3.54)$$

as its two complex roots. By the remark after (3.51), the two real-valued solutions are

$$y_1 = e^{-x} \cos(2x), y_2 = e^{-x} \sin(2x) \quad (3.55)$$

and we have

$$y'_1 = -2e^{-x} \sin(2x) - e^{-x} \cos(2x), y'_2 = 2e^{-x} \cos(2x) - e^{-x} \sin(2x) \quad (3.56)$$

Substituting the I.C.s into the general solution:

$$y = c_1 y_1 + c_2 y_2 = c_1 e^{-x} \cos(2x) + c_2 e^{-x} \sin(2x) \quad (3.57)$$

$$y' = c_1 (-2e^{-x} \sin(2x) - e^{-x} \cos(2x)) + c_2 (2e^{-x} \cos(2x) - e^{-x} \sin(2x)) \quad (3.58)$$

we retrieve

$$\begin{cases} c_1(1) + c_2(0) = -1 \\ c_1(-1) + c_2(2) = 3 \end{cases} \quad (3.59)$$

$$\therefore c_1 = -1, c_2 = 1 \quad (3.60)$$

□

Finally, Table 3.1 below provides a short summary about these three cases of the auxiliary equation.

Roots of Auxiliary Equation	Solution Form
Two distinct real roots	$c_1 e^{r_1 x} + c_2 e^{r_2 x}$
One repeated real root	$c_1 e^{r_1 x} + c_2 x e^{r_1 x}$
Two complex conjugate roots	$c_1 e^{p x} \cos(q x) + c_2 e^{p x} \sin(q x)$

Table 3.1.: *The three cases of auxiliary equation: two distinct real roots, one repeated real root, two complex roots, and their corresponding solution forms.*

3.2.2. Damped Spring-Mass System, Harmonic Oscillators

As mentioned in the introduction, many physical applications essentially belong to the wide class of **harmonic oscillators**. We will use the *spring-mass system* as an illustration. Initially, a mass m is connected to and hung by a spring vertically in mid-air, as demonstrated by Figure 3.1a below. The spring has a spring constant $k > 0$, and the force exerted by it on the mass is $-kz$ by *Hooke's Law*, where z is the vertical displacement (taking downward as positive) of the spring (stretched/compressed) from its unperturbed position. Then, by *Newton's Second Law*, the governing equation is

$$F = -kz + mg = m \frac{d^2 z}{dt^2} \quad (3.61)$$

with g as the gravitational acceleration. We can do a variable transformation $z = x + \frac{mg}{k}$ so that the expression of the net force acting on the mass

$$\begin{aligned} m \frac{d^2}{dt^2} \left(x + \frac{mg}{k} \right) &= -k \left(x + \frac{mg}{k} \right) + mg \\ m \frac{d^2 x}{dt^2} &= -kx - mg + mg = -kx \end{aligned} \quad (3.62)$$

is simplified to being linearly proportional to the *equilibrium position* x , i.e. $\frac{d^2 x}{dt^2} = 0$ when $x = 0$ ($z = \frac{mg}{k}$).

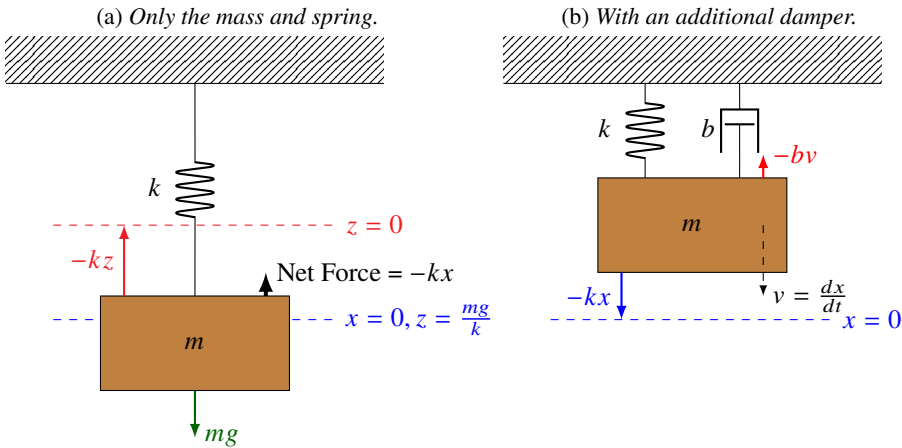


Figure 3.1.: Two schematics about the spring-mass system.

According to the last subsection, the spring-mass system (3.62) can be rearranged into the common

form of a second-order constant-coefficient homogeneous ODE as

$$\frac{d^2x}{dt^2} + \frac{k}{m}x = 0 \quad (3.63)$$

with an auxiliary equation of $r^2 + \frac{k}{m} = 0$, and has a simple general solution of

$$\begin{aligned} x &= c_1 \cos\left(\sqrt{\frac{k}{m}}t\right) + c_2 \sin\left(\sqrt{\frac{k}{m}}t\right) \\ &= c_1 \cos(\omega_0 t) + c_2 \sin(\omega_0 t) \end{aligned} \quad (3.64)$$

where $\omega_0 = \sqrt{\frac{k}{m}}$ is referred to as the *intrinsic frequency*. This kind of purely sinusoidal motion is then called a **simple harmonic oscillator**.

Now we will add a damper to the spring-mass system (see Figure 3.1b). The damping force is assumed to be linearly proportional to the opposite of velocity $v = \frac{dx}{dt}$ so that (3.62) becomes

$$m \frac{d^2x}{dt^2} = -kx - b \frac{dx}{dt} \quad (3.65)$$

where $b > 0$ is the damping coefficient. Rearranging it further to get

$$\begin{aligned} m \frac{d^2x}{dt^2} + b \frac{dx}{dt} + kx &= 0 \\ \frac{d^2x}{dt^2} + \frac{b}{m} \frac{dx}{dt} + \frac{k}{m}x &= 0 \\ \frac{d^2x}{dt^2} + \alpha \frac{dx}{dt} + \omega_0^2 x &= 0 \end{aligned} \quad (3.66)$$

where we set $\alpha = \frac{b}{m}$. We now have the equation of motion for a damped harmonic oscillator as a general second-order constant-coefficient homogeneous ODE. Again, according to the last subsection, there will be three types of solutions/responses depending on the magnitude of α : under-damped, over-damped, and critically-damped, explained as follows.

Under-damped

If the damping is weak, we may expect that there would still be some oscillation, but the amplitude has to decay gradually. This is known as *under-damped* and happens when the discriminant for the auxiliary equation for (3.66), $r^2 + \alpha r + \omega_0^2 = 0$, is negative, i.e. $\alpha^2 - 4\omega_0^2 < 0$ or $\alpha < 2\omega_0$ (α and ω_0 are always positive). The two complex roots for the auxiliary equation are then

$$r_{\pm} = \frac{-\alpha \pm \sqrt{\alpha^2 - 4\omega_0^2}}{2}$$

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$$= -\frac{\alpha}{2} \pm \left(\sqrt{\omega_0^2 - \frac{\alpha^2}{4}} \right) i = -\frac{\alpha}{2} \pm \omega_1 \quad (3.67)$$

where we let $\omega_1 = \sqrt{\omega_0^2 - \frac{\alpha^2}{4}}$. The general solution is hence

$$x = c_1 e^{-\frac{\alpha}{2}t} \cos(\omega_1 t) + c_2 e^{-\frac{\alpha}{2}t} \sin(\omega_1 t) \quad (3.68)$$

By using the trigonometric identity $\cos a \cos b + \sin a \sin b = \cos(a - b)$, we can reorganize the solution as

$$\begin{aligned} x &= A e^{-\frac{\alpha}{2}t} \cos \phi \cos(\omega_1 t) + A e^{-\frac{\alpha}{2}t} \sin \phi \sin(\omega_1 t) \\ &= A e^{-\frac{\alpha}{2}t} \cos(\omega_1 t - \phi) \end{aligned} \quad (3.69)$$

where we have taken the liberty to assign $c_1 = A \cos \phi$, $c_2 = A \sin \phi$, and $A = \sqrt{c_1^2 + c_2^2}$, $\tan \phi = \frac{c_2}{c_1}$. Compared to the simple harmonic oscillator case, the amplitude of an under-damped oscillator decays exponentially with a characteristic rate of $\frac{\alpha}{2}$. Moreover, the damping causes the actual frequency $\omega_1 = \sqrt{\omega_0^2 - \frac{\alpha^2}{4}}$ to be lower than the intrinsic frequency ω_0 .

Over-damped

In contrast, when the discriminant for the auxiliary equation $r^2 + \alpha r + \omega_0^2 = 0$ is positive, so that $\alpha^2 - 4\omega_0^2 > 0$ or $\alpha > 2\omega_0$, it will have two real roots:

$$\begin{aligned} r_{\pm} &= \frac{-\alpha \pm \sqrt{\alpha^2 - 4\omega_0^2}}{2} \\ &= -\frac{\alpha}{2} \pm \sqrt{\frac{\alpha^2}{4} - \omega_0^2} \end{aligned} \quad (3.70)$$

Letting $\frac{\alpha_1}{2} = -r_+ = -\left(-\frac{\alpha}{2} + \sqrt{\frac{\alpha^2}{4} - \omega_0^2}\right)$, $\frac{\alpha_2}{2} = -r_- = -\left(-\frac{\alpha}{2} - \sqrt{\frac{\alpha^2}{4} - \omega_0^2}\right)$, the general solution can be written as

$$x = c_1 e^{-\frac{\alpha_1}{2}t} + c_2 e^{-\frac{\alpha_2}{2}t} \quad (3.71)$$

where $\alpha_1 < \alpha_2$ and the first part of the solution will decay more slowly than the second part. This is referred to as *over-damped*.

Critically-damped

The final case is *critically-damped* where $\alpha = 2\omega_0$ exactly. Let's define this particular value by $2\omega_0 := \alpha_c$, and the auxiliary equation will have only one repeated real root: $r = \frac{\alpha_c}{2}$. The general

solution will then be

$$x = c_1 e^{-\frac{\alpha_c}{2}t} + c_2 t e^{-\frac{\alpha_c}{2}t} \quad (3.72)$$

Given a fixed intrinsic frequency ω_0 (so the spring is always the same one) and a varying damping force α , the decay of displacement amplitude in the critically-damped scenario will be the fastest. To see this, the under-damped case has the same form of decay rate, $\frac{\alpha}{2}$, but with a smaller magnitude than the critically-damped case, as $\alpha < \alpha_c = 2\omega_0$. Meanwhile, in the over-damped case, we can observe that $\frac{\alpha_1}{2} = \frac{\alpha}{2} - \sqrt{\frac{\alpha^2}{4} - \omega_0^2} < \omega_0$, seen by considering a right-angled triangle with sides ω_0 , $\sqrt{\frac{\alpha^2}{4} - \omega_0^2}$, and the hypotenuse $\frac{\alpha}{2}$, so this portion of the over-damped solution will decay slower than that of the critically-damped one.

3.3. Particular Solution for Non-homogeneous ODEs

Previously, we have been dealing with only homogeneous constant-coefficient ODEs. If there is an inhomogeneous term, then a particular solution should arise. This section will explore two methods to obtain a particular solution: undetermined coefficients and variation of parameters.

3.3.1. Method of Undetermined Coefficients

For simple source terms such as powers of x , an exponential, or a sine/cosine, we can guess the particular solution that has the same form but with some other coefficients to be decided. This is known as the **Method of Undetermined Coefficients**. Table 3.2 below shows what the guessed ansatz should look like, given different source terms.

For a linear combination of the above inhomogeneous terms, the particular solution will also be a linear combination of the corresponding ansatzes due to the linearity of constant-coefficient ODEs.

Example 3.4.

Find the full solution for the ODE

$$\frac{d^2 y}{dx^2} + y = 5x e^{2x} \quad (3.73)$$

with the I.C.s $y(0) = -1$, $y'(0) = 1$.

Solution. It is not hard to see that the complementary solution corresponding to the homogeneous part of this second-order constant-coefficient ODE is $y_c = c_1 \cos x + c_2 \sin x$, as the auxiliary

3.3. Particular Solution for Non-homogeneous ODEs ■

R.H.S.	Guess
A polynomial $p(x)$	Another polynomial of the same degree $q(x)$
An exponential e^{ax}	The same exponential Ae^{ax} but with an extra coefficient
A polynomial times an exponential $p(x)e^{ax}$	Another polynomial of the same degree times the exponential $q(x)e^{ax}$
A sine/cosine $\sin(bx), \cos(bx)$	A combination of both the sine and cosine $A \sin(bx) + B \cos(bx)$, even if the R.H.S. only contains one of them
A polynomial times a sine/cosine $p(x) \sin(bx), p(x) \cos(bx)$	Another two polynomials of the same degree times the sine/cosine $q_1(x) \sin(bx) + q_2(x) \cos(bx)$
An exponential times a sine/cosine $e^{ax} \sin(bx), e^{ax} \cos(bx)$	A combination of both the sine and cosine times the exponential $Ae^{ax} \sin(bx) + Be^{ax} \cos(bx)$

Table 3.2.: List of ansatzes for the Method of Undetermined Coefficients.

equation is $r^2 + 1 = 0$ and has the two complex roots $r = \pm i$. Meanwhile, according to the table above, we can make a guess for the particular solution in the following form:

$$y_p = (A_1x + A_0)e^{2x} \quad (3.74)$$

Substituting this into (3.73) gives

$$\begin{aligned}
 \frac{d^2}{dx^2} [(A_1x + A_0)e^{2x}] + (A_1x + A_0)e^{2x} &= 5xe^{2x} \\
 \frac{d}{dx} [(A_1x + A_0)2e^{2x} + A_1e^{2x}] + (A_1x + A_0)e^{2x} &= 5xe^{2x} \\
 [(A_1x + A_0)4e^{2x} + 4A_1e^{2x}] + (A_1x + A_0)e^{2x} &= 5xe^{2x} \\
 5A_1xe^{2x} + (4A_1 + 5A_0)e^{2x} &= 5xe^{2x}
 \end{aligned} \quad (3.75)$$

By comparing the coefficients, we quickly obtain $A_1 = 1$, $A_0 = -\frac{4}{5}$, and so the particular solution is $y_p = (x - \frac{4}{5})e^{2x}$. The general solution is hence

$$y = y_c + y_p = c_1 \cos x + c_2 \sin x + \left(x - \frac{4}{5}\right)e^{2x} \quad (3.76)$$

and its derivative can be checked to be $y' = -c_1 \sin x + c_2 \cos x + (2x - \frac{3}{5})e^{2x}$. Putting the I.C.s into these, we arrive at the final answer:

$$\begin{cases} c_1(1) + c_2(0) - \frac{4}{5}(1) = c_1 - \frac{4}{5} = -1 \\ -c_1(0) + c_2(1) - \frac{3}{5}(1) = c_2 - \frac{3}{5} = 1 \end{cases}$$

$$\therefore c_1 = -\frac{1}{5}, c_2 = \frac{8}{5} \quad (3.77)$$

$$\therefore y = -\frac{1}{5} \cos x + \frac{8}{5} \sin x + \left(x - \frac{4}{5}\right)e^{2x} \quad (3.78)$$

□

However, if the source term coincides with any complementary solution of the ODE, then plugging in the same form, supposedly as the particular solution, will result in zero and fail to match the inhomogeneous part. The remedy is very similar to the case where the auxiliary equation has a repeated root as described in Section 3.2.1 before: We will just add a power of x to the ansatz. To illustrate, consider a special case where

$$\left(\frac{d}{dx} - \alpha\right)\left(\frac{d}{dx} - \beta\right)y = \frac{d^2y}{dx^2} - (\alpha + \beta)\frac{dy}{dx} + \alpha\beta y = Ae^{\alpha x} \quad (3.79)$$

This ODE has the obvious complementary solutions of $e^{\alpha x}$ and $e^{\beta x}$ (assumed that $\alpha \neq \beta$), so if we insert something like $Be^{\alpha x}$ as the test particular solution, the L.H.S. will vanish and it does not work for the source term $Ae^{\alpha x}$ on the R.H.S. This is equivalent to the fact that

$$\left(\frac{d}{dx} - \alpha\right)\left(\frac{d}{dx} - \beta\right)e^{\alpha x} = 0 \quad (3.80)$$

Now we can differentiate this equation with respect to α to get

$$\begin{aligned} \frac{\partial}{\partial \alpha} \left[\left(\frac{d}{dx} - \alpha\right)\left(\frac{d}{dx} - \beta\right)e^{\alpha x} \right] &= 0 \\ -\left(\frac{d}{dx} - \beta\right)e^{\alpha x} + \left(\frac{d}{dx} - \alpha\right)\left(\frac{d}{dx} - \beta\right)(xe^{\alpha x}) &= 0 \\ \left(\frac{d}{dx} - \alpha\right)\left(\frac{d}{dx} - \beta\right)(xe^{\alpha x}) &= \left(\frac{d}{dx} - \beta\right)e^{\alpha x} = (\alpha - \beta)e^{\alpha x} \end{aligned} \quad (3.81)$$

This readily shows that the new particular solution $xe^{\alpha x}$ can reproduce the inhomogeneous term $e^{\alpha x}$ on the R.H.S. with an extra factor $\alpha - \beta$, and it is a viable ansatz, where we can just take the coefficient to be $\frac{A}{\alpha - \beta}$.

To generalize, for a higher-order constant-coefficient ODE, if the inhomogeneous source term $f(x)$ (probably multiplied by some polynomial $p(x)$ of degree k) appears m times (sometimes known as the *multiplicity*) in the complementary solution basis, then for the ansatz, we will have to try $q(x)f(x)$ where $q(x)$ is a polynomial of degree $k + m$ to be determined.

Example 3.5.

Find the full solution for the ODE

$$\frac{d^2y}{dx^2} + y = 2 \sin x \quad (3.82)$$

where $y(0) = 1, y'(0) = 0$.

3.3. Particular Solution for Non-homogeneous ODEs ■

Solution. It is not difficult to see that the appropriate auxiliary equation is $r^2 + 1 = 0$, so $r = \pm i$, and the complementary solution consists of $y_1 = \cos x$ and $y_2 = \sin x$, as in the last example. Since the inhomogeneous term is the same as (a scalar multiple of) y_2 , we need to raise one degree and try $y_p = Ax \sin x + Bx \cos x$. The corresponding derivatives are

$$\frac{dy_p}{dx} = Ax \cos x + A \sin x - Bx \sin x + B \cos x \quad (3.83)$$

$$\frac{d^2 y_p}{dx^2} = -Ax \sin x + 2A \cos x - Bx \cos x - 2B \sin x \quad (3.84)$$

Substituting these into (3.82) yields

$$\begin{aligned} (-Ax \sin x + 2A \cos x - Bx \cos x - 2B \sin x) + (Ax \sin x + Bx \cos x) &= 2 \sin x \\ -2A \cos x - 2B \sin x &= 2 \sin x \end{aligned} \quad (3.85)$$

Matching the coefficients gives $A = 0, B = -1$, and the general solution has the form of $y = c_1 \cos x + c_2 \sin x - x \cos x$, and thus $\frac{dy}{dx} = -c_1 \sin x + c_2 \cos x - \cos x + x \sin x$. Substituting the I.C.s then results in $c_1 = c_2 = 1$, hence the final answer is

$$y = \cos x + \sin x - x \cos x \quad (3.86)$$

□

3.3.2. Variation of Parameters

Another more general way to derive the particular solution for a non-homogeneous second-order ODE is the method of **Variation of Parameters**, which can be applied to a much broader class of forcing terms compared to the method of undetermined coefficients. Assume that the complementary solution takes the form of $y_c = c_1 y_1 + c_2 y_2$, where $y_1(x), y_2(x)$ are linearly independent. The essence of this method is to "vary" the constants c_1, c_2 so now they are some parameters $v_1(x), v_2(x)$ that are functions of x , and the particular solution will look like

$$y_p = v_1(x)y_1 + v_2(x)y_2 \quad (3.87)$$

The corresponding first derivative is

$$\begin{aligned} y'_p &= v'_1 y_1 + v_1 y'_1 + v'_2 y_2 + v_2 y'_2 \\ &= (v'_1 y_1 + v'_2 y_2) + v_1 y'_1 + v_2 y'_2 \\ &= v_1 y'_1 + v_2 y'_2 \end{aligned} \quad (3.88)$$

where we have made a very important assumption that

$$v'_1 y_1 + v'_2 y_2 = 0 \quad (3.89)$$

Chapter 3 – Basic Second-order Ordinary Differential Equations

This will be of use later. Subsequently, the second derivative will be

$$y_p'' = v_1' y_1' + v_1 y_1'' + v_2' y_2' + v_2 y_2'' \quad (3.90)$$

For the original second-order non-homogeneous ODE that supposedly takes the form of:

$$\frac{d^2 y}{dx^2} + \alpha \frac{dy}{dx} + \beta y = f(x) \quad (3.91)$$

where $f(x)$ is the source term, substituting the assumed particular solution (3.87) with (3.88) and (3.90) leads to

$$\begin{aligned} y_p'' + \alpha y_p' + \beta y_p &= f(x) \\ (v_1' y_1' + v_1 y_1'') + v_2' y_2' + v_2 y_2'' + \alpha(v_1 y_1' + v_2 y_2') + \beta(v_1 y_1 + v_2 y_2) &= f(x) \\ v_1 y_1'' + \alpha v_1 y_1' + \beta v_1 y_1 + v_2 y_2'' + \alpha v_2 y_2' + \beta v_2 y_2 + v_1' y_1' + v_2' y_2' &= f(x) \\ v_1(y_1'' + \alpha y_1' + \beta y_1) + v_2(y_2'' + \alpha y_2' + \beta y_2) + v_1' y_1' + v_2' y_2' &= f(x) \\ v_1' y_1' + v_2' y_2' &= f(x) \end{aligned} \quad (3.92)$$

The bracketed terms vanish because y_1, y_2 are exactly the complementary solutions that satisfy the given ODE: $y_1'' + \alpha y_1' + \beta y_1 = 0$ and $y_2'' + \alpha y_2' + \beta y_2 = 0$. Equations (3.89) and (3.92) then form a linear system

$$\begin{cases} v_1' y_1 + v_2' y_2 = 0 \\ v_1' y_1' + v_2' y_2' = f(x) \end{cases} \quad \begin{matrix} (3.93a) \\ (3.93b) \end{matrix}$$

which can be solved for v_1', v_2' (y_1, y_2 are known already). Integrating them respectively then yields v_1, v_2 , and hence we may proceed to retrieve the particular solution from (3.87).

Example 3.6.

Find the particular solution for the non-homogeneous ODE:

$$\frac{d^2 y}{dx^2} + 4y = \tan x \quad (3.94)$$

Solution. The two complementary solutions for the homogeneous part should now be clearly seen to be $y_1 = \cos(2x)$ and $y_2 = \sin(2x)$. Now we let $y_p = v_1(x)y_1 + v_2(x)y_2$ as suggested above, and need to solve, according to (3.93):

$$\begin{cases} v_1' \cos(2x) + v_2' \sin(2x) = 0 \\ v_1' (-2 \sin(2x)) + v_2' (2 \cos(2x)) = \tan x \end{cases} \quad \begin{matrix} (3.95a) \\ (3.95b) \end{matrix}$$

By Cramer's Rule, we get

$$v_1' = \frac{1}{W} \begin{vmatrix} 0 & \sin(2x) \\ \tan x & 2 \cos(2x) \end{vmatrix}$$

3.3. Particular Solution for Non-homogeneous ODEs ■

$$\begin{aligned}
 &= \frac{1}{2}(-\sin(2x)\tan(x)) \\
 &= -\frac{1}{2}(2\sin x \cos x) \frac{\sin x}{\cos x} = -\sin^2 x
 \end{aligned} \tag{3.96}$$

where the Wronskian is

$$\begin{aligned}
 W[y_1, y_2] &= \begin{vmatrix} \cos(2x) & \sin(2x) \\ -2\sin(2x) & 2\cos(2x) \end{vmatrix} \\
 &= 2\cos^2(2x) + 2\sin^2(2x) = 2
 \end{aligned} \tag{3.97}$$

Similarly,

$$\begin{aligned}
 v_2' &= \frac{1}{W} \begin{vmatrix} \cos(2x) & 0 \\ -2\sin(2x) & \tan x \end{vmatrix} \\
 &= \frac{1}{2} \cos(2x) \tan x \\
 &= \frac{1}{2} (2\cos^2 x - 1) \tan x \\
 &= \sin x \cos x - \frac{1}{2} \tan x = \frac{1}{2} \sin 2x - \frac{1}{2} \tan x
 \end{aligned} \tag{3.98}$$

Now what remains is to integrate them to obtain

$$\begin{aligned}
 v_1 &= \int -\sin^2 x dx \\
 &= \int \frac{\cos(2x) - 1}{2} dx = \frac{1}{4} \sin(2x) - \frac{1}{2} x
 \end{aligned} \tag{3.99}$$

$$\begin{aligned}
 v_2 &= \int \left(\frac{1}{2} \sin 2x - \frac{1}{2} \tan x \right) dx \\
 &= -\frac{1}{4} \cos(2x) + \frac{1}{2} \ln(\cos x)
 \end{aligned} \tag{3.100}$$

(No need to include the integration constants, why?) and thus the particular solution will be

$$\begin{aligned}
 y_p &= v_1 y_1 + v_2 y_2 \\
 &= \left[\frac{1}{4} \sin(2x) - \frac{1}{2} x \right] (\cos(2x)) + \left[-\frac{1}{4} \cos(2x) + \frac{1}{2} \ln(\cos x) \right] (\sin(2x)) \\
 &= -\frac{1}{2} x \cos(2x) + \frac{1}{2} \sin(2x) \ln(\cos x)
 \end{aligned} \tag{3.101}$$

□

3.3.3. Forced Harmonic Oscillators, Resonance

In Section 3.2.2, we discussed the damped spring-mass system as a type of harmonic oscillator, without any external force exerted on it. In general, the governing equation is represented by

$$\frac{d^2x}{dt^2} + \alpha \frac{dx}{dt} + \omega_0^2 x = f(t) \quad (3.102)$$

where the source term was zero before, $f(t) = 0$ for (3.66). Now, we will consider the case where an extra forcing $f(t) \neq 0$ is applied to the system. It is further assumed that the forcing is independent of the position x while sinusoidal in time t , particularly consists of a single frequency (*monochromatic*), $f(t) = \frac{\hat{F}}{m} \cos(\omega t)$ for an easier treatment. The governing equation then becomes

$$\frac{d^2x}{dt^2} + \alpha \frac{dx}{dt} + \omega_0^2 x = \frac{\hat{F}}{m} \cos(\omega t) \quad (3.103)$$

The techniques just introduced can be used to derive the particular solution corresponding to this source term. Specifically, the method of undetermined coefficients will be more convenient: let's guess a particular solution in the form of

$$x_p(t) = C \cos(\omega t) + S \sin(\omega t) \quad (3.104)$$

then

$$x'_p = -C\omega \sin(\omega t) + S\omega \cos(\omega t) \quad (3.105)$$

$$x''_p = -C\omega^2 \cos(\omega t) - S\omega^2 \sin(\omega t) \quad (3.106)$$

Plugging all of these into the ODE (3.103) results in

$$\begin{aligned} &(-C\omega^2 \cos(\omega t) - S\omega^2 \sin(\omega t)) + \alpha(-C\omega \sin(\omega t) + S\omega \cos(\omega t)) \\ &+ \omega_0^2(C \cos(\omega t) + S \sin(\omega t)) = \frac{\hat{F}}{m} \cos(\omega t) \end{aligned} \quad (3.107)$$

Grouping the terms and comparing the coefficients for the sine/cosine, we have

$$\begin{cases} (\omega_0^2 - \omega^2)C + \alpha\omega S = \frac{\hat{F}}{m} \\ -\alpha\omega C + (\omega_0^2 - \omega^2)S = 0 \end{cases} \quad (3.108a)$$

$$(3.108b)$$

This linear system can be solved as (e.g. by Cramer's Rule)

$$\begin{cases} C = \frac{\hat{F}}{m} \frac{\omega_0^2 - \omega^2}{(\omega_0^2 - \omega^2)^2 + \alpha^2 \omega^2} \\ S = \frac{\hat{F}}{m} \frac{\alpha \omega}{(\omega_0^2 - \omega^2)^2 + \alpha^2 \omega^2} \end{cases} \quad (3.109a)$$

$$(3.109b)$$

3.3. Particular Solution for Non-homogeneous ODEs ■

Therefore,

$$\begin{aligned}
 x_p(t) &= C \cos(\omega t) + S \sin(\omega t) \\
 &= \frac{\hat{F}}{m} \left(\frac{\omega_0^2 - \omega^2}{(\omega_0^2 - \omega^2)^2 + \alpha^2 \omega^2} \cos(\omega t) + \frac{a\omega}{(\omega_0^2 - \omega^2)^2 + \alpha^2 \omega^2} \sin(\omega t) \right) \\
 &= \frac{\hat{F}}{m \sqrt{(\omega_0^2 - \omega^2)^2 + \alpha^2 \omega^2}} \\
 &\quad \left(\frac{\omega_0^2 - \omega^2}{\sqrt{(\omega_0^2 - \omega^2)^2 + \alpha^2 \omega^2}} \cos(\omega t) + \frac{a\omega}{\sqrt{(\omega_0^2 - \omega^2)^2 + \alpha^2 \omega^2}} \sin(\omega t) \right) \\
 &= A (\cos \theta \cos(\omega t) + \sin \theta \sin(\omega t)) = A \cos(\omega t - \theta)
 \end{aligned} \tag{3.110}$$

where we have assigned

$$\sin \theta = \frac{a\omega}{\sqrt{(\omega_0^2 - \omega^2)^2 + \alpha^2 \omega^2}} \quad \cos \theta = \frac{\omega_0^2 - \omega^2}{\sqrt{(\omega_0^2 - \omega^2)^2 + \alpha^2 \omega^2}} \tag{3.111}$$

$$A = \frac{\hat{F}}{m \sqrt{(\omega_0^2 - \omega^2)^2 + \alpha^2 \omega^2}} \tag{3.112}$$

by manipulating trigonometric relationships, similar to the unforced case. The value A will be the oscillation amplitude of the forced response and θ the phase angle (lagged relative to the input forcing). Usually, the damping α is a relatively smaller number, so the largest amplitude of A will occur when the external vibration coincides with the intrinsic frequency, $\omega = \omega_0$ (where the $(\omega_0^2 - \omega^2)^2$ part becomes zero):

$$A_{\max} = \frac{\hat{F}}{m \sqrt{\alpha^2 \omega_0^2}} = \frac{\hat{F}}{m \alpha \omega_0} \tag{3.113}$$

This is when *resonance* occurs. This sheds some light on the mechanism behind the deadly collapse of buildings from a severe earthquake or why a seemingly strong bridge can be destroyed by some gusty wind, where the structures have a natural intrinsic frequency ω_0 very close to the wind frequency ω .

We may approximate the change in amplitude near the resonance region. Rewriting (3.112), we have

$$A = \frac{\hat{F}}{m \alpha \omega} \left(\frac{(\omega_0^2 - \omega^2)^2}{\alpha^2 \omega^2} + 1 \right)^{-\frac{1}{2}} \tag{3.114}$$

The behavior of A when $\omega \approx \omega_0$ can be inferred by doing a linear approximation via *binomial*

expansion $((1+z)^n \approx 1+nz)$:

$$A \approx \frac{\hat{F}}{m\alpha\omega_0} \left(1 - \frac{1}{2} \frac{(\omega_0^2 - \omega^2)^2}{\alpha^2\omega_0^2} \right) \quad (3.115)$$

(We keep the difference term $\omega_0^2 - \omega^2$ intact.) This shows that the asymptotic behavior of A at $\omega \approx \omega_0$ is a quadratic function in ω^2 that peaks at $\omega = \omega_0$ (of course) and concave downwards. This is illustrated in Figure 3.2 below. The width of the peak can be estimated by locating the *half-amplitude* points:

$$\begin{aligned} A &\approx \frac{\hat{F}}{m\alpha\omega_0} \left(1 - \frac{1}{2} \frac{(\omega_0^2 - \omega^2)^2}{\alpha^2\omega_0^2} \right) = \frac{1}{2} A_{\max} = \frac{1}{2} \frac{\hat{F}}{m\alpha\omega_0} \\ 1 - \frac{1}{2} \frac{(\omega_0^2 - \omega^2)^2}{\alpha^2\omega_0^2} &= \frac{1}{2} \\ (\omega_0^2 - \omega^2)^2 &= \alpha^2\omega_0^2 \\ (\omega_0 - \omega)^2(\omega_0 + \omega)^2 &= \alpha^2\omega_0^2 \\ (\omega_0 - \omega)(\omega_0 + \omega) &= \pm\alpha\omega_0 \\ (\omega_0 - \omega)(2\omega_0) &\approx \pm\alpha\omega_0 \quad (\omega \approx \omega_0) \\ \therefore \Delta\omega = 2|\omega_0 - \omega| &\approx \alpha \end{aligned} \quad (3.116)$$

So the width of the amplitude peak is of the order α .

Amplitude of Forced, Damped Harmonic Oscillator

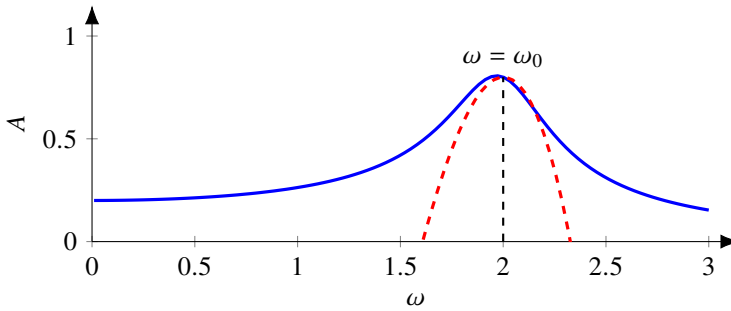


Figure 3.2.: The graph of the forced oscillation amplitude as a function of ω (blue, (3.114)) and the approximation near the resonance region (red, (3.115)), with the setting $\omega_0 = 2$, $\alpha = 0.5$.

3.4. Exercises

3.1) Compute the Wronskian for $y_1 =$ **3.8)** Solve

$e^{-x} \sin(5x)$, $y_2 = e^{-x} \cos(5x)$, and show that they are linearly independent.

$$\frac{d^2 y}{dx^2} - 2 \frac{dy}{dx} + 5y = 0 \quad (3.123)$$

3.2) Solve

$$\frac{d^2 y}{dx^2} - 3 \frac{dy}{dx} - 4y = 0 \quad (3.117)$$

with the I.C.s $y(0) = 2$, $y'(0) = 3$.

with the I.C.s $y(0) = -1$, $y'(0) = 1$.

3.3) Solve

$$\frac{d^2 y}{dx^2} + 4 \frac{dy}{dx} + 4y = 0 \quad (3.118)$$

with the I.C.s $y(0) = 2$, $y'(0) = -5$.

3.9) Show that $y_1 = 1$ and $y_2 = x$ are two linearly independent solutions to the rather simple second-order ODE

$$\frac{d^2 y}{dx^2} = 0 \quad (3.124)$$

3.4) Solve

$$\frac{d^2 y}{dx^2} + 12y = 0 \quad (3.119)$$

where the I.C.s consist of $y(0) = 1$, $y'(0) = 2$.

3.10) Consider the *RLC circuit system*:

$$L \frac{d^2 q}{dt} + R \frac{dq}{dt} + \frac{1}{C} q = 0 \quad (3.125)$$

where q is the amount of charge, L is the inductance, R is the resistance, and C is the capacitance. Derive the intrinsic frequency and determine when the critical damping arises by comparing against the discussion in Section 3.2.2. Hence, also find its resonance frequency/amplitude.

3.5) Solve for the ODE

$$\frac{d^2 y}{dx^2} - 6 \frac{dy}{dx} + y = 0 \quad (3.120)$$

with the I.C.s $y(0) = 1$, $y'(0) = 3$.

3.11) Find the full solution to the ODE

3.6) Solve

$$\frac{d^2 y}{dx^2} + 2 \frac{dy}{dx} + 4y = 0 \quad (3.121)$$

with the I.C.s $y(0) = 1$, $y'(0) = -1$.

$$\frac{d^2 y}{dx^2} + y = \sin(2x) \quad (3.126)$$

with the I.C.s of $y(0) = 0$, $y'(0) = 0$.

3.7) Solve

$$\frac{d^2 y}{dx^2} - 6 \frac{dy}{dx} + 9y = 0 \quad (3.122)$$

if $y(0) = 1$, $y'(0) = 4$.

3.12) Derive the general solution to the ODE below (initial conditions unspecified):

$$\frac{d^2 y}{dx^2} - 3 \frac{dy}{dx} + 2y = x \sin x \quad (3.127)$$

Series Solution

Introduction Despite having talked about how to solve various kinds of well-known ODEs analytically in the last four chapters, more often than not, a given ODE, even just slightly more complicated, will not have a simple closed-form solution. This would be particularly true for many important differential equations in Physics that we shall encounter in later chapters. Motivated by the idea of the Taylor series, we may want to construct a so-called series solution, which has the form of

$$y(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + \cdots = \sum_{n=0}^{\infty} a_nx^n \quad (5.1)$$

in the simplest sense. The logical next step is to substitute this expression into the ODE in question to determine the required coefficients a_n completely. If it can be done, we may expect that we can compute the solution within the degree of accuracy we want as we take increasingly more terms. Such a series solution is often a useful method for *linear* ODEs, and many important ODEs are linear. In the following parts of this chapter, we will see under what situations, (5.1) is indeed a valid solution, and also explore the actual procedure to reach such a series solution.

5.1. Series Solution for First-order ODEs

We will begin our analysis for **series solution** based on the easiest case of first-order ODEs.

5.1.1. Plausibility of Series Solution

Since an first-order ODE obviously involves the derivative y' , we have to first argue that the *term-by-term differentiation* of (5.1) is correct, i.e.

$$y'(x) = a_1 + 2a_2x + 3a_3x^2 + \cdots = \sum_{n=1}^{\infty} na_nx^{n-1} \quad (5.2)$$

and so on for higher-order derivatives if required later. This is guaranteed if (5.1) is **analytic** by **Abel's Theorem**.

Definition 5.1.

(Analyticity) A function $f(x)$ is real-analytic over an open interval (x_-, x_+) if it can be expressed as a power series convergent to $f(x)$ for x in the neighborhood of every $x_0 \in (x_-, x_+)$.

So, it implies that if we can derive a series solution like (5.1) (in this case $x_0 = 0$) by plugging it, together with the term-by-term derivative(s), into the ODE, and the resulting expression turns out to be analytic over an interval that contains $x = x_0$, it will then be a plausible solution of that ODE. The maximum possible interval, within which the series solution is analytic/convergent, is called the *radius of convergence*.

To brush up on the concept of analyticity, we can recall some basic examples from elementary Calculus. The Taylor series of the exponential function

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots = \sum_{n=0}^{\infty} \frac{x^n}{n!} \quad (5.3)$$

is analytic over the entire real x -axis. Meanwhile,

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots = \sum_{n=0}^{\infty} \frac{(-1)^{n+1}x^n}{n} \quad (5.4)$$

has a radius of convergence $|x| < 1$.

Now, for a general first-order linear ODE:

$$\frac{dy}{dx} + P(x)y = Q(x) \quad (5.5)$$

we may propose the series solution suggested by (5.1) and (5.2). To be more flexible, we may take the point of expansion to be about some $x = x_0$ other than $x = 0$, so that they become

$$y(x) = a_0 + a_1(x - x_0) + a_2(x - x_0)^2 + a_3(x - x_0)^3 + \cdots = \sum_{n=0}^{\infty} a_n(x - x_0)^n \quad (5.6)$$

$$y'(x) = a_1 + 2a_2(x - x_0) + 3a_3(x - x_0)^2 + \cdots = \sum_{n=1}^{\infty} na_n(x - x_0)^{n-1} \quad (5.7)$$

Substituting either pair of (5.1) and (5.2), or (5.6) and (5.7) into (5.5), and equating the coefficients, will then produce a **recurrence relation** that can be used to compute the coefficients a_n successively. The complexity depends on the form of $P(x)$: easy if it is just a constant or some power of x alone, difficult if it is other functions that are possibly expressed in a Taylor series with multiple terms; and $Q(x)$, which affects the algebraic process to solve for a_n at each step. Finally, we will have to check for the analyticity of the series solution.

Example 5.1.

Find the series solution to the first-order ODE

$$\frac{dy}{dx} + y = \tan^{-1} x \quad (5.8)$$

Solution. Substituting (5.1) and (5.2) into (5.8) leads to

$$\begin{aligned} \sum_{n=1}^{\infty} na_n x^{n-1} + \sum_{n=0}^{\infty} a_n x^n &= x - \frac{x^3}{3} + \frac{x^5}{5} - \cdots \\ \sum_{n=0}^{\infty} (n+1)a_{n+1} x^n + \sum_{n=0}^{\infty} a_n x^n &= \sum_{m=0}^{\infty} \frac{(-1)^m x^{2m+1}}{2m+1} \end{aligned} \quad (5.9)$$

where we have expressed $\tan^{-1} x$ in terms of Taylor series as well, and also replaced n by $n+1$ in the first summation, at the same time shifting the index by -1 . Unfortunately, due to the "on-and-off" pattern (only odd power terms present) in the Taylor series of $Q(x) = \tan^{-1} x$, we don't have a very clean recurrence relation or a simple closed-form solution. We will therefore only write down the first few terms of the series solution generated by comparing the coefficients of x^n in (5.9):

$$\begin{aligned} a_1 + a_0 &= 0 \Rightarrow a_1 = -a_0 \\ 2a_2 + a_1 &= 1 \Rightarrow a_2 = \frac{1 - a_1}{2} = \frac{1}{2} + \frac{a_0}{2} \\ 3a_3 + a_2 &= 0 \Rightarrow a_3 = \frac{-a_2}{3} = -\frac{1}{6} - \frac{a_0}{6} \\ 4a_4 + a_3 &= -\frac{1}{3} \Rightarrow a_4 = \frac{-\frac{1}{3} - a_3}{4} = -\frac{1}{24} + \frac{a_0}{24} \\ &\vdots \end{aligned} \quad (5.10)$$

Hence, the series solution will be

$$y = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + a_4 x^4 + \cdots$$

$$\begin{aligned}
 &= a_0 - a_0x + \left(\frac{1}{2} + \frac{a_0}{2}\right)x^2 + \left(-\frac{1}{6} - \frac{a_0}{6}\right)x^3 + \left(-\frac{1}{24} + \frac{a_0}{24}\right)x^4 + \cdots \\
 &= a_0\left(1 - x + \frac{1}{2}x^2 - \frac{1}{6}x^3 + \frac{1}{24}x^4 + \cdots\right) + \left(\frac{1}{2}x^2 - \frac{1}{6}x^3 - \frac{1}{24}x^4 + \cdots\right) \\
 &= a_0e^{-x} + \left(\frac{1}{2}x^2 - \frac{1}{6}x^3 - \frac{1}{24}x^4 + \cdots\right)
 \end{aligned} \tag{5.11}$$

Here, the first term associated with a_0 will be the complementary solution, while the second series is the particular solution. With this form of series solution, an initial condition imposed at $x = 0$ can be easily taken care of. For example, if $y(0) = 3$, then we easily conclude that $a_0 = 3$ as the other infinite series just vanish. $\square$

5.2. Series Solution for Second-order ODEs

It is straightforward to generalize the method of series solution to second-order ODEs. Specifically, the series form of the second-order derivative is produced by differentiating (5.7):

$$y''(x) = 2a_2 + 6a_3(x - x_0) + \cdots = \sum_{n=2}^{\infty} n(n-1)a_n(x - x_0)^{n-2} \tag{5.12}$$

When we are expanding about the origin, so that $x_0 = 0$, it is reduced to

$$y''(x) = 2a_2 + 6a_3x + \cdots = \sum_{n=2}^{\infty} n(n-1)a_nx^{n-2} \tag{5.13}$$

5.2.1. Definition of Ordinary Points

The next ingredient is the concept of **ordinary points**.

Definition 5.2.

(Ordinary Points) Ordinary points of a given second-order linear (homogeneous) ODE

$$\frac{d^2y}{dx^2} + P(x)\frac{dy}{dx} + Q(x)y = 0 \tag{5.14}$$

refer to the set of x_0 where $P(x)$ and $Q(x)$ are analytic at $x = x_0$.

Points that are otherwise not ordinary are called **singular points**. It can be shown that at such an ordinary point, the corresponding series solution (5.1) or (5.6) will be valid with a radius of convergence same as the smaller of those for $P(x)$ and $Q(x)$ (see Appendix A.2). For example,

1. $y'' + 3x^2y' + (\sin x)y = 0$: All x are ordinary points;

2. $y'' + \frac{1}{x}y' + \frac{1}{x^2}y = 0$: $x = 0$ is a singular point and all other points are ordinary;
3. $y'' + \frac{1}{x-2}y' + \frac{1}{x^2-1}y = 0$: $x = -1, 1, 2$ are the three singular points;
4. $y'' + \frac{\sin x}{x}y' + y = 0$: All x , including $x = 0$, are ordinary points.

5.2.2. Two-term Recurrence Relation

Although we just talked about ordinary points at any x , for simplicity, we will mostly consider the case of $x = 0$. We shall start with a very simple example.

Example 5.2.

Use the method of series solution to solve

$$\frac{d^2 y}{dx^2} + 4y = 0 \quad (5.15)$$

Solution. By the essence of Section 3.2.1, we know that its general solution is made up of $y_1 = \cos(2x)$ and $y_2 = \sin(2x)$. Now we will try to acquire a series solution to see if they are consistent. Specifically, $x = 0$ is an ordinary point of (5.15). Assuming (5.1) and also plugging (5.13) into (5.15):

$$\begin{aligned} \sum_{n=2}^{\infty} n(n-1)a_n x^{n-2} + 4 \sum_{n=0}^{\infty} a_n x^n &= 0 \\ \sum_{n=0}^{\infty} (n+2)(n+1)a_{n+2} x^n + 4 \sum_{n=0}^{\infty} a_n x^n &= 0 \end{aligned} \quad (5.16)$$

where as before we have replaced n by $n+2$ in the first summation and adjusted the index by -2 . (We will no longer explicitly read this kind of indexing change from now on, since it will be a regular routine.) The recurrence relation is then, by comparing the coefficients:

$$\begin{aligned} (n+2)(n+1)a_{n+2} + 4a_n &= 0 \\ a_{n+2} &= -\frac{4}{(n+2)(n+1)}a_n \end{aligned} \quad (5.17)$$

This form suggests that the coefficients a_n can be generated with a stride of 2 ($n \rightarrow n+2 \rightarrow n+4 \rightarrow \dots$) and the solution can be cleanly separated into two series, one for the odd indices and another for the even indices. For the even indices, we have

$$a_{2m} = -\frac{4}{(2m)(2m-1)}a_{2m-2}$$

$$\begin{aligned}
 &= \frac{(-4)^2}{(2m)(2m-1)(2m-2)(2m-3)} a_{2m-4} \\
 &= \frac{(-1)^m 2^{2m}}{(2m)!} a_0
 \end{aligned} \tag{5.18}$$

inductively. And similarly, for the odd indices, we can deduce that

$$a_{2m+1} = \frac{(-1)^m 2^{2m}}{(2m+1)!} a_1 \tag{5.19}$$

So the full series solution will be

$$\begin{aligned}
 y &= \sum_{n=0}^{\infty} a_n x^n \\
 &= a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \cdots \\
 &= (a_0 + a_2 x^2 + \cdots) + (a_1 x + a_3 x^3 + \cdots) \\
 &= a_0 (1 - 2x^2 + \cdots) + a_1 \left(x - \frac{2}{3} x^3 + \cdots \right) \\
 &= a_0 \sum_{m=0}^{\infty} \frac{(-1)^m (2x)^{2m}}{(2m)!} + \frac{a_1}{2} \sum_{m=0}^{\infty} \frac{(-1)^m (2x)^{2m+1}}{(2m+1)!} \\
 &= a_0 \cos(2x) + \frac{a_1}{2} \sin(2x)
 \end{aligned} \tag{5.20}$$

where we recognize the Taylor series of sine and cosine. So, this coincides with the usual solution except the integration constants are now denoted and scaled differently in terms of a_0, a_1 . $\square$

5.2.3. Three-term Recurrence Relation

In the last part, we dealt with ODEs that take a very simple form of having only the dependent variable and its second derivative with some constant coefficients, hence resulting in a neat two-term recurrence relation. However, it will be seldom the case: If there is also the first derivative involved, the recurrence relation will generally consist of three terms that will be much harder to compute. Here, we will explore a more tractable case of the old second-order constant-coefficient ODEs, about how to retrieve a special expression for the coefficients in the series solution.

Example 5.3.

Derive the series solution to

$$\frac{d^2 y}{dx^2} - 3 \frac{dy}{dx} + 2y = x \tag{5.21}$$

5.2. Series Solution for Second-order ODEs ■

Solution. We should have known that $y_1 = e^x$ and $y_2 = e^{2x}$ are the two complementary solutions from the usual approach. Now put the old (5.1), (5.2), and (5.13) into (5.21) above, to get

$$\begin{aligned} \sum_{n=2}^{\infty} n(n-1)a_n x^{n-2} - 3 \sum_{n=1}^{\infty} n a_n x^{n-1} + 2 \sum_{n=0}^{\infty} a_n x^n &= x \\ \sum_{n=0}^{\infty} (n+2)(n+1)a_{n+2} x^n - 3 \sum_{n=0}^{\infty} (n+1)a_{n+1} x^n + 2 \sum_{n=0}^{\infty} a_n x^n &= x \end{aligned} \quad (5.22)$$

Now we will address the homogeneous part first and ignore x on the R.H.S. to acquire the general recurrence relation first:

$$\begin{aligned} (n+2)(n+1)a_{n+2} - 3(n+1)a_{n+1} + 2a_n &= 0 \\ a_{n+2} &= \frac{3}{n+2}a_{n+1} - \frac{2}{(n+2)(n+1)}a_n \end{aligned} \quad (5.23)$$

Due to the pattern of the denominators, it is natural to guess that the a_n coefficients should contain a factor of $\frac{1}{n!}$ for (5.23) to hold throughout the iteration. To see this, suppose $a_n = \frac{b_n}{n!}$ and we shall obtain

$$\begin{aligned} \frac{b_{n+2}}{(n+2)!} &= \frac{3}{n+2} \frac{b_{n+1}}{(n+1)!} - \frac{2}{(n+2)(n+1)} \frac{b_n}{n!} \\ \frac{b_{n+2}}{(n+2)!} &= 3 \frac{b_{n+1}}{(n+2)!} - 2 \frac{b_n}{(n+2)!} \\ b_{n+2} &= 3b_{n+1} - 2b_n \end{aligned} \quad (5.24)$$

which looks much simpler. Borrowing the knowledge from Linear Algebra, we know that this translates to the matrix system

$$\begin{bmatrix} b_{n+2} \\ b_{n+1} \end{bmatrix} = \begin{bmatrix} 3 & -2 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} b_{n+1} \\ b_n \end{bmatrix} \quad (5.25)$$

This prompts us to compute the diagonalization of the matrix, essentially expressing the coefficients in a new basis so they only contain simple powers. Note that we don't actually have to compute that basis, but only the eigenvalues (refer to Appendix B.2) from the determinant:

$$\begin{aligned} \begin{vmatrix} 3-\lambda & -2 \\ 1 & -\lambda \end{vmatrix} &= 0 \\ -(3-\lambda)\lambda - (1)(-2) &= 0 \\ \lambda^2 - 3\lambda + 2 &= (\lambda-1)(\lambda-2) = 0 \end{aligned} \quad (5.26)$$

This reminds us of the usual auxiliary equation. (Admittedly, we have the benefit of hindsight.) The roots are $\lambda_1, \lambda_2 = 1, 2$ and so the b_n will take the form of $c_1 \lambda_1^n + c_2 \lambda_2^n$, thus

$$a_n = \frac{c_1}{n!} + \frac{c_2 2^n}{n!} \quad (5.27)$$

Therefore, we arrive at the usual complementary solution of

$$\begin{aligned}
 y_c &= \sum_{n=0}^{\infty} \left(\frac{c_1}{n!} + \frac{c_2 2^n}{n!} \right) x^n \\
 &= \sum_{n=0}^{\infty} \frac{c_1 x^n}{n!} + \sum_{n=0}^{\infty} \frac{c_2 (2x)^n}{n!} \\
 &= c_1 e^x + c_2 e^{2x}
 \end{aligned} \tag{5.28}$$

For the particular solution, we consider (5.22) again, but with x added back to the R.H.S. Since x is only a finite polynomial with degree of 1, we only need to consider $n = 0, 1$:

$$\begin{cases} 2a_2 - 3a_1 + 2a_0 = 0 & (5.29a) \\ 6a_3 - 6a_2 + 2a_1 = 1 & (5.29b) \end{cases}$$

Moreover, since this will be the particular solution, the higher-degree coefficients must not depend on the previous ones, and here it implies that a_1 must be exactly $\frac{1}{2}$ by the second equation (so that $a_2, a_3, \dots$ computed from the three-term recurrence relation in the future, can be just set to 0 and not related to a_0, a_1), and $a_0 = \frac{3}{4}$ from the first equation, i.e. $y_p = \frac{x}{2} + \frac{3}{4}$ will be the desired polynomial particular solution. $\square$

5.2.4. Revisiting the Curious Example of Error Functions

Let's consider the simple second-order linear ODE (4.43) again:

$$\frac{d^2 y}{dx^2} + 2x \frac{dy}{dx} = 0 \tag{5.30}$$

We have shown that its solution is

$$y = A \int_0^x e^{-t^2} dx + B = A \operatorname{erf}(x) + B \tag{5.31}$$

Now, if we revisit it by the series solution method, substituting the necessary expressions into (5.30) produces

$$\begin{aligned}
 \sum_{n=2}^{\infty} n(n-1)a_n x^{n-2} + 2x \sum_{n=1}^{\infty} n a_n x^{n-1} &= 0 \\
 \sum_{n=2}^{\infty} n(n-1)a_n x^{n-2} + 2 \sum_{n=1}^{\infty} n a_n x^n &= 0 \\
 \sum_{n=0}^{\infty} (n+2)(n+1)a_{n+2} x^n + 2 \sum_{n=1}^{\infty} n a_n x^n &= 0
 \end{aligned} \tag{5.32}$$

Comparing the coefficients then leads to the recurrence relation of

$$(n+2)(n+1)a_{n+2} + 2na_n = 0$$

$$a_{n+2} = -\frac{2na_n}{(n+2)(n+1)} \quad (5.33)$$

which readily generates the required coefficients of the series solution with a stride of 2, and hence the solution will be split into two series, one for the odd indices and another for the even indices. However, notice that when $n = 0$, (5.32) implies that $(2)(1)a_2 + (0) = 0$ since the second summation only starts from $n = 1$, so it must be $a_2 = 0$ and by (5.33) all even indices $a_{2m} = 0$ vanish, except a_0 which can be arbitrary. For odd indices, we have

$$a_3 = -\frac{2(1)a_1}{(3)(2)} = -\frac{a_1}{3} \quad (5.34a)$$

$$a_5 = -\frac{2(3)a_3}{(5)(4)} = -\frac{3}{10} \left(-\frac{a_1}{3}\right) = \frac{a_1}{10} \quad (5.34b)$$

and keep going, recursively:

$$\begin{aligned} a_{2m+1} &= -\frac{2(2m-1)a_{2m-1}}{(2m+1)(2m)} \\ &= -\frac{(2m-1)a_{2m-1}}{(2m+1)m} = -\frac{1}{m} \frac{2m-1}{2m+1} a_{2m-1} \\ &= -\frac{1}{m} \frac{2m-1}{2m+1} \left(-\frac{1}{m-1} \frac{2m-3}{2m-1} a_{2m-3}\right) \\ &= \frac{(-1)^m}{m!(2m+1)} a_1 \end{aligned} \quad (5.35)$$

So the series solution will be

$$\begin{aligned} y &= a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + a_5x^5 + \dots \\ &= a_0 + a_1x + (0) - \frac{a_1}{3}x^3 + (0) + \frac{a_1}{10}x^5 + \dots \\ &= a_0 + a_1 \left(x - \frac{1}{3}x^3 + \frac{1}{10}x^5 - \dots\right) \\ &= a_0 + a_1 \sum_{m=0}^{\infty} \frac{(-1)^m x^{2m+1}}{m!(2m+1)} \end{aligned} \quad (5.36)$$

The a_0 coefficient clearly corresponds to the constant term B in (5.31), while the infinite series related to a_1 corresponds to the error function $A \int_0^x e^{-t^2} dx$. Hence, we have identified the infinite series expansion of the error function as

$$\operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dx = \frac{2}{\sqrt{\pi}} \sum_{m=0}^{\infty} \frac{(-1)^m x^{2m+1}}{m!(2m+1)} \quad (5.37)$$

To check the consistency, we can try to integrate the Taylor series of e^{-x^2} term by term:

$$\begin{aligned}
 e^{-x^2} &= 1 - x^2 + \frac{x^4}{2} - \cdots = \sum_{m=0}^{\infty} \frac{(-1)^m x^{2m}}{m!} \\
 \int e^{-x^2} dx &= \int \left(1 - x^2 + \frac{x^4}{2} - \cdots \right) dx \\
 &= C + x - \frac{x^3}{3} + \frac{x^5}{10} - \cdots = C + \sum_{m=0}^{\infty} \frac{(-1)^m x^{2m+1}}{m!(2m+1)}
 \end{aligned} \tag{5.38}$$

which matches (5.36). The final caveat is to show either e^{-x^2} or $\int e^{-x^2} dx$ is analytic so the term-by-term differentiation/integration operations are allowed. This is not hard to argue because the Taylor series of the exponential function and hence e^{-x^2} is analytic with an infinite radius of convergence (can be shown by the Ratio Test if needed).

5.3. Oscillation and Airy Equation/Functions

Many commonly studied second-order linear ODEs possess the property of oscillation, which is necessary to describe physical waves. To motivate this section, let's go back and consider the simple harmonic second-order ODE:

$$\frac{d^2 y}{dx^2} + y = 0 \tag{5.39}$$

It is obvious that the two linearly independent solutions are $y_1 = \cos x$ and $y_2 = \sin x$, which have some curious characteristics. First, from elementary trigonometry, we know that they are just sinusoidal waves that have a (constant) wavelength/period of 2π . This means that they will oscillate about the x -axis indefinitely, i.e. they have infinitely many *zeros* (the points on the x -axis where the function equals 0). What is even more special is that these zeros of $\cos x$ and $\sin x$ appear alternately, so that there is exactly one zero for $\cos x$ ($x = \frac{\pi}{2} + n\pi$, for all integers n) between two consecutive zeros for $\sin x$ ($x = n\pi$, also for all integers n) and vice versa. (see Figure 5.1) Below, we will show that this is not a coincidence: the nature of the corresponding second-order linear ODEs (such as (5.39)) will enforce these properties upon their two linearly independent solutions.

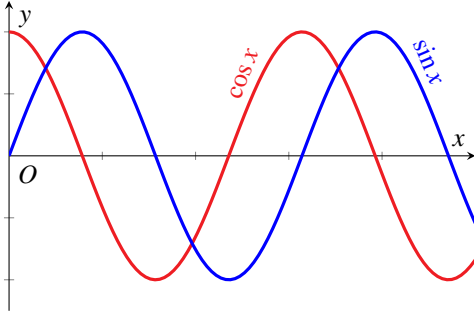


Figure 5.1.: Oscillatory behavior of the sine and cosine functions.

5.3.1. Some Oscillation Theorems

The detailed arguments for the following results are put in Appendix A.2 and mainly taken from Simmons [9]. The first of them is

Theorem 5.3.

If $y_1(x)$ and $y_2(x)$ are two linearly independent solutions to any standard second-order linear ODE:

$$\frac{d^2 y}{dx^2} + P(x) \frac{dy}{dx} + Q(x)y = 0 \quad (5.40)$$

then the two respective sets of zeros for y_1 and y_2 are distinct and occur alternately, i.e. there is exactly one zero of y_1 between any two successive zeros of y_2 and vice versa.

Next,

Theorem 5.4.

Given that $v(x)$ is a non-trivial solution to the canonical second-order linear ODE

$$v'' + g(x)v = 0 \quad (5.41)$$

if $g(x) < 0$ over some interval, then $v(x)$ has at most one zero over the same interval.

Any second-order linear ODE can be cast into the canonical form, as explained in Section 4.1.2, so this result can be extended freely. Therefore, if we want to have an oscillating solution, $g(x)$ in (5.41) must stay positive. However, it is only a necessary requirement. If indeed $g(x) > 0$ is positive but decays at a fast enough rate, then the magnitude of $v''(x) = -g(x)v(x) < 0$ may be too small to cause a reversal in the sign of v' , and v will not traverse back to cut the x -axis again. The theorem below shows a sufficient condition for the oscillation to occur surely if $g(x)$ does not decrease too rapidly.

Theorem 5.5.

For a non-trivial solution $v(x)$ to the canonical second-order linear ODE (5.41) where now $g(x) > 0$ for all $x > 0$, if

$$\int_1^{\infty} g(x)dx = \infty \quad (5.42)$$

then $v(x)$ will have infinitely many zeros on the positive x -axis.

The lower limit of $x = 1$ in the integral may be replaced by any point > 0 , and this theorem can be adapted for the negative x -axis as well.

5.3.2. Series Solution to Airy Equation

To see how the above results play out, we now examine a particular type of second-order linear ODE known as the **Airy equation**.

Definition 5.6.

(Airy Equation) The Airy equation is a second-order ODE in the form of

$$\frac{d^2y}{dx^2} - xy = 0 \quad (5.43)$$

The origin of the Airy equation, in addition to the Airy functions as we will meet soon, came from the study of refraction in rainbows. The two-part article by Bill Casselman (<https://www.ams.org/publicoutreach/feature-column/fcarc-rainbows> and <https://www.ams.org/publicoutreach/feature-column/fc-2015-02>) explains this nicely. A century later, the Airy equation becomes a useful component in the WKB method for Quantum Mechanics, as outlined in Section 11.3.2.

Compared with (5.41), we identify $g(x) = -x$. When $x < 0$, $g(x) = -x > 0$ and clearly $\int_{-\infty}^{-1} -x dx = \infty$, Theorem 5.5 adapted for the negative x -axis then implies that (5.43) is oscillatory to the left. On the other hand, when $x > 0$, $g(x) < 0$, Theorem 5.4 then illustrates that (5.43) will become exponential to the right. This quick analysis points out that the Airy equation demonstrates, in its simple form, a transition from oscillatory to exponential behavior.

The Airy equation (5.43) is easily seen to have an ordinary point at $x = 0$ and hence we can find its series solutions there. Now, substituting the familiar expressions (5.1), (5.2), and (5.13) into (5.43) leads to

$$\sum_{n=2}^{\infty} n(n-1)a_n x^{n-2} - x \sum_{n=0}^{\infty} a_n x^n = 0$$

$$\begin{aligned} \sum_{n=0}^{\infty} (n+2)(n+1)a_{n+2}x^n - \sum_{n=0}^{\infty} a_n x^{n+1} &= 0 \\ \sum_{n=0}^{\infty} (n+2)(n+1)a_{n+2}x^n - \sum_{n=1}^{\infty} a_{n-1}x^n &= 0 \end{aligned} \quad (5.44)$$

By comparing the coefficients, the recurrence relation is then

$$\begin{aligned} (n+2)(n+1)a_{n+2} - a_{n-1} &= 0 \\ a_{n+2} &= \frac{a_{n-1}}{(n+2)(n+1)} \end{aligned} \quad (5.45)$$

for $n \geq 1$, with a stride of 3. When $n = 0$, the second summation does not appear (also, by convention, $a_{-1} = 0$), and we must have $a_2 = 0$, hence $a_2 = a_5 = a_8 = \dots = a_{3m+2} = 0$ for all integers m . For a_{3m} , we have

$$\begin{aligned} a_3 &= \frac{a_0}{(3)(2)} \\ a_6 &= \frac{a_3}{(6)(5)} = \frac{a_0}{(6)(5)(3)(2)} \\ &\vdots \end{aligned} \quad (5.46)$$

and similarly, for a_{3m+1} :

$$\begin{aligned} a_4 &= \frac{a_1}{(4)(3)} \\ a_7 &= \frac{a_4}{(7)(6)} = \frac{a_1}{(7)(6)(4)(3)} \\ &\vdots \end{aligned} \quad (5.47)$$

The patterns keep going on. Therefore, two possible series solutions will then be

$$\begin{aligned} y &= a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + a_5x^5 + a_6x^6 + a_7x^7 + \dots \\ &= a_0 + a_1x + (0) + \frac{a_0}{(3)(2)}x^3 + \frac{a_1}{(4)(3)}x^4 + (0) + \frac{a_0}{(6)(5)(3)(2)}x^6 + \frac{a_1}{(7)(6)(4)(3)}x^7 + \dots \\ &= a_0 \left(1 + \frac{x^3}{(3)(2)} + \frac{x^6}{(6)(5)(3)(2)} + \dots \right) + a_1 \left(x + \frac{x^4}{(4)(3)} + \frac{x^7}{(7)(6)(4)(3)} + \dots \right) \end{aligned} \quad (5.48)$$

These two series are analytic and converge for all x . To see this, we apply the Ratio Test. For the first series:

$$\lim_{m \rightarrow \infty} \left| \frac{\frac{x^{3m+3}}{(3m+3)(3m+2)(3m)(3m-1)\dots}}{\frac{x^{3m}}{(3m)(3m-1)\dots}} \right| = \lim_{m \rightarrow \infty} \left| \frac{x^3}{(3m+3)(3m+2)} \right| = 0 \quad (5.49)$$

and it is similar for the second series. Therefore, any solution to the Airy equation can be expressed as the superposition of these two clearly linearly independent series.

5.3.3. Formulation of Airy Functions

Now we will show how the standard **Airy functions**, the canonical forms of the two solutions to the Airy equation, are constructed. These are built from some specific combinations of the two series solutions in (5.48) so as to abide by a certain convention.

First, we need the observation that any solution y of the Airy equation will either tend to 0 or $\pm\infty$ as $x \rightarrow \infty$. This can be proved by a delta-epsilon argument, and we omit the details. We are then motivated to find a specific solution form whose limit takes the value of zero at the point of infinity. Then, this shall be assigned as the **Airy function of the first kind** $\text{Ai}(x)$. It is known to have the integral form of

$$\text{Ai}(x) = \frac{1}{\pi} \int_0^\infty \cos\left(\frac{t^3}{3} + xt\right) dt \quad (5.50)$$

To develop this expression requires advanced knowledge from Complex Analysis, which is out of our scope. However, we can substitute (5.50) into (5.43) to see that it really satisfies the equation:

$$\begin{aligned} \frac{d^2 y}{dx^2} &= \frac{d^2}{dx^2} (\text{Ai}(x)) = \frac{d^2}{dx^2} \left(\frac{1}{\pi} \int_0^\infty \cos\left(\frac{t^3}{3} + xt\right) dt \right) \\ &= \frac{1}{\pi} \int_0^\infty \frac{\partial^2}{\partial x^2} \cos\left(\frac{t^3}{3} + xt\right) dt \\ &= \frac{1}{\pi} \int_0^\infty -t^2 \cos\left(\frac{t^3}{3} + xt\right) dt \\ &= \frac{1}{\pi} \int_0^\infty -\frac{t^2}{t^2 + x} \cos\left(\frac{t^3}{3} + xt\right) d\left(\frac{t^3}{3} + xt\right) \\ &= \frac{1}{\pi} \int_0^\infty \left(\frac{x}{t^2 + x} - 1 \right) \cos\left(\frac{t^3}{3} + xt\right) d\left(\frac{t^3}{3} + xt\right) \\ &= \frac{1}{\pi} \int_0^\infty \frac{x}{t^2 + x} \cos\left(\frac{t^3}{3} + xt\right) d\left(\frac{t^3}{3} + xt\right) \\ &\quad - \frac{1}{\pi} \int_0^\infty \cos\left(\frac{t^3}{3} + xt\right) d\left(\frac{t^3}{3} + xt\right) \\ &= \frac{x}{\pi} \int_0^\infty \frac{1}{t^2 + x} \cos\left(\frac{t^3}{3} + xt\right) d\left(\frac{t^3}{3} + xt\right) + (0) \\ &= \frac{x}{\pi} \int_0^\infty \cos\left(\frac{t^3}{3} + xt\right) dt = x \text{Ai}(x) = xy \end{aligned} \quad (5.51)$$

We have informally passed the x -derivative into the integral. There is a caveat to argue that the second integral in the third-to-last line has to converge to zero (the most reasonable value), but we will not go too deep into the issue here. However, notice that the integral expression of $\text{Ai}(x)$ in (5.50) can be shown to converge.¹ This partly suggests that what we have done in (5.51) makes

¹By using Dirichlet's test.

5.3. Oscillation and Airy Equation/Functions ■

some sense. Moreover, by the previous observation that an Airy function must either tend to 0 or $\pm\infty$, we immediately know that $\text{Ai}(x) \rightarrow 0$ as $x \rightarrow \infty$.

Further results from Complex Analysis and using the Gamma Function Γ (see Appendix B.1) show that

$$\text{Ai}(0) = \frac{1}{3^{\frac{2}{3}}\Gamma(\frac{2}{3})} \quad \text{Ai}'(0) = -\frac{1}{3^{\frac{1}{3}}\Gamma(\frac{1}{3})} \quad (5.52)$$

and hence by the form of (5.48), the series expansion of $\text{Ai}(x)$ will be just

$$\begin{aligned} \text{Ai}(x) = \text{Ai}(0) & \left(1 + \frac{x^3}{(3)(2)} + \frac{x^6}{(6)(5)(3)(2)} + \cdots \right) \\ & + \text{Ai}'(0) \left(x + \frac{x^4}{(4)(3)} + \frac{x^7}{(7)(6)(4)(3)} + \cdots \right) \end{aligned} \quad (5.53)$$

where we simply take $a_0 = \text{Ai}(0)$ and $a_1 = \text{Ai}'(0)$ to match the set of initial conditions in (5.52).

We can obtain the asymptotic expressions of $\text{Ai}(x)$ as x goes to a positive/negative value at large. This requires the saddle point method, which is quite advanced, and we simply state the results below:

$$\text{Ai}(x) \sim \begin{cases} \frac{1}{\sqrt{\pi}|x|^{\frac{1}{4}}} \sin\left(\frac{2|x|^{\frac{3}{2}}}{3} + \frac{\pi}{4}\right) & \text{as } x \rightarrow -\infty \\ \frac{e^{-\frac{2x^{\frac{3}{2}}}{3}}}{2\sqrt{\pi}x^{\frac{1}{4}}} & \text{as } x \rightarrow \infty \end{cases} \quad (5.54)$$

The two limiting cases correspond to the oscillatory/exponential decaying regimes, respectively.

Meanwhile, the **Airy function of the second kind** $\text{Bi}(x)$ is chosen in a way such that it oscillates $\frac{\pi}{2}$ out of phase relative to $\text{Ai}(x)$ in the negative region. Here we simply state its integral form:

$$\text{Bi}(x) = \frac{1}{\pi} \int_0^\infty \left[\exp\left(-\frac{t^3}{3} + xt\right) + \sin\left(\frac{t^3}{3} + xt\right) \right] dt \quad (5.55)$$

initial conditions:

$$\text{Bi}(0) = \frac{1}{3^{\frac{1}{6}}\Gamma(\frac{2}{3})} \quad \text{Bi}'(0) = \frac{3^{\frac{1}{6}}}{\Gamma(\frac{1}{3})} \quad (5.56)$$

and asymptotic formulas:

$$\text{Bi}(x) \sim \begin{cases} \frac{1}{\sqrt{\pi}|x|^{\frac{1}{4}}} \cos\left(\frac{2|x|^{\frac{3}{2}}}{3} + \frac{\pi}{4}\right) & \text{as } x \rightarrow -\infty \\ \frac{e^{\frac{2x^{\frac{3}{2}}}{3}}}{\sqrt{\pi}x^{\frac{1}{4}}} & \text{as } x \rightarrow \infty \end{cases} \quad (5.57)$$

Note that $\text{Bi}(x) \rightarrow \infty$ as $x \rightarrow \infty$, in contrast to $\text{Ai}(x) \rightarrow 0$. So they must be linearly independent. Moreover, it implies that any other solution to the Airy equation that also decays to zero at $x = \infty$ must be linearly dependent on and hence a constant multiple of $\text{Ai}(x)$ ², indicating the uniqueness of $\text{Ai}(x)$ up to a constant factor. Figure 5.2 below sketches the graphs of $\text{Ai}(x)$ and $\text{Bi}(x)$. It can be observed that in the oscillatory regime, as $x \rightarrow -\infty$ moves to the left, for both of them, the frequency slowly increases while the amplitude slowly decreases.

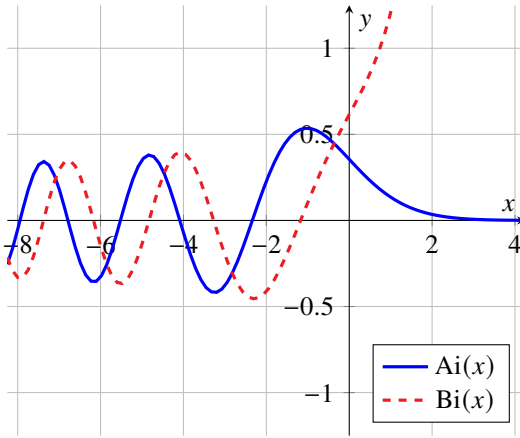


Figure 5.2.: The Airy Functions of the first/second kind, $\text{Ai}(x)$ (blue, solid) and $\text{Bi}(x)$ (red, dashed).

Example 5.4.

(Schrödinger Equation with a Constant Force Field) For a quantum particle that is confined inside a constant force field with a potential of $V(x) = kx$, the corresponding time-independent Schrödinger Equation is

$$\begin{aligned} -\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} + V(x)\psi &= E\psi \\ -\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} + kx\psi &= E\psi \end{aligned} \quad (5.58)$$

where E is the energy level and $\psi(x)$ is the wavefunction. By a suitable change of variable, show that it is actually an Airy Equation and find the physical solution to ψ .

Solution. We first rewrite (5.58) as

$$\frac{d^2\psi}{dx^2} - \frac{2m}{\hbar^2}(kx - E)\psi = 0 \quad (5.59)$$

²Otherwise, these two solutions alone cannot simulate unboundedness at infinity.

By a variable transformation (non-dimensionalization) $x = \left(\frac{\hbar^2}{2mk}\right)^{\frac{1}{3}} \tilde{x}$, it becomes

$$\begin{aligned} \left(\frac{2mk}{\hbar^2}\right)^{\frac{2}{3}} \frac{d^2\psi}{d\tilde{x}^2} - \frac{2m}{\hbar^2} \left(k \left(\frac{\hbar^2}{2mk}\right)^{\frac{1}{3}} \tilde{x} - E\right) \psi &= 0 \\ \frac{d^2\psi}{d\tilde{x}^2} - \left(\tilde{x} - \left(\frac{2m}{\hbar^2 k^2}\right)^{\frac{1}{3}} E\right) \psi &= 0 \end{aligned} \quad (5.60)$$

Denote $\tilde{E} = \left(\frac{2m}{\hbar^2 k^2}\right)^{\frac{1}{3}} E$ as the non-dimensionalized energy level, then (5.60) will appear as an Airy equation with the independent variable $\tilde{x}$ displaced by $\tilde{E}$:

$$\frac{d^2\psi}{d\tilde{x}^2} - (\tilde{x} - \tilde{E})\psi = 0 \quad (5.61)$$

The solution will then be formed by $\text{Ai}(\tilde{x} - \tilde{E})$ and $\text{Bi}(\tilde{x} - \tilde{E})$. However, to have a physically bounded solution, we will just keep $\text{Ai}(\tilde{x} - \tilde{E})$ and discard $\text{Bi}(\tilde{x} - \tilde{E})$, which blows up at infinity. $\tilde{E}$ will then be the turning point of the wavefunction. $\square$

5.4. Exercises

5.1) Revisit the rather simple ODE of

$$\frac{dy}{dx} = \alpha y \quad (5.62)$$

where α is a constant, from a series solution perspective, and compare it with the one obtained via the usual elementary method.

5.2) Solve the ODE

$$\frac{dy}{dx} + xy = \frac{1}{1-x} \quad (5.63)$$

via the method of series solution.

5.3) Find the series solution of

$$\frac{dy}{dx} + (x+1)y = e^{-x} \quad (5.64)$$

5.4) Find the series solution of

$$\frac{dy}{dx} + (x^2+1)y = x \quad (5.65)$$

with the I.C. of $y(0) = 1$.

5.5) Use the series solution method to solve

$$\frac{d^2y}{dx^2} - 9y = 0 \quad (5.66)$$

and compare to the answer that would be obtained by the usual method.

5.6) Solve the ODE

$$\frac{d^2y}{dx^2} + x \frac{dy}{dx} + xy = 0 \quad (5.67)$$

via the method of series solution.

5.7) Similarly, solve the ODE

$$\frac{d^2y}{dx^2} + x \frac{dy}{dx} + x^2 y = 0 \quad (5.68)$$

Laplace Transform

Introduction In this chapter, we shall explore a very useful machinery for solving linear ODEs, the Laplace Transform. We are interested in Laplace Transform, particularly because of its effect of converting an ODE in the time/space (t or x) domain to a simple algebraic equation in the Laplace (frequency) domain s that may be easier to solve, and then we can just apply an inverse transform to retrieve the physical solution for the ODE. This is illustrated by the flowchart shown in Figure 7.1 below. This technique is very popular in many applied scientific fields, like electrical engineering and control theory.

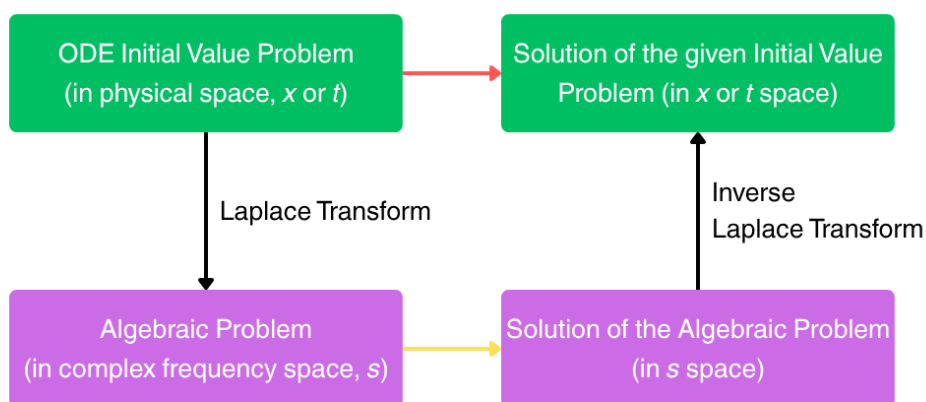


Figure 7.1.: A flowchart outlining the intermediate steps of solving ODEs via Laplace Transform.

7.1. Basic Ideas about Laplace Transform

To build up, we will first discuss several important aspects of the Laplace Transform.

7.1.1. Definition and Motivation of Laplace Transform

Definition 7.1.

(Laplace Transform) The **Laplace Transform** of a function $f(x)$ (or $f(t)$, just replace all x below by t) is defined as

$$\mathcal{L}[f](s) = \lim_{b \rightarrow \infty} \int_0^b e^{-sx} f(x) dx = \int_0^{\infty} e^{-sx} f(x) dx \quad (7.1)$$

where s is another real, positive variable.

To emphasize, such an *integral transform* $\mathcal{L}$ is applied to the function $f(x)$, $x > 0$, to make it become a new function in another real parameter $s > 0$ now.¹ Reasonably, it exists when the improper integral in (7.1) converges.

The form of the Laplace Transform (7.1) can be motivated by rewriting it with $z = e^{-s}$ (and using t instead of x for clarity):

$$\int_0^{\infty} f(t) z^t dt \quad (7.2)$$

that very much looks like a generalized power series of $\sum_{t=0}^{\infty} f(t) z^t$ if we move from a continuous integration to a discrete summation. We want to use a negative exponential so that it is easier for computation and the integral can converge. Essentially, by using the Laplace Transform, we are employing an extended version of the power series solution method to solve ODEs.²

The simplest but often needed properties of the Laplace Transform are about additivity and scalar multiplication. For two functions f and g with both their Laplace Transforms existing, from (7.1):

$$\begin{aligned} \mathcal{L}[f + g](s) &= \int_0^{\infty} e^{-sx} [f(x) + g(x)] dx \\ &= \int_0^{\infty} e^{-sx} f(x) dx + \int_0^{\infty} e^{-sx} g(x) dx = \mathcal{L}[f](s) + \mathcal{L}[g](s) \\ \mathcal{L}[cf](s) &= \int_0^{\infty} e^{-sx} cf(x) dx \end{aligned} \quad (7.3)$$

¹Actually, s can be complex-valued, but it is not of our concern here. As a side note, if s is purely imaginary, then Laplace Transform will be reduced to the famous Fourier Transform.

²For more information, see [this MathOverflow post 383](#).

$$= c \int_0^{\infty} e^{-sx} f(x) dx = c \mathcal{L}[f](s) \quad (7.4)$$

where c can be any constant. This is mainly due to the additivity and scalar multiplication for integrals themselves.

7.1.2. Notable Laplace Transform Values

Here we list some of the commonly seen and used Laplace Transform pairs. First, it is obvious that for a constant function:

$$\begin{aligned} \mathcal{L}[1](s) &= \int_0^{\infty} e^{-sx} (1) dx = \int_0^{\infty} e^{-sx} dx \\ &= \left[-\frac{1}{s} e^{-sx} \right]_0^{\infty} \\ &= -\left[(0) - \left(\frac{1}{s} \right) \right] = \frac{1}{s} \end{aligned} \quad (7.5)$$

For positive integer powers of x , i.e. x^n , $n \geq 1$, recursively using integration by parts leads to

$$\begin{aligned} \mathcal{L}[x^n](s) &= \int_0^{\infty} e^{-sx} x^n dx \\ &= \int_0^{\infty} -\frac{1}{s} x^n d(e^{-sx}) \\ &= \left[-\frac{1}{s} x^n e^{-sx} \right]_0^{\infty} - \int_0^{\infty} \left(-\frac{1}{s} e^{-sx} \right) d(x^n) \\ &= [(0) - (0)] + \int_0^{\infty} \frac{1}{s} e^{-sx} n x^{n-1} dx \\ &= \frac{n}{s} \int_0^{\infty} e^{-sx} x^{n-1} dx \\ &= \frac{n}{s} \mathcal{L}[x^{n-1}](s) \\ &= \frac{n(n-1)}{s^2} \mathcal{L}[x^{n-2}](s) \\ &= \frac{n!}{s^n} \mathcal{L}[1](s) = \frac{n!}{s^{n+1}} \end{aligned} \quad (7.6)$$

where in the last step we just use (7.5).

For the exponential function e^{ax} , we have

$$\mathcal{L}[e^{ax}](s) = \int_0^{\infty} e^{-sx} e^{ax} dx$$

$$\begin{aligned}
 &= \int_0^{\infty} e^{-(s-a)x} dx \\
 &= \left[-\frac{1}{s-a} e^{-(s-a)x} \right]_0^{\infty} \\
 &= -\frac{1}{s-a} [(0) - (1)] = \frac{1}{s-a}
 \end{aligned} \tag{7.7}$$

where we need $s > a$ for this integral to converge.

Meanwhile, for sines and cosines like in the form of $\sin(ax)$, its Laplace Transform is inferred by doing integration by parts twice:

$$\begin{aligned}
 \mathcal{L}[\sin(ax)](s) &= \int_0^{\infty} e^{-sx} \sin(ax) dx \\
 &= \int_0^{\infty} -\frac{1}{a} e^{-sx} d(\cos(ax)) \\
 &= \left[-\frac{1}{a} e^{-sx} \cos(ax) \right]_0^{\infty} + \int_0^{\infty} \frac{1}{a} \cos(ax) d(e^{-sx}) \\
 &= \frac{1}{a} [(0) - (1)] - \int_0^{\infty} \frac{s}{a} e^{-sx} \cos(ax) dx \\
 &= \frac{1}{a} - \int_0^{\infty} \frac{s}{a^2} e^{-sx} d(\sin(ax)) \\
 &= \frac{1}{a} - \frac{s}{a^2} [e^{-sx} \sin(ax)]_0^{\infty} + \frac{s}{a^2} \int_0^{\infty} \sin(ax) d(e^{-sx}) \\
 &= \frac{1}{a} - \frac{s}{a^2} (0) - \frac{s^2}{a^2} \int_0^{\infty} e^{-sx} \sin(ax) dx \\
 &= \frac{1}{a} - \frac{s^2}{a^2} \mathcal{L}[\sin(ax)](s)
 \end{aligned} \tag{7.8}$$

So

$$\begin{aligned}
 \mathcal{L}[\sin(ax)](s) &= \frac{1}{a} - \frac{s^2}{a^2} \mathcal{L}[\sin(ax)](s) \\
 \left(1 + \frac{s^2}{a^2}\right) \mathcal{L}[\sin(ax)](s) &= \frac{1}{a} \\
 \mathcal{L}[\sin(ax)](s) &= \frac{1}{a} \frac{1}{1 + \frac{s^2}{a^2}} = \frac{1}{a} \frac{a^2}{s^2 + a^2} = \frac{a}{s^2 + a^2}
 \end{aligned} \tag{7.9}$$

In the same fashion, we should also get

$$\mathcal{L}[\cos(ax)](s) = \frac{s}{s^2 + a^2} \tag{7.10}$$

For the hyperbolic functions, we simply apply (7.7), (7.3), (7.4):

$$\mathcal{L}[\sinh(ax)](s) = \mathcal{L}\left[\frac{e^{ax} - e^{-ax}}{2}\right](s)$$

7.1. Basic Ideas about Laplace Transform ■

$$\begin{aligned}
 &= \frac{1}{2} \mathcal{L}[e^{ax}](s) - \frac{1}{2} \mathcal{L}[e^{-ax}](s) \\
 &= \frac{1}{2} \frac{1}{s-a} - \frac{1}{2} \frac{1}{s+a} \\
 &= \frac{a}{s^2 - a^2}
 \end{aligned} \tag{7.11}$$

and similarly

$$\mathcal{L}[\cosh(ax)](s) = \mathcal{L}\left[\frac{e^{ax} + e^{-ax}}{2}\right](s) = \frac{s}{s^2 - a^2} \tag{7.12}$$

It is not easy to compute an inverse Laplace transformation directly, which involves complex contour integration using the *Bromwich formula*. Rather, we usually consult a table of Laplace transform pairs that includes like (7.5)-(7.12) above and just recognize the expressions. This is given as Table 7.1 below.

Functions	Laplace Transforms
1	$\frac{1}{s}$
x^n	$\frac{n!}{s^{n+1}}$
e^{ax}	$\frac{1}{s-a}$
$\sin(ax)$	$\frac{a}{s^2 + a^2}$
$\cos(ax)$	$\frac{s}{s^2 + a^2}$
$\sinh(ax)$	$\frac{a}{s^2 - a^2}$
$\cosh(ax)$	$\frac{s}{s^2 - a^2}$
$x^{-\frac{1}{2}}$	$\sqrt{\frac{\pi}{s}}$ (see Exercise 7.7)

Table 7.1.: A table summarizing commonly used Laplace Transform pairs.

Finally, we note that the Laplace Transform $\mathcal{L}[f](s)$ of a function f , if it exists, will always tend to 0 as $s \rightarrow \infty$ under some mild condition imposed. This can be taken advantage of as a sanity check. [This StackExchange post 2949958](#) shows how to prove this.

Example 7.1.

Find the Laplace Transforms of the following functions (in t):

1. $t^3 - 4t^2 + 1$;
2. $5 \sin(3t) - 3 \cos(5t)$;
3. $e^{-2t} + \sinh(2t)$.

as well as the inverse Laplace Transforms of:

1. $\frac{5}{s^4} + \frac{3}{s^2} - \frac{2}{s-1}$;
2. $\frac{4}{s^2+9}$;
3. $\frac{1}{s^4-1}$ (Hint: Partial Fractions).

Solution. For the first part, the answers are

1. $\frac{3!}{s^4} - 4\left(\frac{2!}{s^3}\right) + \frac{1}{s} = \frac{6}{s^4} - \frac{8}{s^3} + \frac{1}{s}$;
2. $5\left(\frac{3}{s^2+3^2}\right) - 3\left(\frac{s}{s^2+5^2}\right) = \frac{15}{s^2+9} - \frac{3s}{s^2+25}$;
3. $\frac{1}{s-(-2)} + \frac{2}{s^2-2^2} = \frac{1}{s+2} + \frac{2}{(s-2)(s+2)} = \frac{s}{s^2-4}$ (notice that the original expression is just $\cosh(2t)$).

For the second part, we have

1. $5\left(\frac{t^3}{3!}\right) + 3\left(\frac{t}{1!}\right) - 2e^t = \frac{5t^3}{6} + 3t - 2e^t$;
2. Notice that it is just $\frac{4}{s^2+3^2}$, so the inverse Laplace transform is $\frac{4}{3} \sin(3t)$;
3. Following the hint, we decompose the expression into partial fractions:

$$\begin{aligned} \frac{1}{s^4-1} &= \frac{1}{2} \frac{1}{s^2-1} - \frac{1}{2} \frac{1}{s^2+1} \\ &= \frac{1}{4} \frac{1}{s-1} - \frac{1}{4} \frac{1}{s+1} - \frac{1}{2} \frac{1}{s^2+1} \end{aligned} \quad (7.13)$$

Hence the desired inverse Laplace transform will be $\frac{1}{4}e^t - \frac{1}{4}e^{-t} - \frac{1}{2} \sin t$.

□

7.2. Properties of Laplace Transform

We also need to know about some convenient properties that Laplace Transform possesses and that facilitate computation.

7.2.1. Heaviside Step/Dirac Delta Function

Before getting into using Laplace Transform to solve ODEs, we shall study two special kinds of functions that frequently show up in such a context. The first one is the **Heaviside step function**, which, as its name suggests, appears as a flat unit rise to the right like a step (see Figure 7.2 below). The definition of the step function is naturally

Definition 7.2.

(Heaviside Step Function) The Heaviside step function is given as

$$H(x) = \begin{cases} 1 & \text{when } x > 0 \\ 0 & \text{when } x < 0 \end{cases} \quad (7.14)$$

The exact value at $x = 0$ depends on the scenario, usually it will be $\frac{1}{2}$ if defined.

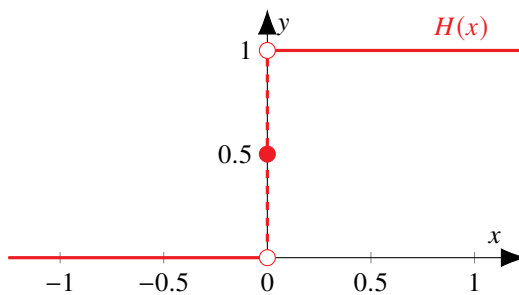


Figure 7.2.: Plotting the Heaviside step function.

Subsequently, the step function $H(x - a)$ (shifted horizontally by a) is then like a switch that turns on after $x = a$. For example, the value of $3xH(x - 2)$ remains 0 for $x < 2$, but becomes just $3x$ for $x > 2$. In this case, at $x = 2$, $3xH(x - 2)$ is discontinuous where the left limit is 0 but the right limit is $3(2) = 6$.

To generalize, we can use the step function to define a piecewise continuous function, formed by gluing different pieces of functions that are themselves continuous, end-to-end at finitely many points. If at such a point it is discontinuous, then at least it should have a finite left limit when approaching from the left, and similarly a finite right limit, although they may not be equal. This is

known as a *simple discontinuity*. Mathematically, if

$$f(x) = \begin{cases} g_1(x) & x < x_1 \\ g_2(x) & x_1 < x < x_2 \\ g_3(x) & x_2 < x < x_3 \\ \vdots & \\ g_n(x) & x_{n-1} < x < x_n \\ \vdots & \end{cases} \quad (7.15)$$

provided all the $g_n(x)$ are continuous over each corresponding interval, then we can rewrite it using the Heaviside step functions as either

$$f(x) = (1 - H(x - x_1))g_1(x) + (H(x - x_1) - H(x - x_2))g_2(x) + \cdots + (H(x - x_{n-1}) - H(x - x_n))g_n(x) + \cdots \quad (7.16a)$$

$$= g_1(x) + H(x - x_1)(g_2(x) - g_1(x)) + \cdots + H(x - x_{n-1})(g_n(x) - g_{n-1}(x)) + \cdots \quad (7.16b)$$

Figure 7.3 below shows an example of a piecewise continuous function that consists of three smaller functions. The Laplace Transform of the step function centered at $x = a$ is then, by definition,

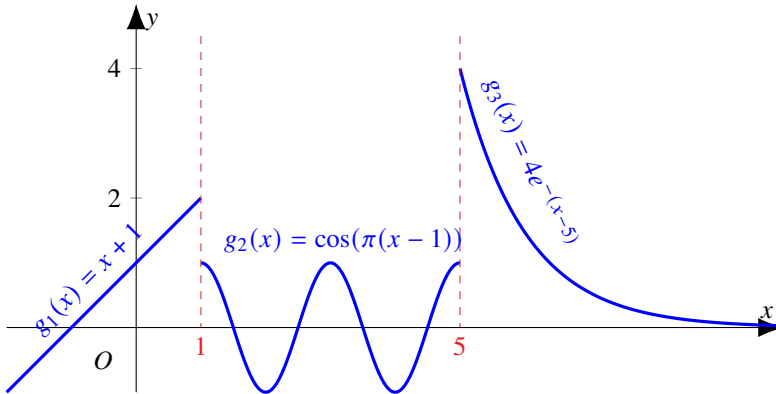


Figure 7.3.: Plotting a piecewise continuous function $f(x) = (1 - H(x - 1))(x + 1) + (H(x - 5) - H(x - 1))\cos(\pi(x - 1)) + 4H(x - 5)e^{-(x-5)}$.

(7.1):

$$\begin{aligned} \mathcal{L}[H(x - a)](s) &= \int_0^{\infty} e^{-sx} H(x - a) dx \\ &= \int_0^a e^{-sx} H(x - a) dx + \int_a^{\infty} e^{-sx} H(x - a) dx \end{aligned}$$

$$\begin{aligned}
 &= \int_0^a e^{-sx}(0)dx + \int_a^\infty e^{-sx}(1)dx && \text{(by (7.14))} \\
 &= 0 + \left[-\frac{1}{s} e^{-sx} \right]_a^\infty \\
 &= \frac{e^{-sa}}{s}
 \end{aligned} \tag{7.17}$$

Another interesting function is the **Dirac Delta function**.

Definition 7.3.

(Dirac Delta) The Dirac Delta function is described as

$$\delta(x) = \begin{cases} \infty & \text{when } x = 0 \\ 0 & \text{else if } x \neq 0 \end{cases} \tag{7.18}$$

and having the property of

$$\int_{-\infty}^{\infty} \delta(x)f(x)dx = f(0) \tag{7.19}$$

While (7.18) is not a really rigorous way to define the delta function, (7.19) is a better definition that characterizes it. From (7.19), we can easily derive two more results: by taking $f(x) = 1$,

$$\int_{-\infty}^{\infty} \delta(x)dx = 1 \tag{7.20}$$

and writing $\tilde{f}(x) = f(x + a)$:

$$\begin{aligned}
 \int_{-\infty}^{\infty} \delta(x)\tilde{f}(x)dx &= \tilde{f}(0) \\
 \int_{-\infty}^{\infty} \delta(x)f(x+a)dx &= f(a) \\
 \int_{-\infty}^{\infty} \delta(x-a)f(x)dx &= f(a) && (x \rightarrow x-a)
 \end{aligned} \tag{7.21}$$

From (7.18) and (7.20), we can visualize the delta function as physically a unit pulse/signal over an infinitesimally small time window, see the schematic in Figure 7.4 below. It can be thought of as a rectangle that has a width of ε and a height of $\frac{1}{\varepsilon}$, where $\varepsilon \rightarrow 0$.

Meanwhile, (7.21) implies that the role of the delta function $\delta(x - a)$ in an integral is to extract the value of $f(x)$ at a fixed point $x = a$, where the integration interval can be any one that includes this point. Hence, sometimes the delta function is also called the **sampling function** in areas like signal processing.

The Laplace Transform of the delta function located at $x = a$ is thus, by (7.1) and (7.21):

$$\mathcal{L}[\delta(x-a)](s) = \int_0^\infty e^{-sx}\delta(x-a)dx = e^{-sa} \tag{7.22}$$

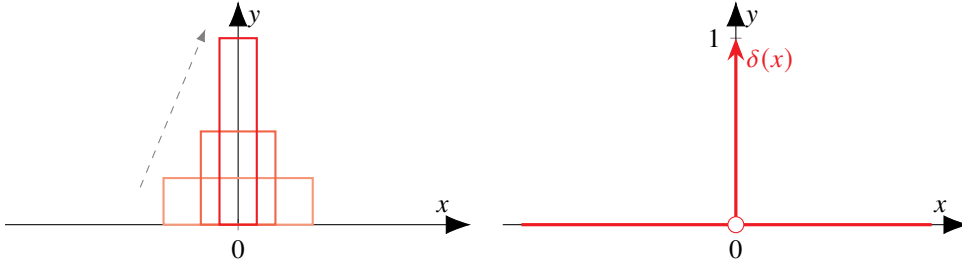


Figure 7.4.: A schematic for the Dirac Delta function.

Relationship between Heaviside Step/Dirac Functions

The derivative of a Heaviside step function readily results in a Dirac function. A rigorous treatment requires distribution theory, but here we will just show it heuristically. Consider some suitable test function $\varphi(x)$ and

$$\begin{aligned} \int_{-\infty}^{\infty} H(x)\varphi'(x)dx &= \int_0^{\infty} \varphi'(x)dx \\ &= -\varphi(0) = -\int_{-\infty}^{\infty} \delta(x)\varphi(x)dx \end{aligned} \quad (7.23)$$

assumed that $\varphi \rightarrow 0$ at $x = \infty$. Recognizing the form of integration by parts, we formally conclude that $\frac{d}{dx}(H(x)) = \delta(x)$.

7.2.2. First and Second Shifting Theorems

There are two **shifting theorems** for the Laplace Transform. Let's denote the Laplace Transform of $f(x)$ as $F(s)$ for convenience. The **first shifting theorem** is relatively easy to get:

$$\begin{aligned} \mathcal{L}[e^{ax}f](s) &= \int_0^{\infty} e^{-sx}e^{ax}f(x)dx \\ &= \int_0^{\infty} e^{-(s-a)x}f(x)dx = F(s-a) \end{aligned} \quad (7.24)$$

and the **second shifting theorem** considers translating the original function $f(x)$, $x > 0$ to the right, truncated to the left by the Heaviside step function:

$$\begin{aligned} \mathcal{L}[H(x-a)f(x-a)](s) &= \int_0^{\infty} e^{-sx}H(x-a)f(x-a)dx \\ &= \int_a^{\infty} e^{-sx}f(x-a)dx \end{aligned} \quad (\text{by (7.14)})$$

$$\begin{aligned}
&= \int_0^{\infty} e^{-s(x+a)} f(x) d(x+a) && (x \rightarrow x+a) \\
&= e^{-sa} \int_0^{\infty} e^{-sx} f(x) dx = e^{-sa} F(s) && (7.25)
\end{aligned}$$

The interpretations for these two theorems are as follows. For (7.25), if we want to shift a function over the physical space, then we have to multiply by e^{-sa} in the Laplace frequency space. On the other hand, regarding (7.24), if we would like to shift a function in the Laplace space, then we ought to multiply the physical function by e^{ax} . These combined are essentially a duality between the two spaces.

Example 7.2.

Find the Laplace Transforms of the following functions:

1. $3e^{2t}t^2$;
2. $e^t \cos(4t)$;
3. $tH(t-2)$.

and the inverse Laplace Transforms of:

1. $\frac{5}{(s+1)^2} + \frac{1}{s-3}$;
2. $\frac{2}{s^2-2s+3}$ (Hint: Completing square);
3. $\frac{e^{-s}}{(s+2)^3}$.

Solution. For the first part,

1. By (7.24), the Laplace Transform will be shifted by 2 to become $\frac{6}{(s-2)^3}$;
2. Again by (7.24), it will be $\frac{s-1}{(s-1)^2+4^2} = \frac{s-1}{s^2-2s+17}$;
3. To use (7.25), here $a=2$ and $t=(t-2)+2$, so the Laplace Transform will be $e^{-2s}\left(\frac{1}{s^2} + \frac{2}{s}\right)$.
(can also be obtained via (7.26), see next subsection).

For the second part,

1. With (7.24), the inverse Laplace Transform is $5te^{-t} + e^{3t}$;
2. Note that $\frac{2}{s^2-2s+3} = \frac{2}{(s-1)^2+(\sqrt{2})^2}$, so the required inverse is $\sqrt{2} \sin(\sqrt{2}t)e^t$;

3. Consider the first shifting theorem (7.24) first, the inverse Laplace transform of $\frac{1}{(s+2)^3}$ is $e^{-2t} \frac{t^2}{2}$. Then, using the second shifting theorem (7.25), the required inverse Laplace Transform for $\frac{e^{-s}}{(s+2)^3}$ will be $H(t-1)e^{-2(t-1)} \frac{(t-1)^2}{2}$. $\square$

7.2.3. Derivative/Integral of Laplace Transform

Next, we will show what happens if we differentiate a given Laplace Transform. By Leibniz Integral Rule, we may differentiate under the integral sign to obtain

$$\begin{aligned} \frac{d}{ds} F(s) &= \frac{d}{ds} \int_0^{\infty} e^{-st} f(t) dt \\ F'(s) &= \int_0^{\infty} \frac{\partial}{\partial s} (e^{-st} f(t)) dt \\ &= \int_0^{\infty} -te^{-st} f(t) dt \\ &= \mathcal{L}[-tf(t)](s) \end{aligned} \quad (7.26)$$

which means that the inverse transform of a derivative in the Laplace domain corresponds to the original physical function being multiplied by $-t$. Extending (7.26) in the opposite direction, we can also inquire about the result of integrating a Laplace transform³:

$$\begin{aligned} -\frac{d}{ds} \int_s^{\infty} F(s) ds &= F(s) = \mathcal{L}[f(t)](s) = -\mathcal{L}\left[-\frac{tf(t)}{t}\right](s) \\ \int_s^{\infty} F(s) ds &= \mathcal{L}\left[\frac{f(t)}{t}\right](s) \end{aligned} \quad (7.27)$$

7.2.4. Convolution Theorem

Finally, the most powerful result regarding Laplace Transform is possibly the **Convolution Theorem**. Of course, we first need to know what a **convolution** is.

Definition 7.4.

(**Convolution**) Convolution is carried out for two functions, let's say $f(x)$ and $g(x)$, and is defined to be

$$[f * g](x) = \int_{-\infty}^{\infty} f(u)g(x-u)du \quad (7.28)$$

³It is not hard to show that for the integration constant to be consistent, the upper limit of the integral must be ∞ .

To understand why the form of convolution is like this, we can imagine $f(x)$ is the distribution of mass/signal (e.g. electric charge, sea surface temperature), while $g(x)$ is the physical response to a unit amount of the signal (impulse) (e.g. electric field due to an electric charge, atmospheric convection due to SST heating). The convolution $[f * g](x)$ is subsequently the overall large-scale physical response pattern arising from all the masses/signals combined (e.g. the net resulting electric field/atmospheric flow throughout the space).

At a fixed location x , the output will consist of all the responses, due to impulses over different u and the reaction to each of them at a displacement of $x - u$ away ($u + (x - u) = x$). This is illustrated in Figure 7.5 below. Notice that convolution is symmetric, so that we can exchange the roles of f and

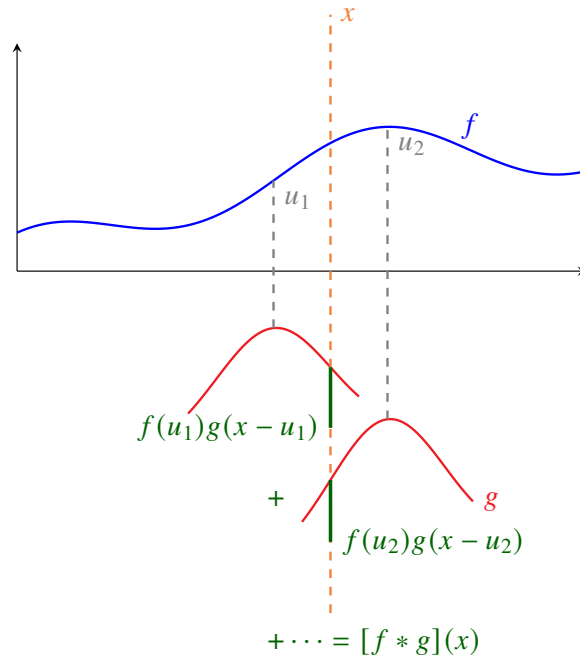


Figure 7.5.: A schematic showing the idea of a convolution.

g as we see fit:

$$\begin{aligned}
 [f * g](x) &= \int_{u=-\infty}^{u=\infty} f(u)g(x-u)du \\
 &= \int_{v=\infty}^{v=-\infty} f(x-v)g(v)d(x-v) & (v = x-u) \\
 &= - \int_{\infty}^{-\infty} f(x-v)g(v)dv
 \end{aligned}$$

$$= \int_{-\infty}^{\infty} f(x-v)g(v)dv \quad (7.29)$$

where u and v are just dummy variables. Notice that, since for the Laplace Transform, the input functions are confined to $x \geq 0$, the lower and upper limit of a convolution (7.28) in the context of Laplace Transform will be truncated to 0 and x , i.e.

$$[f * g](x) = \int_0^x f(u)g(x-u)du \quad (7.30)$$

Now we are ready to show the desired theorem.

Theorem 7.5.

(Convolution Theorem) Denote the Laplace Transforms of $f(t)$ and $g(t)$ by $F(s)$ and $G(s)$. Then, we have

$$\mathcal{L}[f * g](s) = F(s)G(s) \quad (7.31)$$

This implies that the Laplace Transform of a convolution between two functions f, g is simply the product of their respective Laplace Transforms F, G . To show this, start from the R.H.S. (writing the dummy physical integration variables as τ, τ' for the beginning):

$$\begin{aligned} F(s)G(s) &= \left(\int_0^{\infty} e^{-s\tau'} f(\tau') d\tau' \right) \left(\int_0^{\infty} e^{-s\tau} g(\tau) d\tau \right) \\ &= \int_0^{\infty} \int_0^{\infty} e^{-s(\tau'+\tau)} f(\tau') g(\tau) d\tau' d\tau \\ &= \int_0^{\infty} \left(\int_0^{\infty} e^{-s(\tau'+\tau)} f(\tau') d\tau' \right) g(\tau) d\tau \end{aligned} \quad (7.32)$$

Let $\tau' + \tau = t$, so that $\tau' = t - \tau$, where τ is fixed during integration in the inner integral, and $d\tau' = dt$, hence

$$\begin{aligned} \int_0^{\infty} \left(\int_0^{\infty} e^{-s(\tau'+\tau)} f(\tau') d\tau' \right) g(\tau) d\tau &= \int_0^{\infty} \left(\int_{\tau}^{\infty} e^{-st} f(t-\tau) dt \right) g(\tau) d\tau \\ &= \int_0^{\infty} \int_{\tau}^{\infty} e^{-st} f(t-\tau) g(\tau) dt d\tau \end{aligned} \quad (7.33)$$

According to the schematic in Figure 7.6 below, we can switch the order of integration as

$$\begin{aligned} \int_{\tau=0}^{\tau=\infty} \int_{t=\tau}^{t=\infty} e^{-st} f(t-\tau) g(\tau) dt d\tau &= \int_{t=0}^{t=\infty} \int_{\tau=0}^{\tau=t} e^{-st} f(t-\tau) g(\tau) d\tau dt \\ &= \int_{t=0}^{t=\infty} e^{-st} \left(\int_{\tau=0}^{\tau=t} f(t-\tau) g(\tau) d\tau \right) dt \\ &= \int_{t=0}^{t=\infty} e^{-st} (f(t) * g(t)) dt = \mathcal{L}[f * g](s) \end{aligned} \quad (7.34)$$

and we are done.

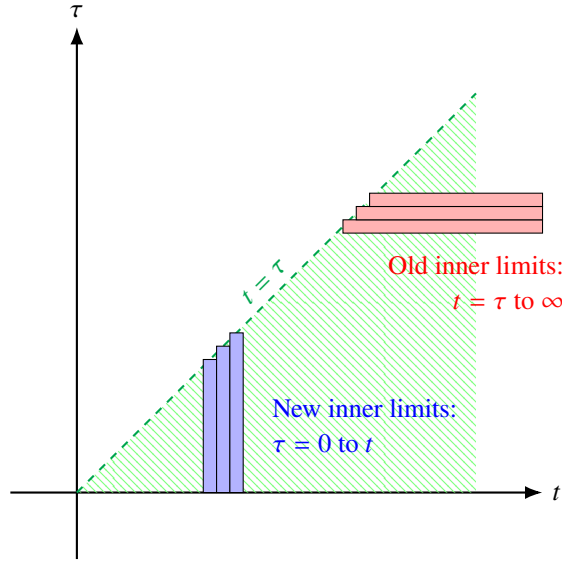


Figure 7.6.: Illustration of switching the order of integration during establishing the convolution theorem.

Laplace Transform of an Integral

By the Convolution Theorem, we can swiftly derive the Laplace Transform of an integral in general: putting $g = 1$ in (7.30) and applying (7.31) gives

$$\mathcal{L}\left[\int_0^t f(\tau) d\tau\right](s) = \mathcal{L}[f * g](s) = F(s)G(s) = \frac{F(s)}{s} \quad (7.35)$$

by recalling that the Laplace transform of 1 is just $\frac{1}{s}$.

Example 7.3.

Find the Inverse Laplace Transform of

$$\frac{s}{(s^2 + 1)^2} = \left(\frac{1}{s^2 + 1}\right)\left(\frac{s}{s^2 + 1}\right) \quad (7.36)$$

by the Convolution Theorem. Hence, deduce the Laplace Transform of $t \sin t$.

Solution. The two factors on the R.H.S. are obviously the Laplace Transforms for $\sin t$ and $\cos t$. By (7.31), the required inverse transform will be

$$\int_0^t \sin(\tau) \cos(t - \tau) d\tau = \int_0^t \frac{1}{2} [\sin(\tau + (t - \tau)) + \sin(\tau - (t - \tau))] d\tau$$

$$\begin{aligned}
 &= \frac{1}{2} \int_0^t (\sin t + \sin(2\tau - t)) d\tau \\
 &= \frac{1}{2} \left[\tau \sin t - \frac{1}{2} \cos(2\tau - t) \right]_0^t \\
 &= \frac{1}{2} (t \sin t + 0) = \frac{1}{2} t \sin t
 \end{aligned} \tag{7.37}$$

where at the start we have used the product-to-sum formula $\sin a \cos b = \frac{1}{2}(\sin(a + b) + \sin(a - b))$. Hence

$$\begin{aligned}
 \mathcal{L}\left[\frac{1}{2}t \sin t\right] &= \frac{s}{(s^2 + 1)^2} \\
 \mathcal{L}[t \sin t] &= \frac{2s}{(s^2 + 1)^2}
 \end{aligned} \tag{7.38}$$

□

7.3. Solving ODEs by Laplace Transform

Now we will finally delve into how Laplace Transform can be applied to solve ODEs, particularly those that are linear.

7.3.1. Laplace Transform of Derivatives

To do so, we first acquire the expressions for the Laplace Transform of first/second-order derivatives: Integrating by parts on the definition of Laplace Transform (7.1) gives

$$\begin{aligned}
 \mathcal{L}[y'](s) &= \int_0^\infty e^{-sx} y' dx \\
 &= \int_0^\infty e^{-sx} d(y) \\
 &= [e^{-sx} y]_0^\infty - \int_0^\infty y d(e^{-sx}) \\
 &= -y(0) + s \int_0^\infty e^{-sx} y dx = sY(s) - y(0)
 \end{aligned} \tag{7.39}$$

and then

$$\begin{aligned}
 \mathcal{L}[y''](s) &= \int_0^\infty e^{-sx} y'' dx \\
 &= \int_0^\infty e^{-sx} d(y')
 \end{aligned}$$

$$\begin{aligned}
 &= [e^{-sx} y']_0^\infty - \int_0^\infty y' d(e^{-sx}) \\
 &= -y'(0) + s \int_0^\infty e^{-sx} y' dx \\
 &= -y'(0) + s(sY(s) - y(0)) = s^2 Y(s) - sy(0) - y'(0) \quad (7.40)
 \end{aligned}$$

where $Y(s)$ is the Laplace transform of y . They suffice for most usages and are worthy to be noted as a standalone property:

Properties 7.6.
(Laplace Transform of First/Second-order Derivatives)

$$\mathcal{L}[y'](s) = sY(s) - y(0) \quad (7.41)$$

$$\mathcal{L}[y''](s) = s^2 Y(s) - sy(0) - y'(0) \quad (7.42)$$

For general n -th order derivatives $y^{(n)}$, by recursion we similarly have

$$\begin{aligned}
 &\mathcal{L}[y^{(n)}](s) \\
 &= \int_0^\infty e^{-sx} y^{(n)} dx \\
 &= \int_0^\infty e^{-sx} d(y^{(n-1)}) \\
 &= [e^{-sx} y^{(n-1)}]_0^\infty - \int_0^\infty y^{(n-1)} d(e^{-sx}) \\
 &= -y^{(n-1)}(0) + s \int_0^\infty e^{-sx} y^{(n-1)} dx \\
 &= -y^{(n-1)}(0) + s(-y^{(n-2)}(0) + s \int_0^\infty e^{-sx} y^{(n-2)} dx) \\
 &= s^{n-1} \int_0^\infty e^{-sx} y' dx - s^{n-2} y'(0) - \dots - sy^{(n-2)}(0) - y^{(n-1)}(0) \\
 &= s^{n-1}(-y(0) + s \int_0^\infty e^{-sx} y dx) - s^{n-2} y'(0) - \dots - sy^{(n-2)}(0) - y^{(n-1)}(0) \\
 &= s^n Y(s) - s^{n-1} y(0) - s^{n-2} y'(0) - \dots - sy^{(n-2)}(0) - y^{(n-1)}(0) \quad (7.43)
 \end{aligned}$$

where $y^{(n)}(0)$ denotes the value of the n -times derivative at $x = 0$.

Then, for example, given a second-order constant-coefficient ODE in the form of

$$y'' + \alpha y' + \beta y = f(x) \quad (7.44)$$

Carrying out Laplace Transform on both sides yields

$$\mathcal{L}[y'' + \alpha y' + \beta y](s) = \mathcal{L}[f](s)$$

$$\begin{aligned}(s^2Y(s) - sy(0) - y'(0)) + \alpha(sY(s) - y(0)) + \beta Y(s) &= \mathcal{L}[f](s) \\ (s^2 + \alpha s + \beta)Y(s) - (s + \alpha)y(0) - y'(0) &= \mathcal{L}[f](s)\end{aligned}\quad (7.45)$$

So, the original ODE is readily converted to an algebraic equation in s that may be easier to solve, and also with the initial conditions automatically incorporated. Consequently, the physical solution is then recovered by an inverse Laplace Transform. If the initial conditions are given at $x = a$ instead where $a \neq 0$, then we can simply make a change of variable $\tilde{x} = x - a$ and the technique will remain the same.

Example 7.4.

Solve the second-order ODE

$$y'' - y = e^{-t} \quad (7.46)$$

by Laplace Transform, where the I.C.s are $y(0) = 0, y'(0) = 1$.

Solution. Applying Laplace Transform on both sides with (7.42) and recalling (7.7), we get

$$\begin{aligned}(s^2Y(s) - sy(0) - y'(0)) - Y(s) &= \frac{1}{s+1} \\ (s^2 - 1)Y(s) - 1 &= \frac{1}{s+1} \\ Y(s) &= \frac{1}{(s^2 - 1)} + \frac{1}{(s+1)(s^2 - 1)}\end{aligned}\quad (7.47)$$

By partial fractions, we have

$$\begin{aligned}\frac{1}{(s^2 - 1)} + \frac{1}{(s+1)(s^2 - 1)} &= \frac{1}{(s-1)(s+1)} + \frac{1}{(s-1)(s+1)^2} \\ &= \frac{1}{2} \left(\frac{1}{s-1} - \frac{1}{s+1} \right) + \left(\frac{1}{4} \frac{1}{s-1} - \frac{1}{4} \frac{1}{s+1} - \frac{1}{2} \frac{1}{(s+1)^2} \right) \\ &= \frac{3}{4} \frac{1}{s-1} - \frac{3}{4} \frac{1}{s+1} - \frac{1}{2} \frac{1}{(s+1)^2}\end{aligned}\quad (7.48)$$

Thus by (7.7) again and (7.24), the recovered solution is

$$y = \frac{3}{4}e^t - \frac{3}{4}e^{-t} - \frac{1}{2}te^{-t} \quad (7.49)$$

□

7.3.2. Solving ODEs with Step/Delta Functions

In many physical scenarios, the ODEs that we are solving may have a source term/forcing on the R.H.S. that is not simply a continuous function but rather consists of step/delta functions, e.g. in the form of a sequence of impulses. Then, exploiting the Laplace Transform of those step/delta functions, as suggested in Section 7.2.1, can save us a lot of trouble. Let's now take a look at the example below.

Example 7.5.

Solve the given ODE

$$\frac{dy}{dx} + \alpha y = (1 - H(x - 1)) \sin(\pi x) \quad (7.50)$$

where $y(0) = 0$ and α is a fixed constant.

Solution. The source term can be viewed as the first half of a sinusoidal wave that is switched off afterwards. By applying Laplace Transform on both sides, following all the previously discussed results, including the second shifting theorem (7.25) (note that $\sin(\pi x) = -\sin(\pi(x - 1))$), we have

$$\begin{aligned} (sY(s) - y(0)) + \alpha Y(s) &= (s + \alpha)Y(s) = \frac{\pi}{s^2 + \pi^2} + \frac{e^{-s}\pi}{s^2 + \pi^2} \\ Y(s) &= \frac{\pi}{(s + \alpha)(s^2 + \pi^2)} + \frac{e^{-s}\pi}{(s + \alpha)(s^2 + \pi^2)} \end{aligned} \quad (7.51)$$

Note that we have the partial fractions:

$$\frac{\pi}{(s + \alpha)(s^2 + \pi^2)} = \frac{\pi}{\pi^2 + \alpha^2} \left(\frac{1}{s + \alpha} - \frac{s - \alpha}{s^2 + \pi^2} \right) \quad (7.52)$$

The inverse Laplace transform of which is

$$\begin{aligned} &\frac{\pi}{\pi^2 + \alpha^2} \left(e^{-\alpha x} - \cos(\pi x) + \frac{\alpha}{\pi} \sin(\pi x) \right) \\ &= \frac{\pi}{\pi^2 + \alpha^2} (e^{-\alpha x} - \cos(\pi x)) + \frac{\alpha}{\pi^2 + \alpha^2} \sin(\pi x) \end{aligned} \quad (7.53)$$

Hence, the answer will be, by applying the second shifting theorem again:

$$\begin{aligned} y &= \frac{\pi}{\pi^2 + \alpha^2} (e^{-\alpha x} - \cos(\pi x)) + \frac{\alpha}{\pi^2 + \alpha^2} \sin(\pi x) \\ &\quad + \frac{H(x - 1)\pi}{\pi^2 + \alpha^2} (e^{-\alpha(x-1)} - \cos(\pi(x - 1))) + \frac{H(x - 1)\alpha}{\pi^2 + \alpha^2} \sin(\pi(x - 1)) \end{aligned} \quad (7.54)$$

□

7.3.3. Example: RC Circuit Impulses

We now look at one more physical example, about switching on/off an *RC circuit*.

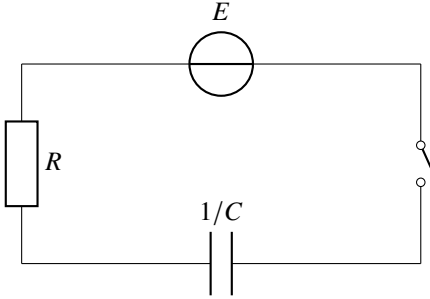


Figure 7.7.: The RC circuit as described in Example 7.6.

Example 7.6.

(Impulses in RC Circuit System) A simple RC circuit can be described

by the ODE

$$R \frac{dq}{dt} + \frac{1}{C} q = E(t) \quad (7.55)$$

This is the same as described by Exercise 3.10 but with the L component discarded and a new source of voltage $E(t)$ present. Consider a case in which initially the switch is open, and there is neither charge nor current (see Figure 7.7). After some time $t = t_1$, the switch is closed, and a constant voltage V_0 is applied, and at a later time $t = t_2$, the switch is open again, and the voltage supply is cut. Find how the charge q and current $I = \frac{dq}{dt}$ change in time. How about if there is a short electric potential pulse $E(t) = V_d \delta(t - t_3)$ instead? Important: the unit of a Dirac delta is the reciprocal of that of the argument inside, hence the Dirac delta here has a unit of s^{-1} , and V_d will have a unit of Vs for consistency.

Solution. For the first case,

$$E(t) = V_0(H(t - t_1) - H(t - t_2)) \quad (7.56)$$

After applying Laplace Transform, the equation becomes

$$\begin{aligned} R(s\mathcal{L}[q]) + \frac{1}{C}\mathcal{L}[q] &= V_0\left(\frac{e^{-st_1}}{s} - \frac{e^{-st_2}}{s}\right) \\ (RCs + 1)\mathcal{L}[q] &= CV_0\left(\frac{e^{-st_1}}{s} - \frac{e^{-st_2}}{s}\right) \\ \mathcal{L}[q] &= CV_0\left(\frac{e^{-st_1}}{s(RCs + 1)} - \frac{e^{-st_2}}{s(RCs + 1)}\right) \end{aligned}$$

$$= CV_0 \left[e^{-st_1} \left(\frac{1}{s} - \frac{1}{s + \frac{1}{RC}} \right) - e^{-st_2} \left(\frac{1}{s} - \frac{1}{s + \frac{1}{RC}} \right) \right] \quad (7.57)$$

Thus, the answer is

$$q = CV_0 \left[H(t - t_1) \left(1 - e^{-\frac{t-t_1}{RC}} \right) - H(t - t_2) \left(1 - e^{-\frac{t-t_2}{RC}} \right) \right] \quad (7.58)$$

For the pulse case, we have

$$\begin{aligned} \left(Rs + \frac{1}{C} \right) \mathcal{L}[q] &= V_d e^{-st_3} \\ \mathcal{L}[q] &= \frac{V_d}{R} \frac{e^{-st_3}}{s + \frac{1}{RC}} \end{aligned} \quad (7.59)$$

Therefore,

$$q = \frac{V_d}{R} H(t - t_3) e^{-\frac{t-t_3}{RC}} \quad (7.60)$$

□

7.4. First Take on Green's Function

Finally, we will introduce the **Green's Function** approach to solving linear ODEs. For any given n -th order linear ODE in the form of

$$a_n(x) \frac{d^n y}{dx^n} + \cdots + a_2(x) \frac{d^2 y}{dx^2} + a_1(x) \frac{dy}{dx} + a_0(x)y = f(x) \quad (7.61)$$

where $a_n(x)$ are the linear coefficients that depend on x , and $f(x)$ is the source term, we may condense the L.H.S. by assigning the linear operator:

$$\mathcal{L} = a_n(x) \frac{d^n}{dx^n} + \cdots + a_2(x) \frac{d^2}{dx^2} + a_1(x) \frac{d}{dx} + a_0(x) \quad (7.62)$$

So that (7.61) simply becomes

$$\mathcal{L}[y](x) = f(x) \quad (7.63)$$

A Green's Function of (7.63) is then some function that also accepts an extra variable, let's say z , so that this $G(x, z)$ satisfies (7.63) if its R.H.S. is replaced by a delta function $\delta(x - z)$:

$$\mathcal{L}[G](x, z) = \delta(x - z) \quad (7.64)$$

with all the differential operations contained in $\mathcal{L}$ accordingly changed to partial derivatives with respect to x . Physically, it means that $G(x, z)$ is the solution to the ODE, as a response of the system when the source term/forcing is solely a unit impulse at $x = z$ while holding z fixed. Of course, the

actual form of $G(x, z)$ also needs to be consistent with the initial/boundary conditions. Particularly, for example, when they are in the homogeneous form of, like $y(a) = y(b) = 0$ or $y(a) = y'(a) = 0$, it will also be $G(a, z) = G(b, z) = 0$ or $G(a, z) = G'(a, z) = 0$.

Subsequently, motivated by the idea of convolution in Section 7.2.4, we propose a solution that is constructed from the sum of signals $f(x)$ and the appropriate physical response to a unit signal $G(x, z)$ as just developed:

$$y = \int_{\Omega} G(x, z) f(z) dz \quad (7.65)$$

where Ω denotes the domain/interval. Then

$$\begin{aligned} \mathcal{L}[y](x) &= \mathcal{L}\left[\int_{\Omega} G(x, z) f(z) dz\right] \\ &= \int_{\Omega} \mathcal{L}[G](x, z) f(z) dz \\ &= \int_{\Omega} \delta(x - z) f(z) dz = f(x) \end{aligned} \quad (7.66)$$

where we assume that the differential operator $\mathcal{L}$ can be put inside the integral and have used (7.64) and the defining property of the delta function (7.21). This verifies that (7.65) is indeed a valid solution to (7.63), given that the initial/boundary conditions are obeyed properly.

Remember that the delta function in (7.64) can be conveniently dealt with the Laplace Transform as suggested in Section 7.2.1. This also circumvents a potential problem regarding the discontinuity of the Green's function, which we will concern ourselves with in the next chapter. A final note is that the Green's function method is actually equivalent to the so-called *Duhamel's Principle*.

Example 7.7.

Solve the second-order ODE

$$\frac{d^2 y}{dx^2} + y = \tan x \quad (7.67)$$

by finding its Green's function, where it obeys the I.C. $y(0) = y'(0) = 0, x \geq 0$.

Solution. The Green's function is computed according to (7.64) and then by applying Laplace Transform on the x argument:

$$\begin{aligned} \frac{d^2}{dx^2} [G(x, z)] + G(x, z) &= \delta(x - z) \\ s^2 \mathcal{L}[G](s, z) - sG(0, z) - G'(0, z) + \mathcal{L}[G](s, z) &= e^{-sz} \\ (s^2 + 1) \mathcal{L}[G](s, z) &= e^{-sz} \\ \mathcal{L}[G](s, z) &= \frac{e^{-sz}}{s^2 + 1} \end{aligned} \quad (7.68)$$

Inverting then yields

$$G(x, z) = H(x - z) \sin(x - z) \quad (7.69)$$

or equivalently

$$G(x, z) = \begin{cases} 0 & \text{when } x < z \\ \sin(x - z) & \text{when } x > z \end{cases} \quad (7.70)$$

Therefore, by (7.65), the desired solution is

$$\begin{aligned} y &= \int_0^\infty G(x, z) f(z) dz \\ &= \int_0^x \sin(x - z) \tan z dz \\ &= \int_0^x (\sin x \cos z - \cos x \sin z) \tan z dz \\ &= \int_0^x \left(\sin x \sin z - \cos x \frac{\sin^2 z}{\cos z} \right) dz \\ &= \int_0^x \left(\sin x \sin z - \cos x \frac{1 - \cos^2 z}{\cos z} \right) dz \\ &= [-\sin x \cos z]_0^x - \int_0^x \cos x (\sec z - \cos z) dz \\ &= \sin x (1 - \cos x) - \cos x [\ln |\sec z + \tan z| - \sin z]_0^x \\ &= \sin x (1 - \cos x) - \cos x (\ln |\sec x + \tan x| - \sin x) \\ &= \sin x - \cos x \ln |\sec x + \tan x| \end{aligned} \quad (7.71)$$

where we have made use of several trigonometric identities. □

7.5. Exercises

7.1) Find the Laplace Transforms of the following functions. where

(a) $3t^5 + \frac{1}{2}t^3 - 5e^{-\frac{t}{2}};$

(b) $\sinh(3t) \cos t;$

(c) $H(t-3)(\delta(t-1) + \delta(t-5));$

$$f(t) = \begin{cases} 0 & 0 \leq t < 2 \\ 1 & 2 \leq t < 5 \\ 4 & t \geq 5 \end{cases} \quad (7.76)$$

with the I.C.s $y(0) = 1, y'(0) = 0$.

7.2) Find the inverse Laplace Transforms of the following functions.

(a) $\frac{1}{s^3 - 1};$

(b) $\frac{e^{-s}}{s^2 - 4s + 5};$

(c) $\tan^{-1}\left(\frac{s}{3}\right)$ (Hint: Consider (7.27)).

7.7) Derive the Laplace Transform of $t^{-\frac{1}{2}} = \frac{1}{\sqrt{t}}$. Hint: by definition

$$\mathcal{L}[t^{-\frac{1}{2}}](s) = \int_0^{\infty} e^{-st} t^{-\frac{1}{2}} dt \quad (7.77)$$

You want to show that by making the substitution $z = \sqrt{t}$, we have

$$\int_0^{\infty} e^{-st} t^{-\frac{1}{2}} dt = 2 \int_0^{\infty} e^{-sz^2} dz \quad (7.78)$$

7.3) Similar to Example 7.3, by calculating the inverse Laplace Transform of

$$\frac{s^2}{(s^2 + 1)^2} = \left(\frac{s}{s^2 + 1}\right)^2 \quad (7.72)$$

deduce the Laplace Transform of $t \cos t$.

Now, refer to Appendix B.1 for the value of the Gaussian Integral. Hence derive the formula for Laplace Transform of $t^{n-\frac{1}{2}}$, where n is a positive integer, using the property (7.26).

7.4) Solve the ODE

$$y'' - 4y' + 5y = 4 \cos t \quad (7.73)$$

with the initial conditions $y(0) = \frac{1}{2}, y'(0) = 0$ by Laplace Transform.

7.8) Use the method of Green's function to derive the form of the particular solution of

$$\frac{d^2 y}{dx^2} - y = f(x) \quad (7.79)$$

where $f(x)$ can be any arbitrary function in x , with the initial conditions of $y(0) = 0, y'(0) = 1$ that are inhomogeneous, and hence a suitable change of variable is needed.

7.5) Find the full solution to the ODE

$$y'' + 2y' + 2y = e^{-t} \sin t \quad (7.74)$$

with the initial conditions $y(0) = 1, y'(0) = 1$ using Laplace Transform.

7.9) Solve the integral equation

$$y'' + y = 1 + \int_0^t y(\tau) \sin(t - \tau) d\tau \quad (7.80)$$

7.6) Find the solution to

$$y'' + \frac{9}{4}y = f(t) \quad (7.75) \quad \text{with the I.C.s } y(0) = y'(0) = 0.$$